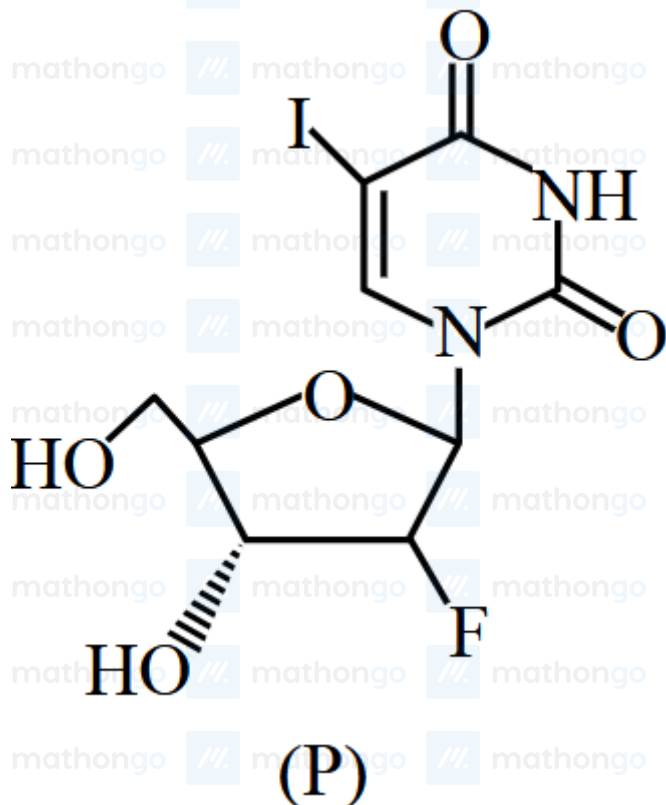


Q1. The amount of calcium oxide produced on heating 150 kg limestone (75% pure) is _____ kg. (Nearest integer)
Given : Molar mass (in g mol^{-1}) of Ca-40, O-16, C-12

Q2. 20 mL of sodium iodide solution gave 4.74 g silver iodide when treated with excess of silver nitrate solution. The molarity of the sodium iodide solution is _____ M. (Nearest Integer value) (Given : Na = 23, I = 127, Ag = 108, N = 14, O = 16 g mol^{-1})

Q3. 0.1 mol of the following given antiviral compound (P) will weigh _____ $\times 10^{-1}$ g



(Given : molar mass in g mol^{-1} H : 1, C : 12, N : 14, O : 16, F : 19, I : 127)

Q4. Among 10^{-9} g (each) of the following elements, which one will have the highest number of atoms? Element : Pb, Po, Pr and Pt

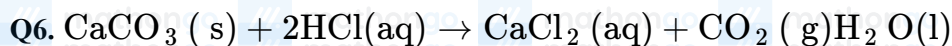
(1) Po

(2) Pr

(3) Pb

(4) Pt

Q5. Fortification of food with iron is done using $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$. The mass in grams of the $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ required to achieve 12 ppm of iron in 150 kg of wheat is _____ (Nearest integer)
[Given : Molar mass of Fe, S and O respectively are 56, 32 and 16 g mol^{-1}]



Consider the above reaction, what mass of CaCl_2 will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of CaCO_3 ?

(Given : Molar mass of Ca, C, O, H and Cl are 40, 12, 16, 1 and 35.5 g mol^{-1} , respectively)

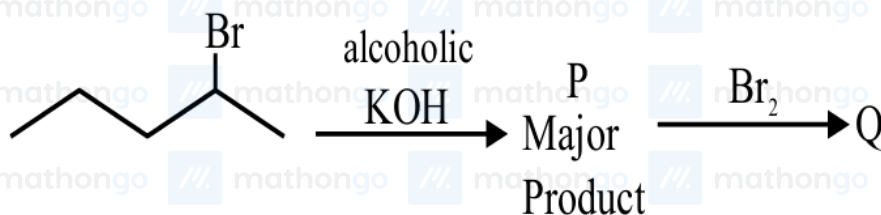
(1) 3.908 g

(2) 2.636 g

(3) 10.545 g

(4) 5.272 g

Q7.



Consider the above sequence of reactions. 151 g of 2-bromopentane is made to react. Yield of major product P is 80% whereas Q is 100%.

Mass of product Q obtained is ____ g.

(Given molar mass in gmol^{-1} H : 1, C : 12, O : 16, Br : 80)

Q8. Mass of magnesium required to produce 220 mL of hydrogen gas at STP on reaction with excess of dil. HCl is

Given : Molar mass of Mg is 24 g mol^{-1} .

(1) 235.7 g

(2) 0.24 mg

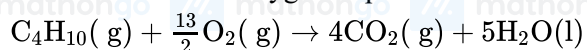
(3) 236 mg

(4) 2.444 g

Q9. An organic compound weighing 500 mg, produced 220 mg of CO_2 on complete combustion. The percentage composition of carbon in the compound is ____ %. (nearest integer)

(Given molar mass in gmol^{-1} of C : 12, O : 16)

Q10. Butane reacts with oxygen to produce carbon dioxide and water following the equation given below



If 174.0 kg of butane is mixed with 320.0 kg of O_2 , the volume of water formed in litres is _____. (Nearest integer)

[Given : (a) Molar mass of C, H, O are 12, 1, 16 g mol^{-1} respectively, (b) Density of water = 1 g mL^{-1}]

Q11. On combustion 0.210 g of an organic compound containing C, H and O gave 0.127 gH₂O and 0.307 gCO₂. The percentages of hydrogen and oxygen in the given organic compound respectively are:

(1) 53.41, 39.6

(2) 6.72, 53.41

(3) 7.55, 43.85

(4) 6.72, 39.87

Q12. On complete combustion 1.0 g of an organic compound (X) gave 1.46 g of CO₂ and 0.567 g of H₂ O. The empirical formula mass of compound (X) is _____ g.

(Given molar mass in g mol⁻¹ C : 12, H : 1, O : 16)

(1) 30

(2) 45

(3) 60

(4) 15

Q1. According to Bohr's model of hydrogen atom, which of the following statement is incorrect?

- (1) Radius of 3rd orbit is nine times larger than that of 1st orbit.
- (2) Radius of 8th orbit is four times larger than that of 4th orbit.
- (3) Radius of 6th orbit is three times larger than that of 4th orbit.
- (4) Radius of 4th orbit is four times larger than that of 2nd orbit.

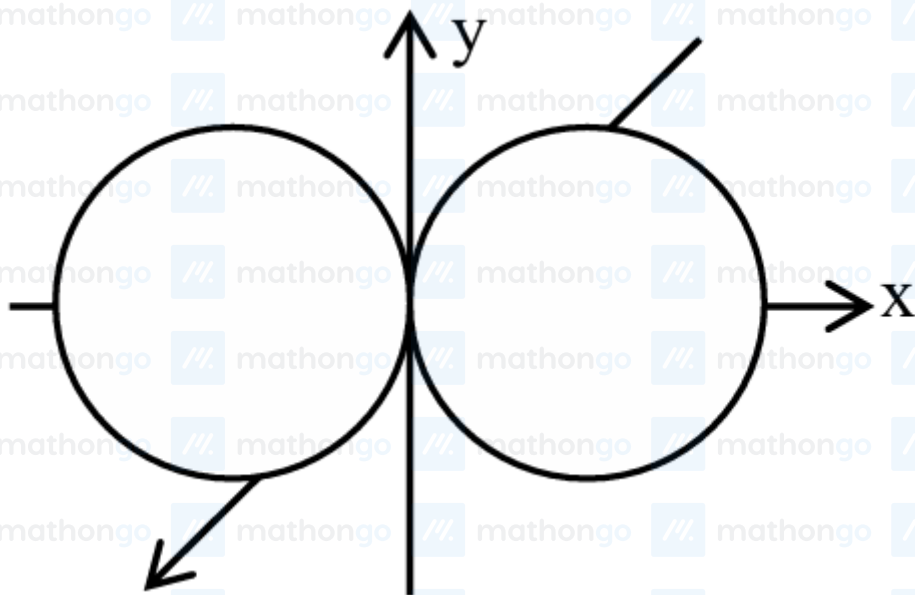
Q2. For electron in '2 s' and '2 p' orbitals, the orbital angular momentum values, respectively are :

- (1) $\sqrt{2} \frac{h}{2\pi}$ and 0
- (2) $\frac{h}{2\pi}$ and $\sqrt{2} \frac{h}{2\pi}$
- (3) 0 and $\sqrt{6} \frac{h}{2\pi}$
- (4) 0 and $\sqrt{2} \frac{h}{2\pi}$

Q3. The energy of an electron in first Bohr orbit of H -atom is -13.6 eV. The magnitude of energy value of electron in the first excited state of Be³⁺ is _____ eV. (nearest integer value)

Q4. Which of the following statements are true?

- (A) The subsidiary quantum number l describes the shape of the orbital occupied by the electron.
- (B)



is the boundary surface diagram of the $2p_x$ orbital.

- (C) The + and - signs in the wave function of the $2p_x$ orbital refer to charge.
- (D) The wave function of $2p_x$ orbital is zero everywhere in the xy plane.

- (1) (B) and (D) only

(2) (A), (B) and (C) only

(3) (C) and (D) only

(4) (A) and (B) only

Q5. Which of the following postulate of Bohr's model of hydrogen atom is not in agreement with quantum mechanical model of an atom ?

(1) An atom in a stationary state does not emit electromagnetic radiation as long as it stays in the same state

(2) An atom can take only certain distinct energies E_1, E_2, E_3 , etc. These allowed states of constant energy are called the stationary states of atom

(3) When an electron makes a transition from a higher energy stationary state to a lower energy stationary state, then it emits a photon of light

(4) The electron in a H atom's stationary state moves in a circle around the nucleus

Q6. Which one of the following about an electron occupying the 1 s orbital in a hydrogen atom is incorrect ? (Bohr's radius is represented by a a_0)

(1) The probability density of finding the electron is maximum at the nucleus

(2) The electron can be found at a distance $2a_0$ from the nucleus

(3) The 1 s orbital is spherically symmetrical

(4) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus

Q7. Consider the ground state of chromium atom ($Z = 24$). How many electrons are with Azimuthal quantum number $l = 1$ and $l = 2$ respectively ?

(1) 12 and 4

(2) 16 and 4

(3) 12 and 5

(4) 16 and 5

Q8. Correct statements for an element with atomic number 9 are

A. There can be 5 electrons for which $m_s = +\frac{1}{2}$ and 4 electrons for which $m_s = -\frac{1}{2}$

B. There is only one electron in p_z orbital

C. The last electron goes to orbital with $n = 2$ and $l = 1$

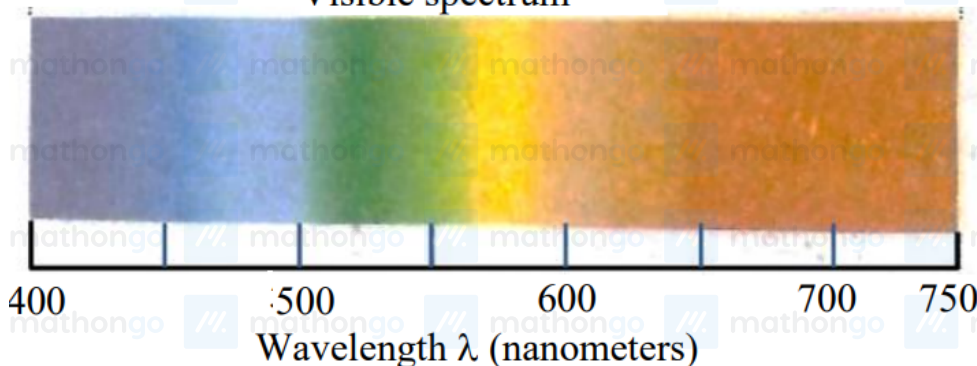
4. The sum of angular nodes of all the atomic orbitals is 1.

Choose the correct answer from the options given below:

- (1) C and D Only
(2) A and C Only
(3) A, C and D Only
(4) A and B Only

Q9.

Visible spectrum



Which of the following statements are correct, if the threshold frequency of caesium is 5.16×10^{14} Hz ?

- A. When Cs is placed inside a vacuum chamber with an ammeter connected to it and yellow light is focused on Cs the ammeter shows the presence of current.
B. When the brightness of the yellow light is dimmed, the value of the current in the ammeter is reduced.
C. When a red light is used instead to the yellow light, the current produced is higher with respect to the yellow light.
D. When a blue light is used, the ammeter shows the formation of current.
E. When a white light is used, the ammeter shows formation of current.

Choose the correct answer from the options given below :

- (1) A, D and E Only
(2) B, C and D Only
(3) A, C, D and E Only
(4) A, B, D and E Only

Q10. The extra stability of half-filled subshell is due to

- (A) Symmetrical distribution of electrons
(B) Smaller coulombic repulsion energy
(C) The presence of electrons with the same spin in non-degenerate orbitals
(D) Larger exchange energy
(E) Relatively smaller shielding of electrons by one another

Identify the correct statements

- (1) (B), (D) and (E) only
(2) (A), (B), (D) and (E) only

(3) (B), (C) and (D) only

(4) (A), (B) and (D) only

Q1. At the sea level, the dry air mass percentage composition is given as nitrogen gas : 70.0, oxygen gas : 27.0 and argon gas : 3.0. If total pressure is 1.15 atm , then calculate the ratio of followings respectively :

(i) partial pressure of nitrogen gas to partial pressure of oxygen gas

(ii) partial pressure of oxygen gas to partial pressure of argon gas

(Given : Molar mass of N, O and Ar are 14, 16, and 40 g mol⁻¹ respectively)

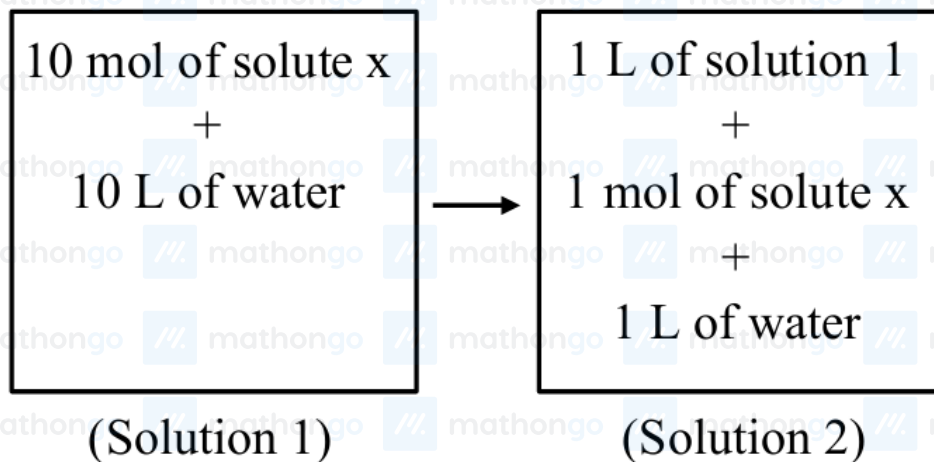
(1) 4.26, 19.3

(2) 2.59, 11.85

(3) 5.46, 17.8

(4) 2.96, 11.2

Q1. Which of the following properties will change when system containing solution 1 will become solution 2?



(1) Molar heat capacity

(2) Density

(3) Concentration

(4) Gibbs free energy

Q2. The hydration energies of K^+ and Cl^- are $-x$ and $-y$ kJ/mol respectively. If lattice energy of KCl is $-z$ kJ/mol, then the heat of solution of KCl is :

(1) $+x - y - z$

(2) $x + y + z$

(3) $z - (x + y)$

(4) $-z - (x + y)$

Q3. Arrange the following in order of magnitude of work done by the system / on the system at constant temperature :

(a) $|w_{\text{reversible}}|$ for expansion in infinite stage.

(b) $|w_{\text{irreversible}}|$ for expansion in single stage.

(c) $|w_{\text{reversible}}|$ for compression in infinite stage.

(d) $|w_{\text{irreversible}}|$ for compression in single stage.

Choose the correct answer from the options given below:

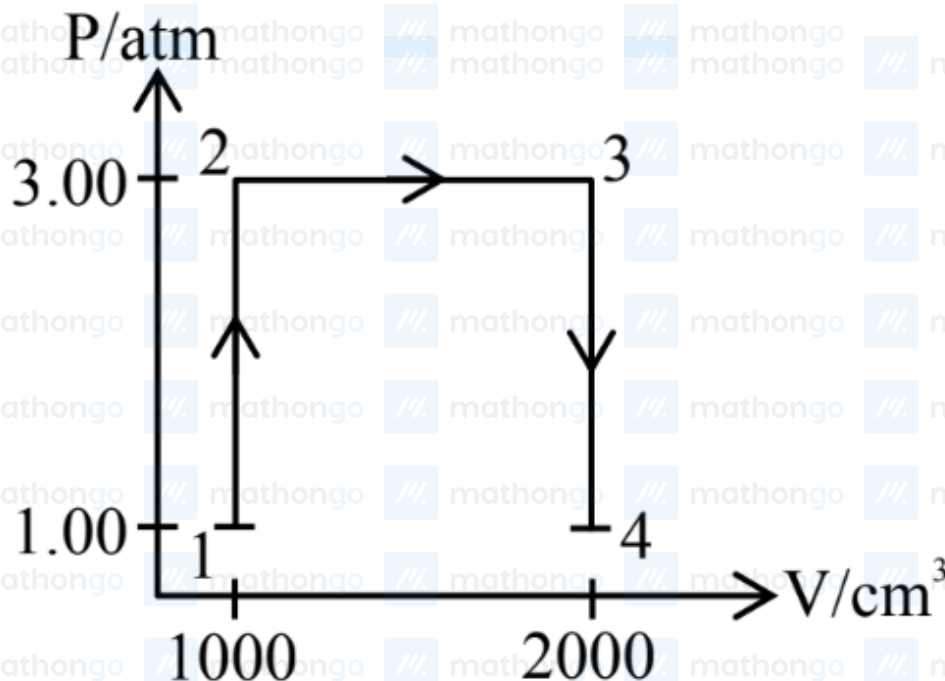
(1) $a > b > c > d$

(2) $d > c = a > b$

(3) $c = a > d > b$

(4) $a > c > b > d$

Q4.



A perfect gas (0.1 mol) having $\bar{C}_v = 1.50R$ (independent of temperature) undergoes the above transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is
(-) _____ J (nearest integer) [Given : $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

Q5. One mole of an ideal gas expands isothermally and reversibly from 10dm^3 to 20dm^3 at 300 K. ΔU , q and work done in the process respectively are :

Given : $R = 8.3\text{JK}^{-1} \text{ mol}^{-1}$

$\ln 10 = 2.3$

$\log 2 = 0.30$

$\log 3 = 0.48$

(1) 0, 21.84 kJ, -1.26 kJ

(2) 0, -17.18 kJ, 1.718 J

(3) 0, 21.84 kJ, 21, 84 kJ

(4) 0, 178 kJ, -1.718 kJ

Q6. Let us consider a reversible reaction at temperature, T.

In this reaction, both ΔH and ΔS were observed to have positive values. If the equilibrium temperature is T_e , then the reaction becomes spontaneous at :

(1) $T = T_e$

(2) $T_e > T$

(3) $T > T_e$

(4) $T_e = 5 T$

Q7. Given :

$$\Delta H_{\text{sub}}^{\ominus} [\text{C}(\text{graphite})] = 710 \text{ kJ mol}^{-1}$$

$$\Delta_{\text{C-H}}^{\ominus} = 414 \text{ kJ mol}^{-1}$$

$$\Delta_{\text{H-H}}^{\ominus} = 436 \text{ kJ mol}^{-1}$$

$$\Delta_{\text{C=C}}^{\ominus} = 611 \text{ kJ mol}^{-1}$$

The $\Delta H_{\text{f}}^{\ominus}$ for $\text{CH}_2 = \text{CH}_2$ is _____ kJ mol^{-1} (nearest integer value)

Q8. Given below are two statements :

Statement I : When a system containing ice in equilibrium with water (liquid) is heated, heat is absorbed by the system and there is no change in the temperature of the system until whole ice gets melted.

Statement II : At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of molecules is not increased at melting point.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

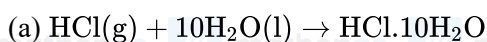
(2) Both Statement I and Statement II are false

(3) Both Statement I and Statement II are true

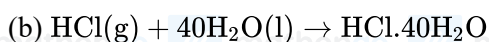
(4) Statement I is false but Statement II is true

Q9. A sample of n-octane (1.14 g) was completely burnt in excess of oxygen in a bomb calorimeter, whose heat capacity is 5 kJ K^{-1} . As a result of combustion reaction, the temperature of the calorimeter is increased by 5 K. The magnitude of the heat of combustion of octane at constant volume is _____ kJ mol^{-1} (nearest integer).

Q10. Consider the given data :



$$\Delta H = -69.01 \text{ kJ mol}^{-1}$$



$$\Delta H = -72.79 \text{ kJ mol}^{-1}$$

Choose the correct statement :

(1) Dissolution of gas in water is an endothermic process

(2) The heat of solution depends on the amount of solvent.

(3) The heat of dilution for the HCl ($\text{HCl} \cdot 10\text{H}_2\text{O}$ to $\text{HCl} \cdot 40\text{H}_2\text{O}$) is 3.78 kJ mol^{-1} .

(4) The heat of formation of HCl solution is represented by both (a) and (b)

Q11. Total enthalpy change for freezing of 1 mol of water at 10°C to ice at -10°C is _____

(Given : $\Delta_{\text{fus}} H = x \text{ kJ/mol}$)

$$C_p [\text{H}_2\text{O}(\text{l})] = y \text{ J mol}^{-1} \text{ K}^{-1}$$

$$C_p [\text{H}_2\text{O}(\text{s})] = z \text{ J mol}^{-1} \text{ K}^{-1}$$

(1) $-x - 10y - 10z$

(2) $-10(100x + y + z)$

(3) $10(100x + y + z)$

(4) $x - 10y - 10z$

Q12. The correct statement amongst the following is :

(1) The term 'standard state' implies that the temperature is 0°C

(2) The standard state of pure gas is the pure gas at a pressure of 1 bar and temperature 273 K

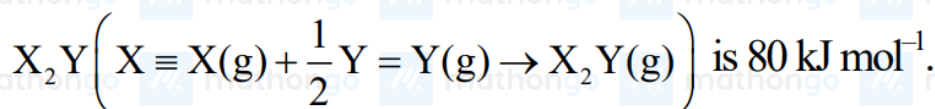
(3) $\Delta_f H_{298}^\theta$ is zero for $\text{O}(\text{g})$

(4) $\Delta_f H_{500}^\theta$ is zero for $\text{O}_2(\text{g})$

Q13. Resonance in X_2Y can be represented as



The enthalpy of formation of



The magnitude of resonance energy of X_2Y is

_____ kJ mol^{-1} (nearest integer value)

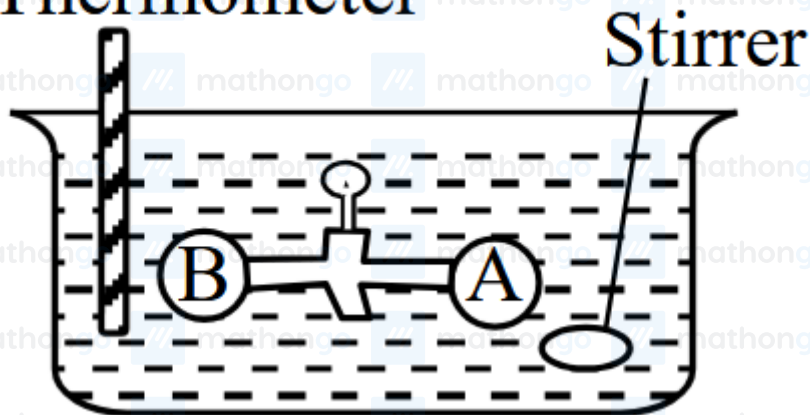
Given : Bond energies of $\text{X} \equiv \text{X}$, $\text{X} = \text{X}$, $\text{Y} = \text{Y}$ and

$\text{X} = \text{Y}$ are 940, 410, 500 and 602 kJ mol^{-1}

respectively.

valence X : 3, Y : 2

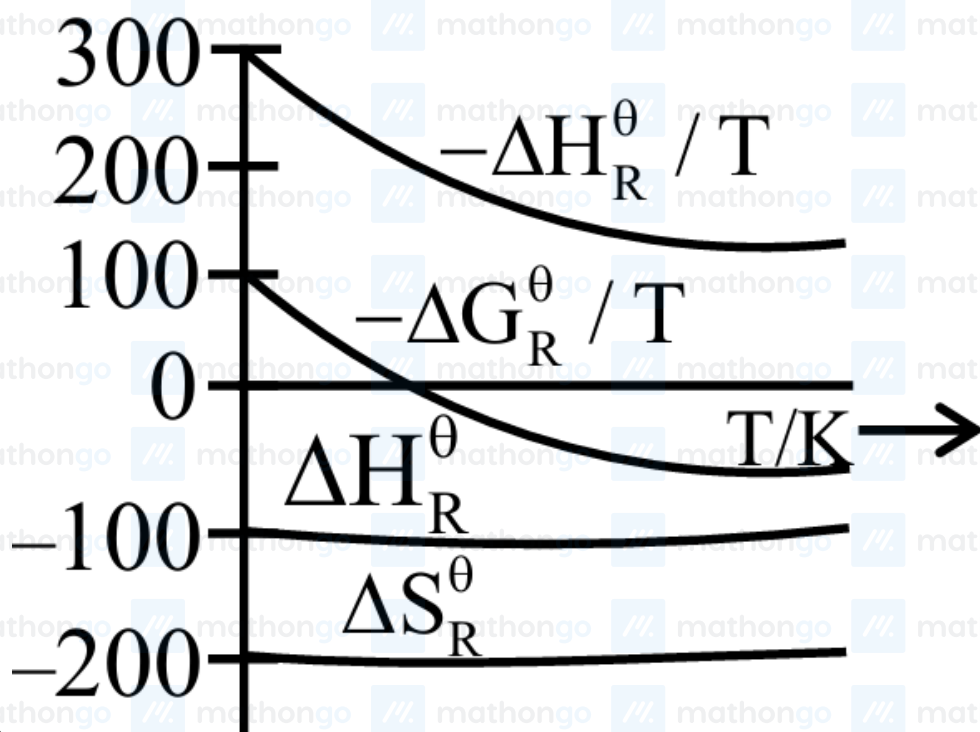
Q14. Thermometer



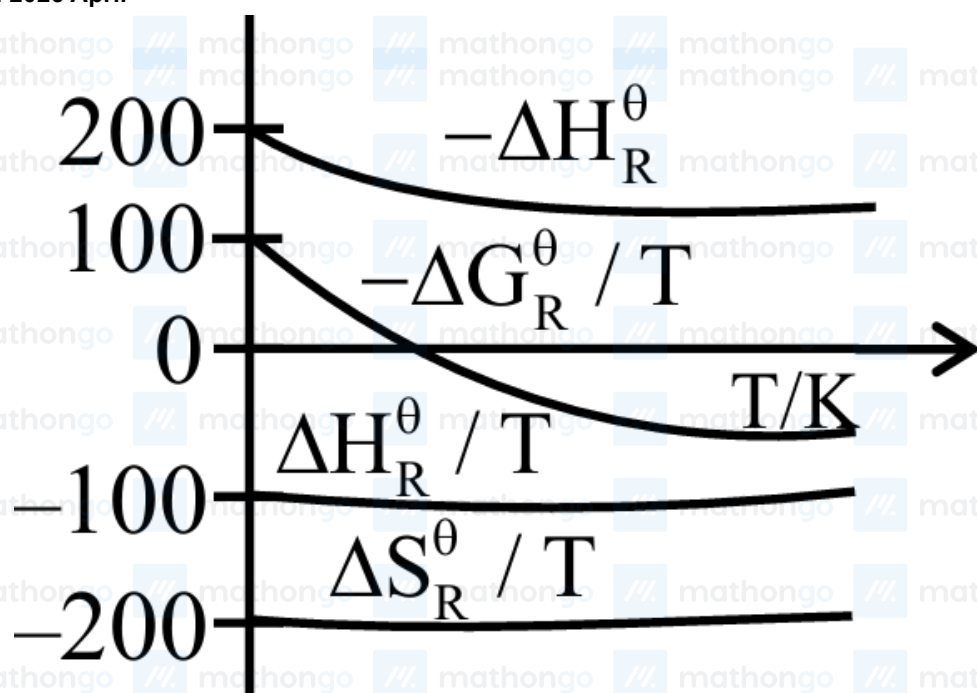
Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true?

- (1) $dw' = 0$
- (2) $dq' = 0$
- (3) $dU = 0$
- (4) The pressure in the vessel B before opening the stopcock is zero.

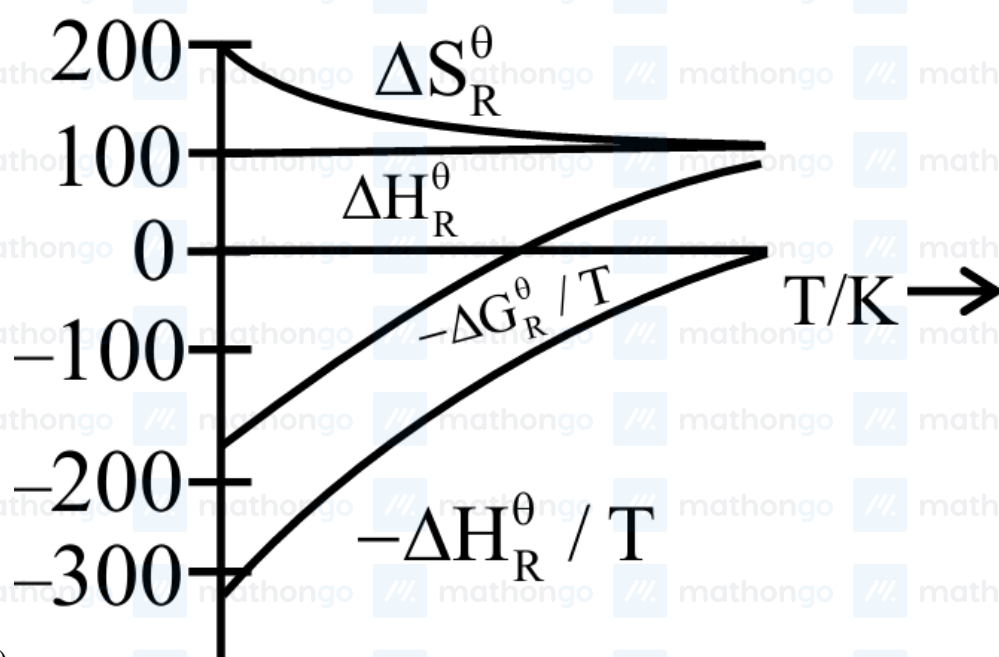
Q15. Which of the following graphs correctly represents the variation of thermodynamic properties of Haber's process?



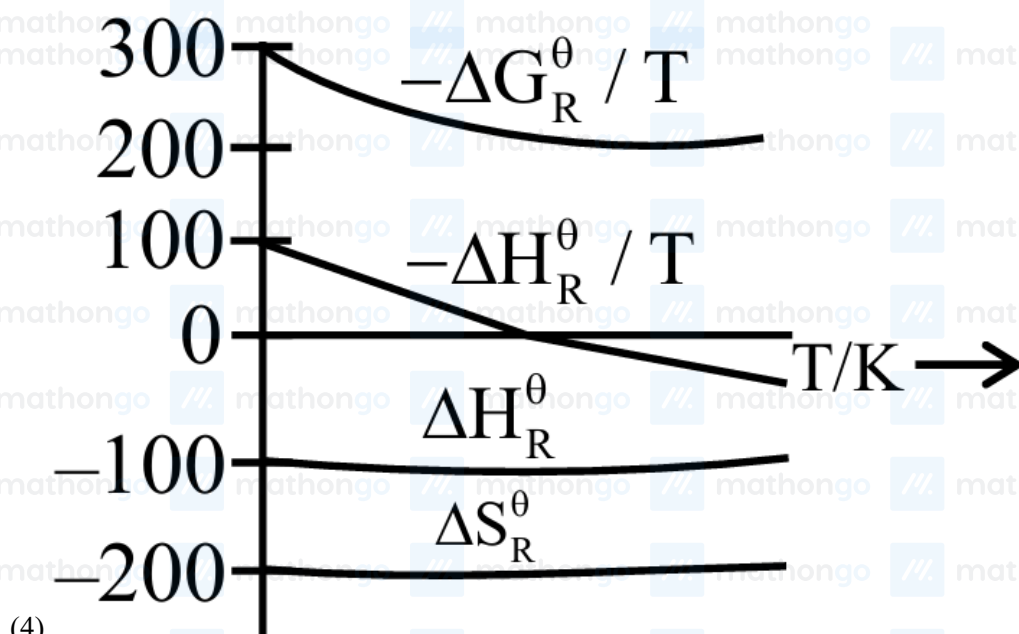
(1)



(2)



(3)



Q1. Given below are two statements

Statement I : A catalyst cannot alter the equilibrium constant (K_c) of the reaction, temperature remaining constant

Statement II : A homogenous catalyst can change the equilibrium composition of a system temperature remaining constant

In the light of the above statements, choose the correct answer from the options given below

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II is false

(4) Statement I is true but Statement II is false

Q2. In the following system, $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ at equilibrium, upon addition of xenon gas at constant T&p, the concentration of

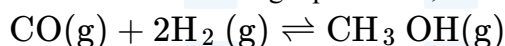
(1) PCl_5 will increase

(2) Cl_2 will decrease

(3) PCl_5 , PCl_3 & Cl_2 remain constant

(4) PCl_3 will increase

Q3. Consider the following equilibrium,

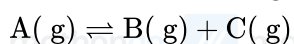


0.1 mol of CO along with a catalyst is present in a 2dm^3 flask maintained at 500 K. Hydrogen is introduced into the flask until the pressure is 5 bar and 0.04 mol of CH_3OH is formed. The K_p^0 is $\text{_____} \times 10^{-3}$ (nearest integer).

Given : $R = 0.08\text{dm}^3 \text{ bar K}^{-1} \text{ mol}^{-1}$

Assume only methanol is formed as the product and the system follows ideal gas behaviour.

Q4. Consider the following chemical equilibrium of the gas phase reaction at a constant temperature :



If p being the total pressure, K_p is the pressure equilibrium constant and α is the degree of dissociation, then which of the following is true at equilibrium?

(1) If p value is extremely high compared to K_p , $\alpha \approx 1$

(2) When p increases α decreases

(3) If K_p value is extremely high compared to p , α becomes much less than unity

(4) When p increases α increases

Q1. x mg of $\text{Mg}(\text{OH})_2$ (molar mass = 58) is required to be dissolved in 1.0 L of water to produce a pH of 10.0 at 298 K. The value of x is _____ mg. (Nearest integer)

(Given : $\text{Mg}(\text{OH})_2$ is assumed to dissociate completely in H_2O)

Q2. An aqueous solution of HCl with pH 1.0 is diluted by adding equal volume of water (ignoring dissociation of water). The pH of HCl solution would

(Given $\log 2 = 0.30$)

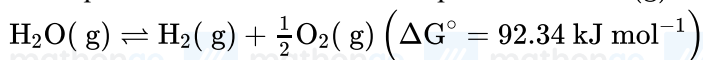
(1) reduce to 0.5

(2) increase to 1.3

(3) remain same

(4) increase to 2

Q3. The equilibrium constant for decomposition of $\text{H}_2\text{O}(\text{g})$



is 8.0×10^{-3} at 2300 K and total pressure at equilibrium is 1 bar. Under this condition, the degree of dissociation (α) of water is _____ $\times 10^{-2}$ (nearest integer value).

[Assume α is negligible with respect to 1]

Q4. 10 mL of 2 M NaOH solution is added to 20 mL of 1 M HCl solution kept in a beaker. Now, 10 mL of this

mixture is poured into a volumetric flask of 100 mL containing 2 moles of HCl and made the volume upto the mark with distilled water. The solution in this flask is :

(1) 0.2 M NaCl solution

(2) 20 M HCl solution

(3) 10 M HCl solution

(4) Neutral solution

Q5. The pH of a 0.01 M weak acid HX ($K_a = 4 \times 10^{-10}$) is found to be 5. Now the acid solution is diluted with excess of water so that the pH of the solution changes to 6. The new concentration of the diluted weak acid is given as $x \times 10^{-4}$ M. The value of x is _____ (nearest integer)

Q6. Only litre buffer solution was prepared by adding 0.10 mol each of NH_3 and NH_4Cl in deionised water. The change in pH on addition of 0.05 mol of HCl to the above solution is _____ $\times 10^{-2}$, (Nearest integer) (Given : pK_b of $\text{NH}_3 = 4.745$ and $\log_{10} 3 = 0.477$)

Q7. If equal volumes of AB_2 and XY (both are salts) aqueous solutions are mixed, which of the following combination will give a precipitate of AY_2 at 300 K ?

(Given K_{sp} (at 300 K) for $\text{AY}_2 = 5.2 \times 10^{-7}$)

(1) $3.6 \times 10^{-3} \text{ M AB}_2$, $5.0 \times 10^{-4} \text{ M XY}$

(2) $2.0 \times 10^{-4} \text{ M AB}_2$, $0.8 \times 10^{-3} \text{ M XY}$

(3) $2.0 \times 10^{-2} \text{ M AB}_2$, $2.0 \times 10^{-2} \text{ M XY}$

(4) $1.5 \times 10^{-4} \text{ M AB}_2$, $1.5 \times 10^{-3} \text{ M XY}$

Q1. Compounds that should not be used as primary standards in titrimetric analysis are :

A. $\text{Na}_2\text{Cr}_2\text{O}_7$

B. Oxalic acid

C. NaOH

D. $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$

E. Sodium tetraborate

Choose the most appropriate answer from the options given below:

(1) B and D Only

(2) D and E Only

(3) C, D and E Only

(4) A, C and D Only

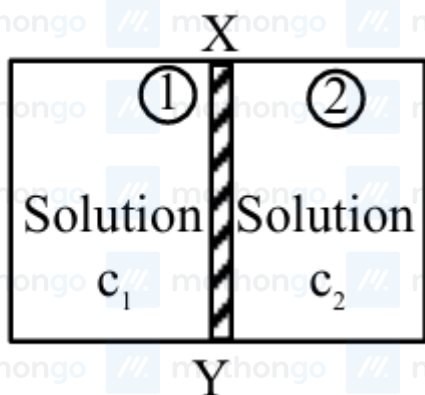
- Q1.** The percentage dissociation of a salt (MX_3) solution at given temperature (van't Hoff factor $i = 2$) is _____ %
(Nearest integer)
- Q2.** Sea water, which can be considered as a 6 molar (6M) solution of NaCl , has a density of 2 g mL^{-1} . The concentration of dissolved oxygen (O_2) in sea water is 5.8 ppm. Then the concentration of dissolved oxygen (O_2) in sea water, is $x \times 10^{-4} \text{ m}$. $x =$ _____. (Nearest integer)
Given: Molar mass of NaCl is 58.5 g mol^{-1} Molar mass of O_2 is 32 g mol^{-1}
- Q3.** A solution is made by mixing one mole of volatile liquid A with 3 moles of volatile liquid B . The vapour pressure of pure A is 200 mm Hg and that of the solution is 500 mm Hg. The vapour pressure of pure B and the least volatile component of the solution, respectively, are :
- (1) 1400 mmHg, A
 - (2) 1400 mmHg, B
 - (3) 600 mmHg, B
 - (4) 600 mmHg, A
- Q4.** Liquid A and B form an ideal solution. The vapour pressure of pure liquids A and B are 350 and 750 mm Hg respectively at the same temperature. If x_A and x_B are the mole fraction of A and B in solution while y_A and y_B are the mole fraction of A and B in vapour phase then :
- (1) $\frac{x_A}{x_B} < \frac{y_A}{y_B}$
 - (2) $\frac{x_A}{x_B} = \frac{y_A}{y_B}$
 - (3) $\frac{x_A}{x_B} > \frac{y_A}{y_B}$
 - (4) $(x_A - y_A) < (x_B - y_B)$
- Q5.** Which of the following binary mixture does not show the behaviour of minimum boiling azeotropes?
- (1) $\text{H}_2\text{O} + \text{CH}_3\text{COC}_2\text{H}_5$
 - (2) $\text{C}_6\text{H}_5\text{OH} + \text{C}_6\text{H}_5\text{NH}_2$
 - (3) $\text{CS}_2 + \text{CH}_3\text{COCH}_3$
 - (4) $\text{CH}_3\text{OH} + \text{CHCl}_3$
- Q6.** When 1 g each of compounds AB and AB_2 are dissolved in 15 g of water separately, they increased the boiling point of water by 2.7 K and 1.5 K respectively. The atomic mass of A (in amu) is _____ $\times 10^{-1}$ (Nearest integer)
(Given : Molal boiling point elevation constant is $0.5 \text{ K kg mol}^{-1}$)

Q7. 2 moles each of ethylene glycol and glucose are dissolved in 500 g of water. The boiling point of the resulting solution is :

(Given : Ebullioscopic constant of water = $0.52 \text{ K kg mol}^{-1}$)

- (1) 379.2 K
- (2) 377.3 K
- (3) 375.3 K
- (4) 277.3 K

Q8. XY is the membrane / partition between two chambers 1 and 2 containing sugar solutions of concentration c_1 and c_2 ($c_1 > c_2$) mol L^{-1} . For the reverse osmosis to take place identify the correct condition (Here p_1 and p_2 are pressures applied on chamber 1 and 2)



- (A) Membrane/Partition; Cellophane, $p_1 > \pi$
- (B) Membrane/Partition ; Porous. $p_2 > \pi$
- (C) Membrane/Partition ; Parchment paper, $p_1 > \pi$
- (D) Membrane/Partition : Cellophane, $p_2 > \pi$

Choose the correct answer from the option given below :

- (1) B and D only
- (2) A and D only
- (3) A and C only
- (4) C only

Q9. Given below are two statements :

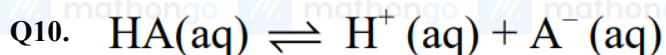
Statement (I) : Molal depression constant K_f is given by $\frac{M_1 RT_f}{\Delta S_{fus}}$, where symbols have their usual meaning.

Statement (II) : K_f for benzene is less than the K_f for water.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I is incorrect but Statement II is correct

- (2) Both Statement I and Statement II are incorrect
(3) Both Statement I and Statement II are correct
(4) Statement I is correct but Statement II is incorrect



The freezing point depression of a 0.1 m aqueous solution of a monobasic weak acid HA is 0.20°C .

The dissociation constant for the acid is

Given :

$$K_f(\text{H}_2\text{O}) = 1.8 \text{ K kg mol}^{-1}, \text{ molality} \equiv \text{molarity}$$

(1) 1.38×10^{-3}

(2) 1.1×10^{-2}

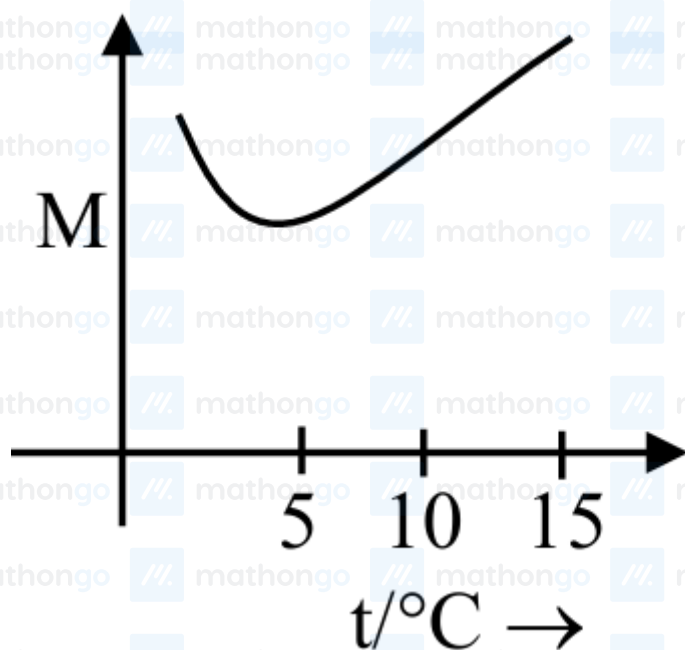
(3) 1.90×10^{-3}

(4) 1.89×10^{-1}

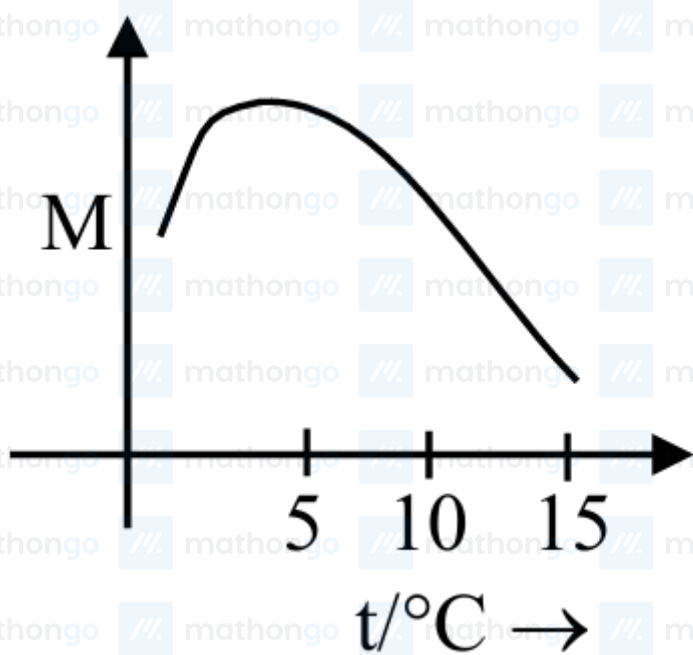
Q11. 'x' g of NaCl is added to water in a beaker with a lid. The temperature of the system is raised from 1°C to 25°C .

Which out of the following plots, is best suited for the change in the molarity (M) of the solution with respect to temperature?

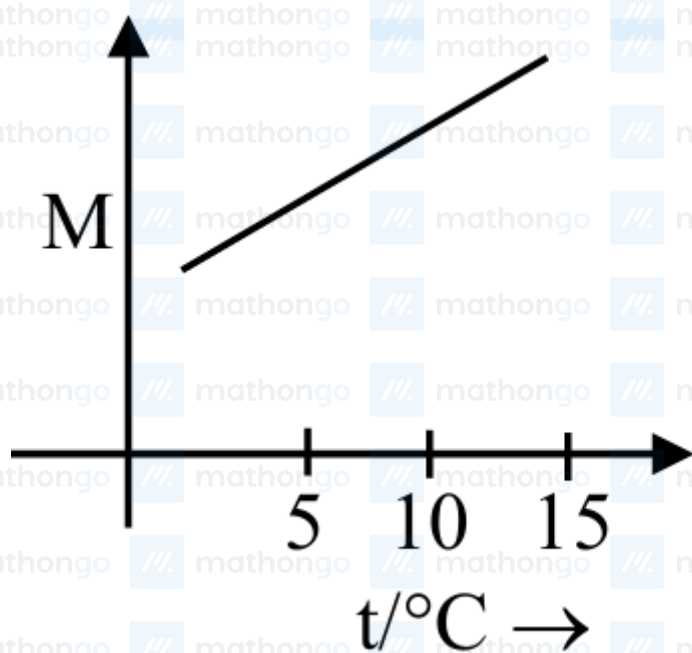
[Consider the solubility of NaCl remains unchanged over the temperature range]



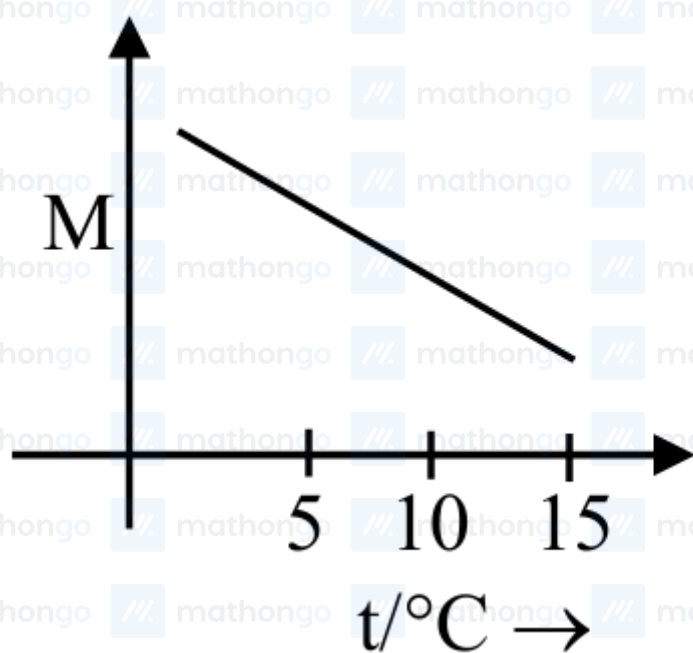
(1)



(2)



(3)



(4)

Q12. Match List-I with List-II

	List-I		List-II
(A)	Solution of chloroform and acetone	(I)	Minimum boiling azeotrope
(B)	Solution of ethanol and water	(II)	Dimerizes
(C)	Solution of benzene and toluene	(III)	Maximum boiling azeotrope
(D)	Solution of acetic acid in benzene	(IV)	$\Delta V_{\text{mix}} = 0$

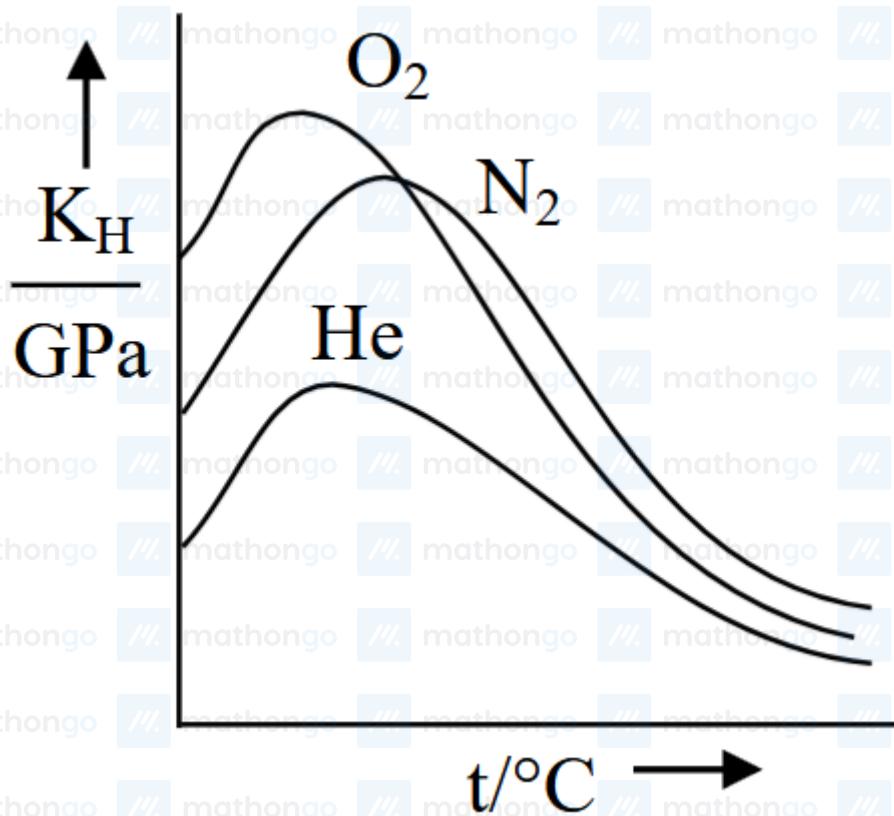
Choose the correct answer from the options given below :

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

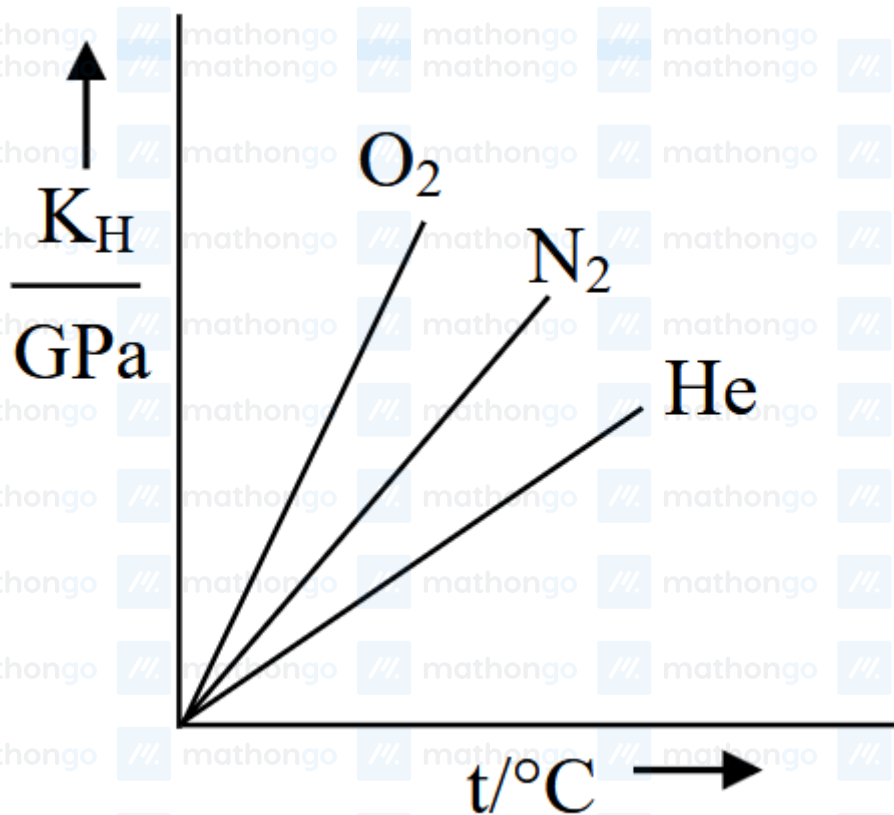
(2) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(3) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

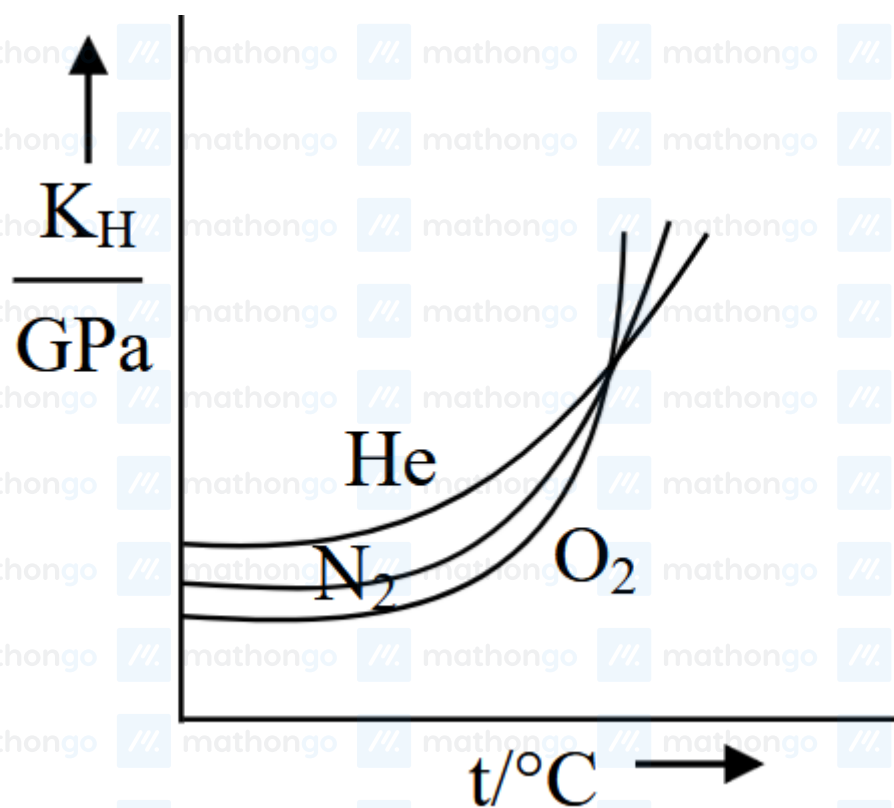
(4) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Q13. Which of the following graph correctly represents the plots of K_H at 1 bar gases in water versus temperature?

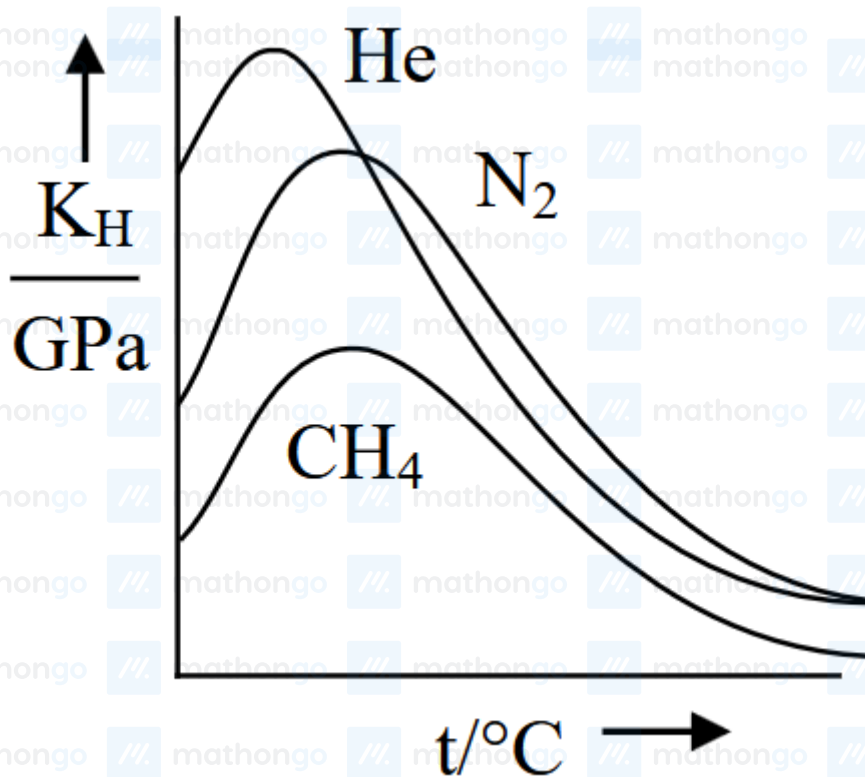
(1)



(2)



(3)



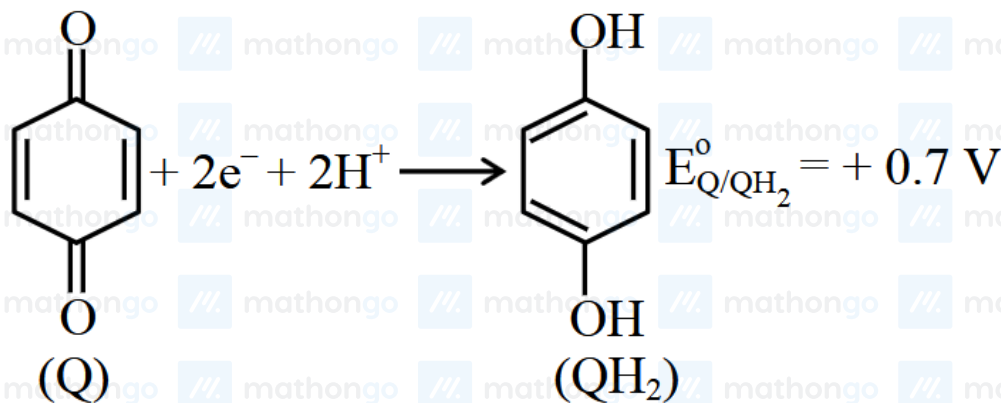
(4)

Q1. Consider the following electrochemical cell at standard condition.



$$E_{\text{cell}} = +0.4 \text{ V}$$

The couple QH_2 / Q represents quinhydrone electrode, the half cell reaction is given below



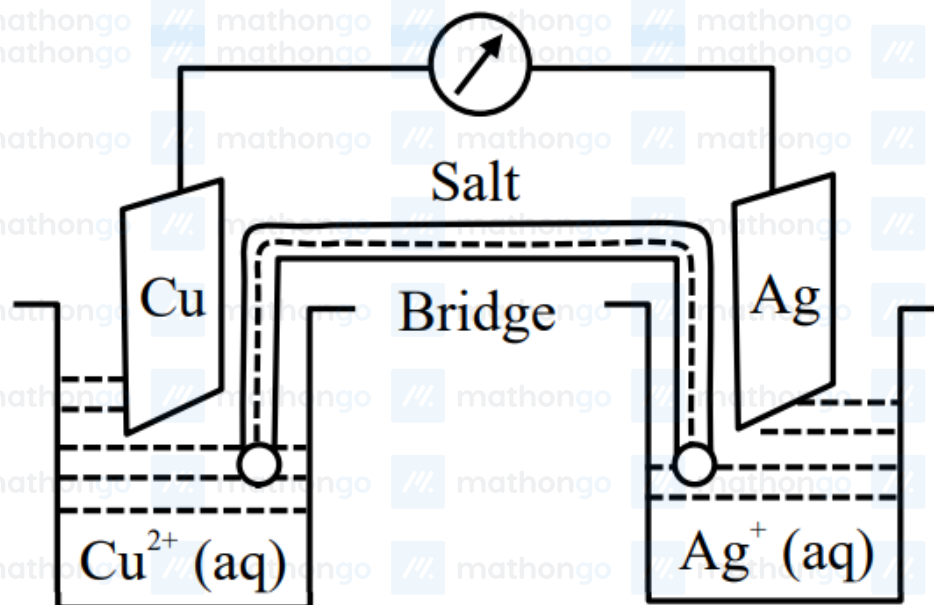
The pK_{b} value of the ammonium halide salt (NH_4X) used here is _____. (nearest integer)

Q2. The standard cell potential ($E_{\text{cell}}^\ominus$) of a fuel cell based on the oxidation of methanol in air that has been used to power television relay station is measured as 1.21 V. The standard half cell reduction potential for O_2 ($E_{\text{O}_2/\text{H}_2\text{O}}^0$) is 1.229 V.

Choose the correct statement:

- (1) The standard half cell reduction potential for the reduction of CO_2 ($E_{\text{CO}_2/\text{CH}_3\text{OH}}^0$) is 19 mV
- (2) Oxygen is formed at the anode.
- (3) Reactants are fed at one go to each electrode.
- (4) Reduction of methanol takes place at the cathode.

Q3. 1 Faraday electricity was passed through $\text{Cu}^{2+}(1.5 \text{ M}, 1 \text{ L})/\text{Cu}$ and 0.1 Faraday was passed through $\text{Ag}^+(0.2\text{M}, 1 \text{ L})/\text{Ag}$ electrolytic cells. After this the two cells were connected as shown below to make an electrochemical cell. The emf of the cell thus formed at 298 K is-



Given: $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$

$E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8 \text{ V}$

$\frac{2.303RT}{F} = 0.06 \text{ V}$

Q4. Given below are two statements :

1 M aqueous solution of each of $\text{Cu}(\text{NO}_3)_2$, AgNO_3 , $\text{Hg}_2(\text{NO}_3)_2$; $\text{Mg}(\text{NO}_3)_2$ are electrolysed using inert electrodes,

Given : $E^\theta_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, $E^\theta_{\text{Hg}_2^{2+}/\text{Hg}} = 0.79 \text{ V}$,

$E^\theta_{\text{Cu}^{2+}/\text{Cu}} = 0.24 \text{ V}$ and $E^\theta_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}$

Statement (I) : With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu

Statement (II) : Magnesium will not be deposited at cathode instead oxygen gas will be evolved at the cathode.

In the light of the above statement, choose the most appropriate answer from the options given below

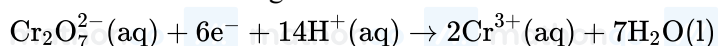
(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

(4) Statement I is incorrect but statement II is correct

Q5. Consider the following half cell reaction



The reaction was conducted with the ratio of $\frac{[\text{Cr}^{3+}]^2}{[\text{Cr}_2\text{O}_7^{2-}]} = 10^{-6}$. The pH value at which the EMF of the half cell will become zero is _____. (nearest integer value)

[Given : standard half cell reduction potential

$$E_{\text{Cr}_2\text{O}_7^{2-}, \text{H}^+ / \text{Cr}^{3+}}^{\circ} = 1.33 \text{ V}, \frac{2.303RT}{F} = 0.059 \text{ V}]$$

Q6. On charging the lead storage battery, the oxidation state of lead changes from x_1 to y_1 at the anode and from x_2 to y_2 at the cathode. The values of x_1, y_1, x_2, y_2 are respectively :

(1) +4, +2, 0, +2

(2) +2, 0, +2, +4

(3) 0, +2, +4, +2

(4) +2, 0, 0, +4

Q7. 0.2%(w/v) solution of NaOH is measured to have resistivity $870.0 \text{ m}\Omega \text{ m}$. The molar conductivity of the solution will be $\text{_____} \times 10^2 \text{ mS dm}^2 \text{ mol}^{-1}$. (Nearest integer)

Q8. Correct order of limiting molar conductivity for cations in water at 298 K is :

(1) $\text{H}^+ > \text{Na}^+ > \text{K}^+ > \text{Ca}^{2+} > \text{Mg}^{2+}$

(2) $\text{H}^+ > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{K}^+ > \text{Na}^+$

(3) $\text{Mg}^{2+} > \text{H}^+ > \text{Ca}^{2+} > \text{K}^+ > \text{Na}^+$

(4) $\text{H}^+ > \text{Na}^+ > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{K}^+$

Q9. The molar conductance of an infinitely dilute solution of ammonium chloride was found to be $185 \text{ S cm}^2 \text{ mol}^{-1}$ and the ionic conductance of hydroxyl and chloride ions are 170 and $70 \text{ S cm}^2 \text{ mol}^{-1}$, respectively. If molar conductance of 0.02 M solution of ammonium hydroxide is $85.5 \text{ S cm}^2 \text{ mol}^{-1}$, its degree of dissociation is given by $x \times 10^{-1}$. The value of x is _____ (Nearest integer)

Q10. Given below are two statements :

Statement I : Mohr's salt is composed of only three types of ions-ferrous, ammonium and sulphate.

Statement II : If the molar conductance at infinite dilution of ferrous, ammonium and sulphate ions are x_1, x_2 and $x_3 \text{ S cm}^2 \text{ mol}^{-1}$, respectively then the molar conductance for Mohr's salt solution at infinite dilution would be given by $x_1 + x_2 + 2x_3$

In the light of the given statements, choose the **correct** answer from the options given below :

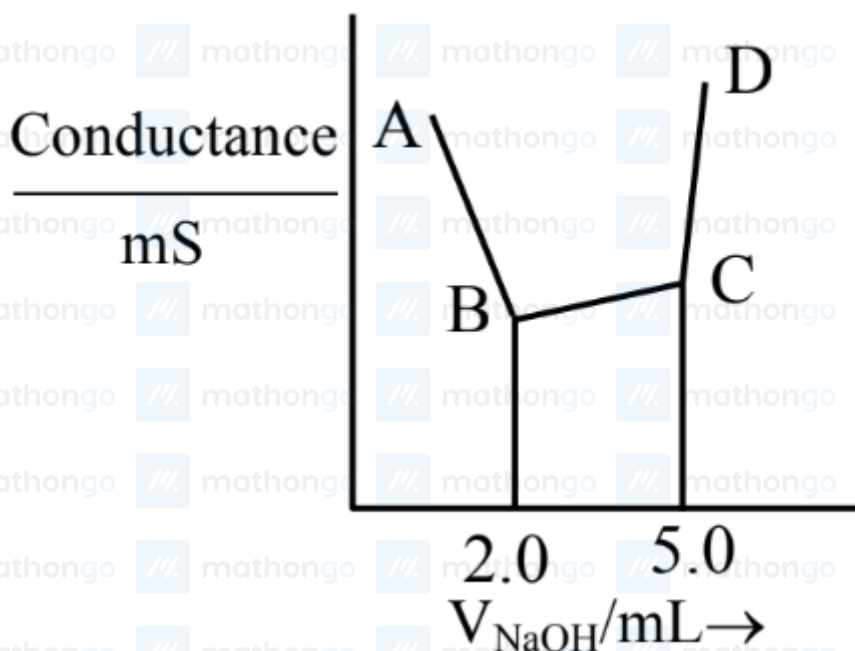
(1) Both statements I and Statement II are false

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II are false

(4) Both statements I and Statement II are true

Q11. 40 mL of a mixture of CH_3COOH and HCl (aqueous solution) is titrated against 0.1 M NaOH solution conductometrically. Which of the following statement is correct?



- (1) The concentration of CH_3COOH in the original mixture is 0.005 M
- (2) The concentration of HCl in the original mixture is 0.005 M
- (3) CH_3COOH is neutralised first followed by neutralisation of HCl
- (4) Point 'C' indicates the complete neutralisation HCl

Q1. Rate law for a reaction between A and B is given by

$$R = k[A]^n[B]^m$$

If concentration of A is doubled and concentration of B is halved from their initial value, the ratio of new rate of reaction to the initial rate of reaction $\left(\frac{r_2}{r_1}\right)$ is

(1) $2^{(n-m)}$

(2) $(n - m)$

(3) $(m + n)$

(4) $\frac{1}{2^{m+n}}$

Q2. Reaction $A(g) \rightarrow 2B(g) + C(g)$ is a first order reaction. It was started with pure A

t/min	Pressure of system at time t/mm Hg
10	160
∞	240

Which of the following option is incorrect?

(1) Initial pressure of A is 80 mm Hg

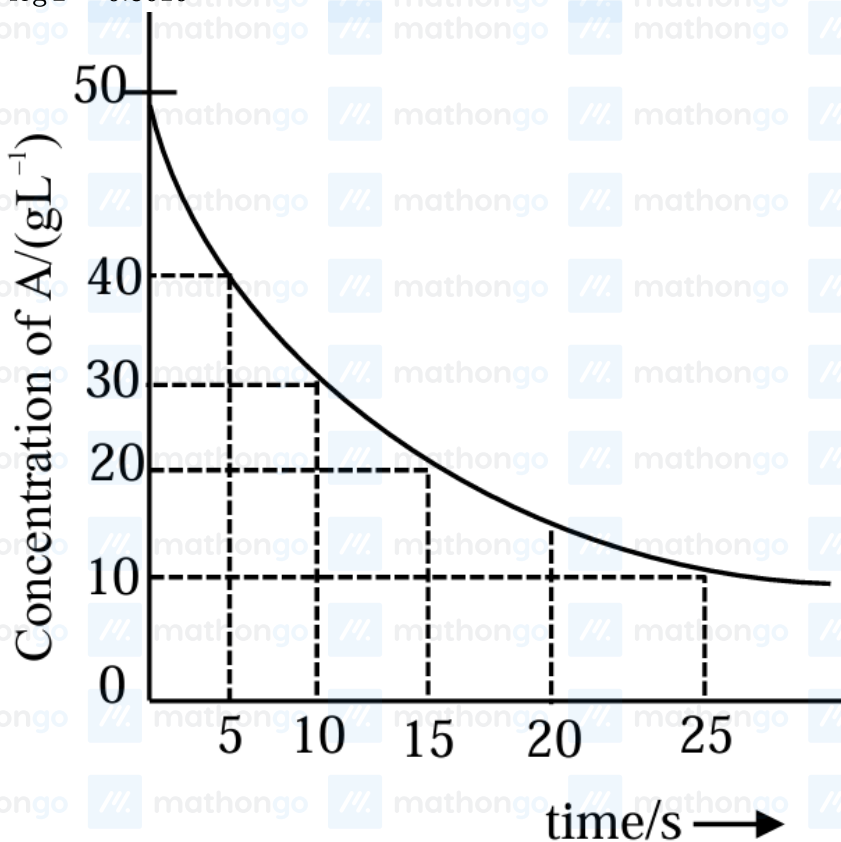
(2) The reaction never goes to completion

(3) Rate constant of the reaction is 1.693 min^{-1}

(4) Partial pressure of A after 10 minute is 40 mm Hg

Q3. For the reaction $A \rightarrow B$ the following graph was obtained. The time required (in seconds) for the concentration of A to reduce to 2.5 g L^{-1} (if the initial concentration of A was 50 g L^{-1}) is _____ (Nearest integer)

Given : $\log 2 = 0.3010$



Q4. $A(g) \rightarrow B(g) + C(g)$ is a first order reaction.

Time	T	∞
P_{system}	P_t	P_{∞}

The reaction was started with reactant A only. Which of the following expression is correct for rate constant k ?

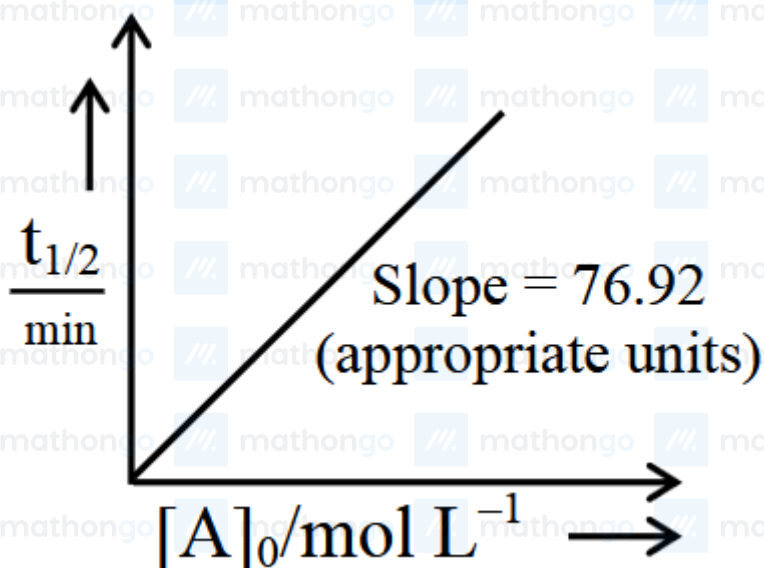
(1) $k = \frac{1}{t} \ln \frac{2(p_{\infty} - P_t)}{P_t}$

(2) $k = \frac{1}{t} \ln \frac{P_{\infty}}{P_t}$

(3) $k = \frac{1}{t} \ln \frac{P_{\infty}}{2(p_{\infty} - P_t)}$

(4) $k = \frac{1}{t} \ln \frac{P_{\infty}}{(p_{\infty} - P_t)}$

Q5. For the reaction $A \rightarrow \text{products}$.



The concentration of A at 10 minutes is $\text{_____} \times 10^{-3} \text{ mol L}^{-1}$ (nearest integer).

The reaction was started with 2.5 mol L^{-1} of A.

Q6. In a reaction $A + B \rightarrow C$, initial concentrations of A and B are related as $[A]_0 = 8[B]_0$. The half lives of A and B are 10 min and 40 min. respectively. If they start to disappear at the same time, both following first order kinetics, after how much time will the concentration of both the reactants be same?

- (1) 60 min
- (2) 80 min
- (3) 20 min
- (4) 40 min

Q7. Half life of zero order reaction $A \rightarrow \text{product}$ is 1 hour, when initial concentration of reaction is 2.0 mol L^{-1} .

The time required to decrease concentration of A from 0.50 to 0.25 mol L^{-1} is:

- (1) 0.5 hour
- (2) 4 hour
- (3) 15 min
- (4) 60 min

Q8. In a first order decomposition reaction, the time taken for the decomposition of reactant to one fourth and one eighth of its initial concentration are t_1 and $t_2(s)$, respectively. The ratio t_1/t_2 will :

- (1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{3}{4}$

(4) $\frac{2}{3}$

Q9. Consider the following statements related to temperature dependence of rate constants.

Identify the correct statements,

A. The Arrhenius equation holds true only for an elementary homogenous reaction.

B. The unit of A is same as that of k in Arrhenius equation.

C. At a given temperature, a low activation energy means a fast reaction.

D. A and E_a as used in Arrhenius equation depend on temperature.

E. When $E_a \gg RT$, A and E_a become interdependent.

Choose the correct answer from the options given below :

(1) A, C and D Only

(2) B, D and E Only

(3) B and C Only

(4) A and B Only

Q10. For $A_2 + B_2 \rightleftharpoons 2AB$

E_a for forward and backward reaction are 180 and 200 kJ mol^{-1} respectively

If catalyst lowers E_a for both reaction by 100 kJ mol^{-1} .

Which of the following statement is correct?

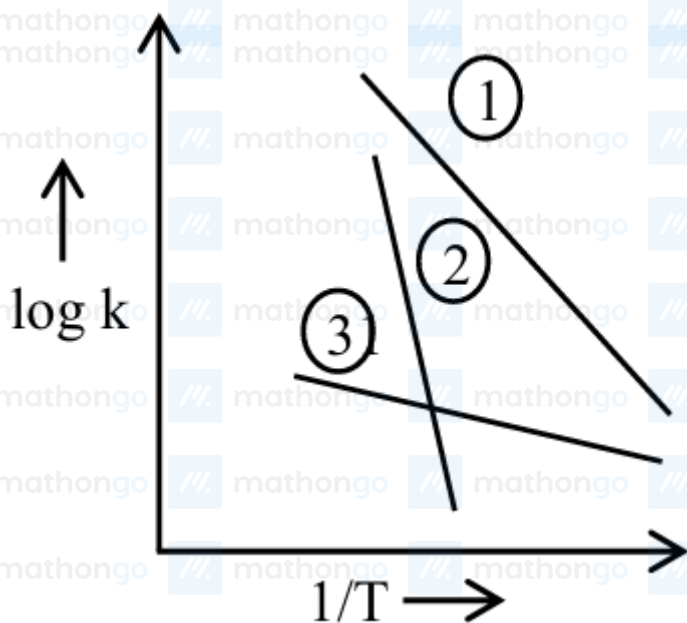
(1) Catalyst does not alter the Gibbs energy change of a reaction.

(2) Catalyst can cause non-spontaneous reactions to occur.

(3) The enthalpy change for the reaction is +20 kJ mol^{-1} .

(4) The enthalpy change for the catalysed reaction is different from that of uncatalysed reaction.

Q11. Consider the following plots of $\log$ of rate constant $k(\log k)$ vs $\frac{1}{T}$ for three different reactions. The correct order of activation energies of these reactions is



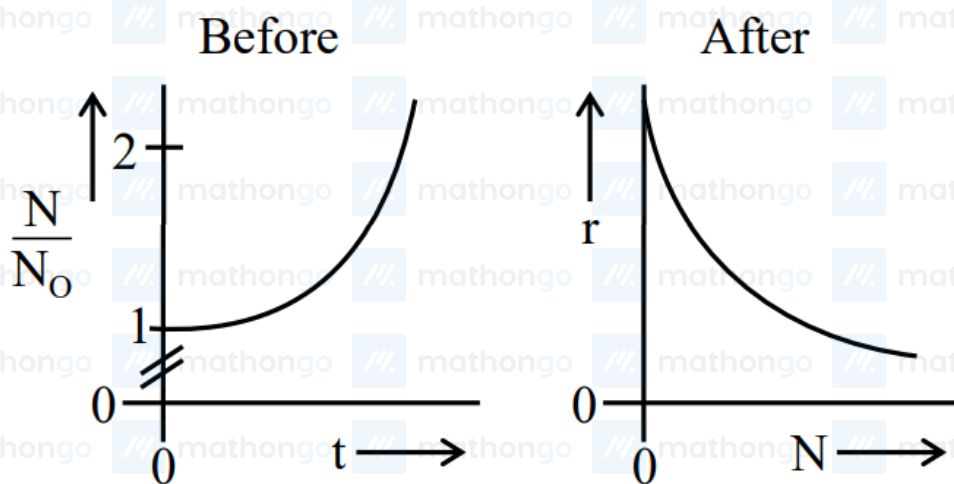
(1) $E_{a2} > E_{a1} > E_{a3}$

(2) $E_{a1} > E_{a3} > E_{a2}$

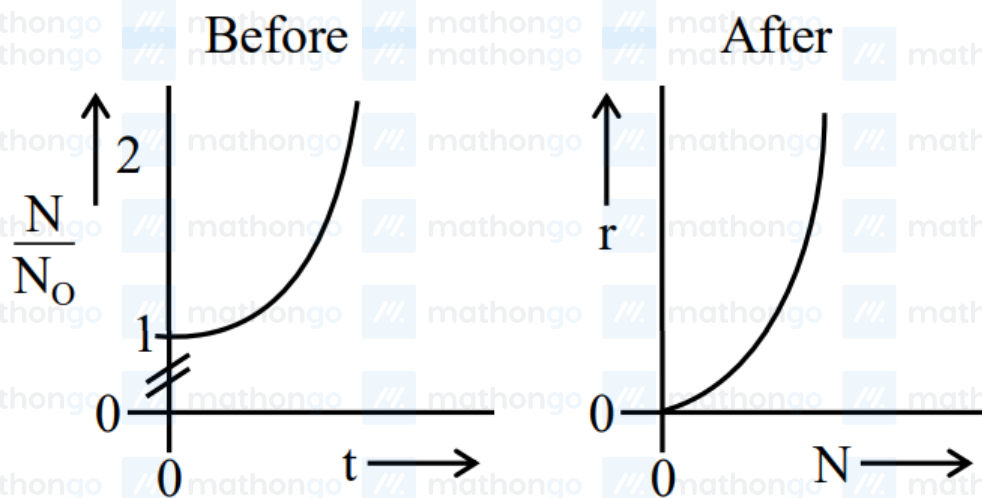
(3) $E_{a1} > E_{a2} > E_{a3}$

(4) $E_{a3} > E_{a2} > E_{a1}$

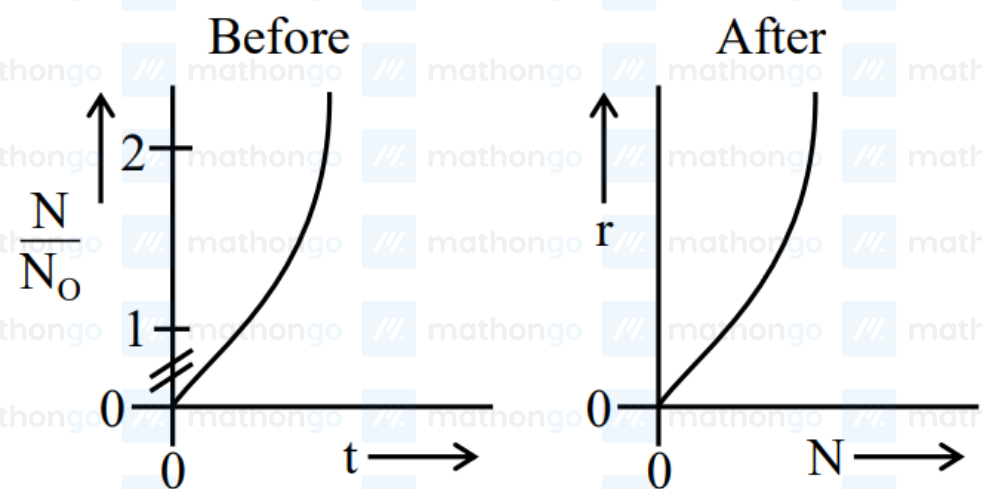
Q12. A person's wound was exposed to some bacteria and then bacteria growth started to happen at the same place. The wound was later treated with some antibacterial medicine and the rate of bacterial decay (r) was found to be proportional with the square of the existing number of bacteria at any instance. Which of the following set of graphs correctly represents the 'before' and 'after' situation of the application of the medicine?
[Given : N = No. of bacteria, t = time, bacterial growth follows I^{st} order kinetics.]



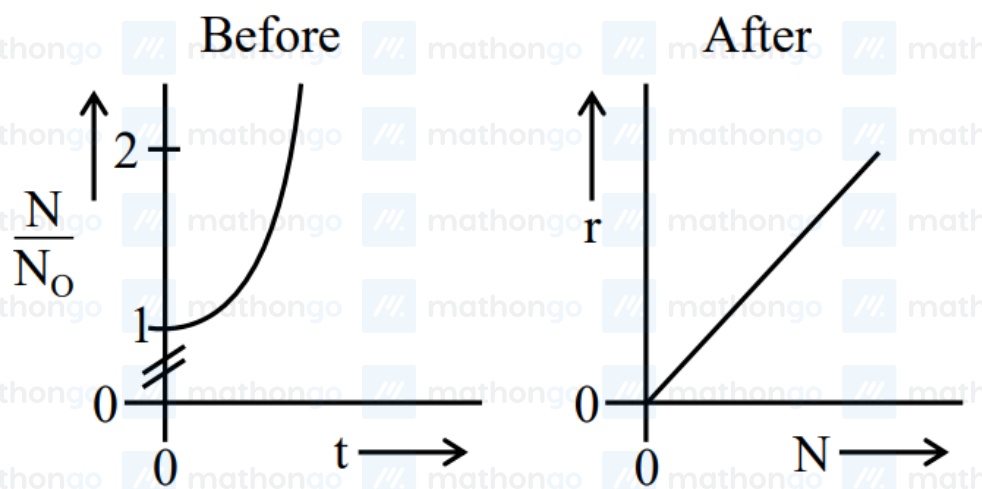
(1)



(2)

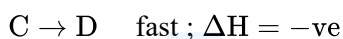
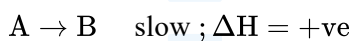


(3)

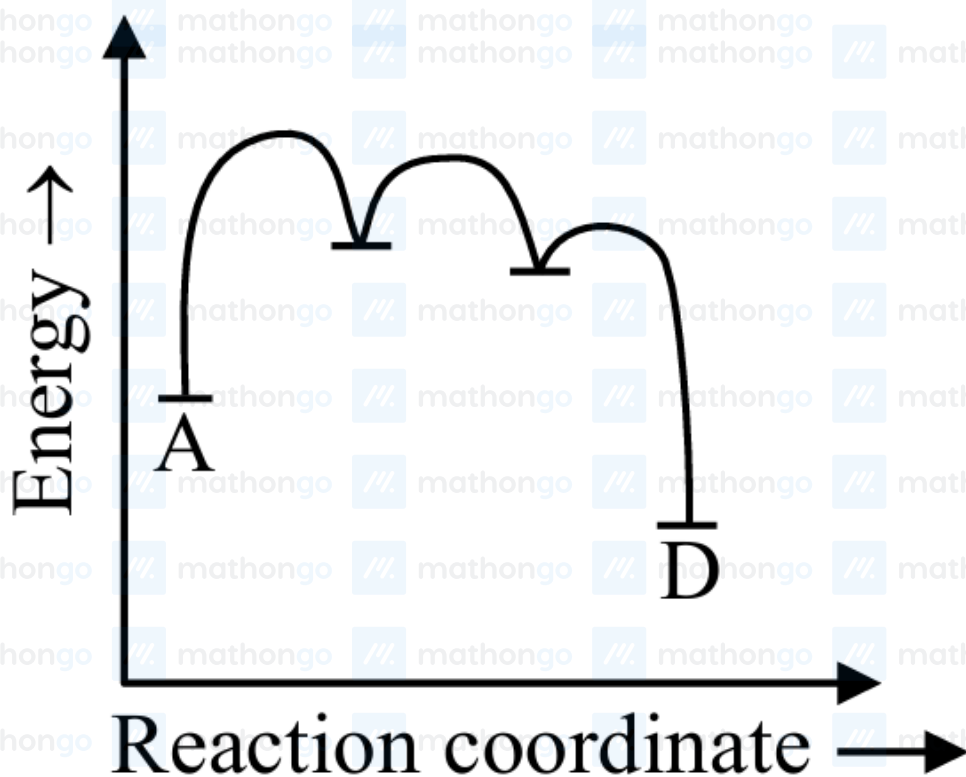


(4)

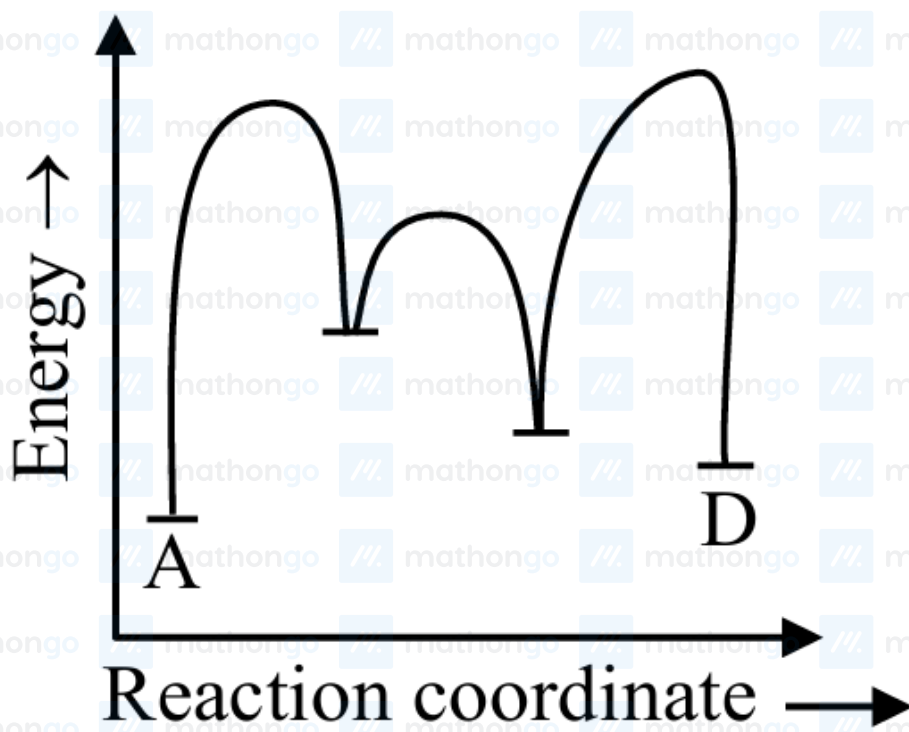
Q13. Reactant A converts to product D through the given mechanism (with the net evolution of heat) :



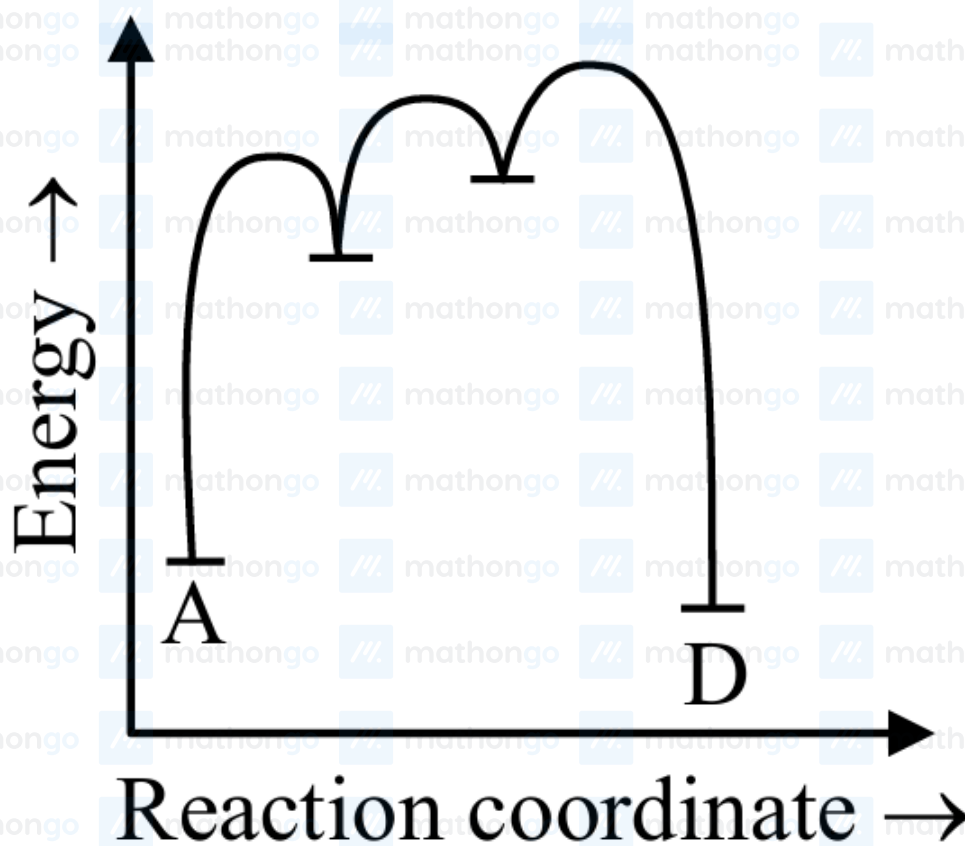
Which of the following represents the above reaction mechanism?



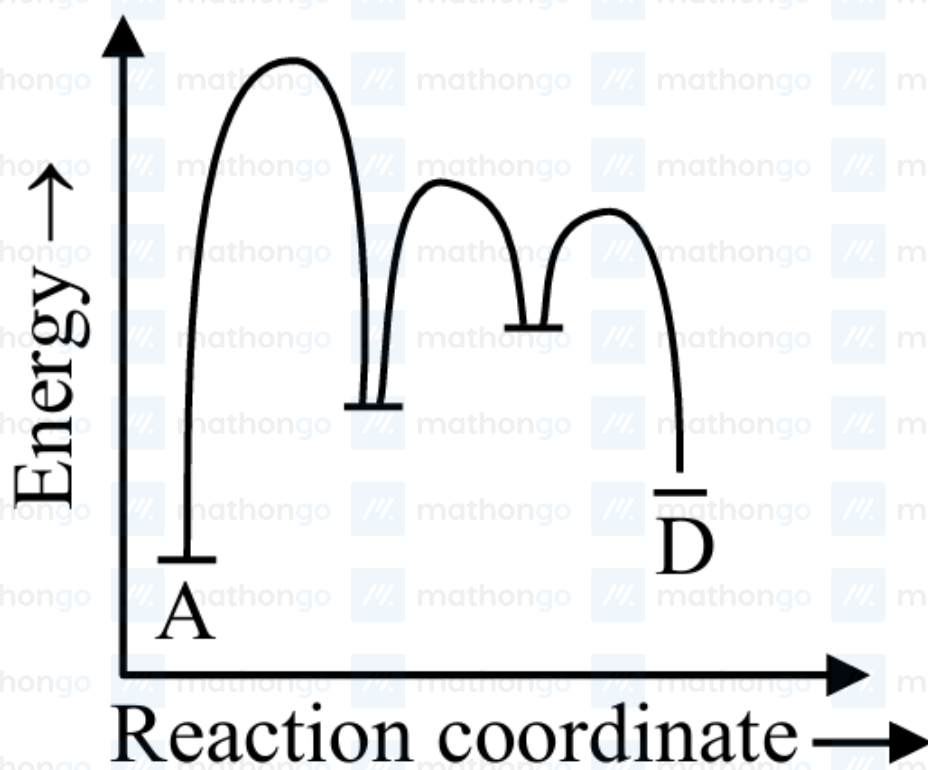
(1)



(2)



(3)



(4)

Q1. Match the LIST-I with LIST-II.

	LIST-I (Family)		LIST-II (Symbol of Element)
A.	Pnicogen (group 15)	I.	Ts
B.	Chalcogen	II.	Og
C.	Halogen	III.	Lv
D.	Noble gas	IV.	Mc

Choose the correct answer from the options given below :

(1) A-IV, B-I, C-II, D-III

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-II, B-III, C-IV, D-I

Q2. Given below are two statements :

Statement (I): The metallic radius of Al is less than that of Ga.

Statement (II) : The ionic radius of Al^{3+} is less than that of Ga^{3+} .

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Statement I is correct but Statement II is incorrect

(4) Both Statement I and Statement II are correct

Q3. Given below are the pairs of group 13 element showing their relation in terms of atomic radius.

(B < Al), (Al < Ga), (Ga < In) and (In < Tl)

Identify the elements present in the incorrect pair and in that pair find out the element (X) that has higher ionic radius (M^{3+}) than the other one. The atomic number of the element (X) is

(1) 31

(2) 49

(3) 13

(4) 81

Q4. The atomic number of the element from the following with lowest 1st ionisation enthalpy is :

(1) 32

(2) 35

(3) 87

(4) 19

Q5. The property/properties that show irregularity in first four elements of group-17 is/are :

(A) Covalent radius

(B) Electron affinity

(C) Ionic radius

(D) First ionization energy

Choose the correct answer from the options given below:

(1) B and D only

(2) A and C only

(3) B only

(4) A, B, C and D

Q6. The correct statements from the following are :

(A) Tl^{3+} is a powerful oxidising agent

(B) Al^{3+} does not get reduced easily

(C) Both Al^{3+} and Tl^{3+} are very stable in solution

(D) Tl^{+} is more stable than Tl^{3+}

(E) Al^{3+} and Tl^{+} are highly stable

Choose the correct answer from the options given below :

(1) (A), (B), (C), (D) and (E)

(2) (A), (B), (D) and (E) only

(3) (B), (D) and (E) only

(4) (A), (C) and (D) only

Q7. Electronic configuration of four elements A, B, C and D are given below :

(A) $1s^2 2s^2 2p^3$

(B) $1s^2 2s^2 2p^4$

(C) $1s^2 2s^2 2p^5$

(D) $1s^2 2s^2 2p^2$

Which of the following is the correct order of increasing electronegativity (Pauling's scale)?

(1) $A < D < B < C$

(2) $A < C < B < D$

(3) $A < B < C < D$

(4) $D < A < B < C$

Q8. Which of the following statements are correct?

A. The process of the addition an electron to a neutral gaseous atom is always exothermic

B. The process of removing an electron from an isolated gaseous atom is always endothermic

C. The 1st ionization energy of the boron is less than that of the beryllium

D. The electronegativity of C is 2.5 in CH_4 and CCl_4

E. Li is the most electropositive among elements of group I

Choose the correct answer from the options gives below

(1) B & C only

(2) A, C and d only

(3) B and D only

(4) B,C and E only

Q9. The correct orders among the following are

Atomic radius : $B < Al < Ga < In < Tl$

Electronegativity : $Al < Ga < In < Tl < B$

Density : $Tl < In < Ga < Al < B$

1st Ionisation Energy : $In < Al < Ga < Tl < B$

Choose the correct answer from the options given below :

(1) B and D Only

(2) A and C Only

(3) C and D Only

(4) A and B Only

Q10. The elements of Group 13 with highest and lowest first ionisation enthalpies are respectively:

(1) B & Ga

(2) B & Tl

(3) Tl & B

(4) B & In

Q11. Choose the incorrect trend in the atomic radii (r) of the elements :

(1) $r_{Br} < r_K$

Chapter-wise Question Bank

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(2) $r_{\text{Mg}} < r_{\text{Al}}$

(3) $r_{\text{Rb}} < r_{\text{Cs}}$

(4) $r_{\text{Rb}} < r_{\text{Cs}}$

Q1. In 3, 3-dimethylhex-1-ene-4-yne, there are _____ sp^3 , _____ sp^2 and _____ sp hybridised carbon atoms respectively :

(1) 4,2,2

(2) 3,3,2

(3) 2,4,2

(4) 2,2,4

Q2. Among SO_2 , NF_3 , NH_3 , XeF_2 , ClF_3 and SF_4 , the hybridization of the molecule with non-zero dipole moment and highest number of lone-pairs of electrons on the central atom is

(1) sp^3

(2) dsp^2

(3) $sp^3 d^2$

(4) $sp^3 d$

Q3. A molecule with the formula AX_4Y has all its elements from p-block. Element A is rarest, monoatomic, non-radioactive from its group and has the lowest ionization enthalpy value among A, X and Y. Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is :

(1) Square pyramidal

(2) Octahedral

(3) Pentagonal planar

(4) Trigonal bipyramidal

Q4. Which among the following molecules is (a) involved in $sp^3 d$ hybridization, (b) has different bond lengths and (c) has lone pair of electrons on the central atom?

(1) PF_5

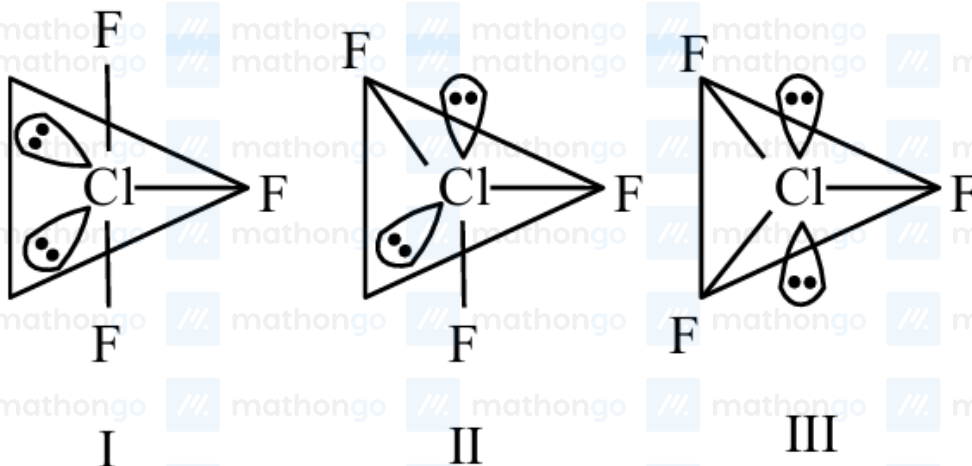
(2) XeF_4

(3) SF_4

(4) XeF_2

Q5. Given below are two statements:

Statement (I) : for F_3 , all three possible structures may be drawn as follows.



Statement (II) : Structure III is most stable, as the orbitals having the lone pairs are axial, where the $\ell p - bp$ repulsion is minimum.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is incorrect but statement II is correct.
- (2) Statement I is correct but statement II is incorrect.
- (3) Both Statement I and statement II are correct.
- (4) Both Statement I and statement II are incorrect.

Q6.

LIST-I		LIST-II	
Molecule/ion		Bond pair : lone pair (on the central atom)	
A.	ICl_2^-	I.	4 : 2
B.	H_2O	II.	4 : 1
C.	SO_2	III.	2 : 3
D.	XeF_4	IV.	2 : 2

Match the LIST-I with LIST-II.

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-IV, C-II, D-I
- (3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

Q7. In SO_2 , NO_2^- and N_3^- the hybridizations at the central atom are respectively :

(1) sp^2 , sp^2 and sp

(2) sp^2 , sp and sp

(3) sp^2 , sp^2 and sp^2

(4) sp , sp^2 and sp

Q8. Which of the following molecules(s) show/s paramagnetic behavior?

(A) O_2

(B) N_2

(C) F_2

(D) S_2

(E) Cl_2

Choose the correct answer from the options given below :

(1) B only

(2) A & C only

(3) A & E only

(4) A & D only

Q9. A metal complex with a formula $\text{MCl}_4 \cdot 3\text{NH}_3$ is involved in $\text{sp}^3 \text{d}^2$ hybridisation. It upon reaction with excess of AgNO_3 solution gives 'x' moles of AgCl . Consider 'x' is equal to the number of lone pairs of electron present in central atom of BrF_5 . Then the number of geometrical isomers exhibited by the complex is _____.

Q1. Given below are two statements :

Statement (I) : The first ionisation enthalpy of group 14 elements is higher than the corresponding elements of group 13.

Statement (II) : Melting points and boiling points of group 13 elements are in general much higher than those of the corresponding elements of group 14. In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are incorrect

(4) Both Statement I and Statement II are correct

Q2. The group 14 elements A and B have the first ionisation enthalpy values of 708 and 715 kJ mol⁻¹ respectively.

The above values are lowest among their group members. The nature of their ions A²⁺ B⁴⁺ respectively is

(1) both reducing

(2) both oxidising

(3) reducing and oxidising

(4) oxidising and reducing

Q1. Given below are two statements

Statement I : The N-N single bond is weaker and longer than that of P – P single bond

Statement II : Compounds of group 15 elements in +3 oxidation states readily undergo disproportionation reactions.

In the light of above statements, choose the correct answer from the options given below

- (1) Statement I is true but statement II is false
- (2) Both statement I and statement II are false
- (3) Statement I is false but statement II is true
- (4) Both statement I and statement II are true

Q2. Given below are two statements :

Statement I : Nitrogen forms oxides with +1 to +5 oxidation states due to the formation of $p\pi - p\pi$ bond with oxygen.

Statement II : Nitrogen does not form halides with +5 oxidation state due to the absence of d-orbital in it.

In the light of given statements, choose the correct answer from the options given below.

- (1) Statement I is true but Statement II is false
- (2) Both Statement I and Statement II are false
- (3) Statement I is false but Statement II is true
- (4) Both Statement I and Statement II are true

Q3. The nature of oxide (TeO_2) and hydride (TeH_2) formed by Te , respectively are :

- (1) Oxidising and acidic
- (2) Reducing and basic
- (3) Reducing and acidic
- (4) Oxidising and basic

Q4. Given below are two statements :

Statement I: H_2Se is more acidic than H_2Te .

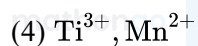
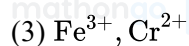
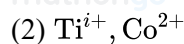
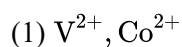
Statement II : H_2Se has higher bond enthalpy for dissociation than H_2Te .

In the light of the above statements, choose the correct answer from the options given below.

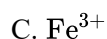
- (1) Both statement I and Statement II are false.
- (2) Both statement I and Statement II are true.
- (3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

Q1. Pair of transition metal ions having the same number of unpaired electrons is :



Q2. The metal ions that have the calculated spin only magnetic moment value of 4.9 B.M. are



Choose the correct answer from the options given below

(1) A, C and E only

(2) A, D and E only

(3) B and E only

(4) A, B and E only

Q3. Given below are two statements :

Statement I : CrO_3 is a stronger oxidizing agent than MoO_3

Statement II : $Cr(VI)$ is more stable than $Mo(VI)$

In the light of the above statements, choose the correct answer from the options given below

(1) Statement I is false but Statement II is true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Q4. Among, Sc, Mn, Co and Cu, identify the element with highest enthalpy of atomisation. The spin only magnetic moment value of that element in its +2 oxidation state is _____ BM (in nearest integer).

Q5. The number of valence electrons present in the metal among Cr, Co, Fe and Ni which has the lowest enthalpy of atomisation is

(1) 8

(2) 9

(3) 6

(4) 10

Q6. The first transition series metal 'M' has the highest enthalpy of atomisation in its series. One of its aquated ion (M^{n+}) exists in green colour. The nature of the oxide formed by the above M^{n+} ion is :

(1) neutral

(2) acidic

(3) basic

(4) amphoteric

Q7. 'X' is the number of acidic oxides among VO_2 , V_2O_3 , CrO_3 , V_2O_5 and Mn_2O_7 . The primary valency of cobalt in $[Co(H_2NCH_2CH_2NH_2)_3](SO_4)_3$ is Y. The value of $X + Y$ is :

(1) 5

(2) 4

(3) 2

(4) 3

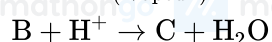
Q8. The correct decreasing order of spin only magnetic moment values (BM) of Cu^+ , Cu^{2+} , Cr^{2+} and Cr^{3+} ions is :

(1) $Cu^+ > Cu^{2+} > Cr^{3+} > Cr^{2+}$ (2) $Cu^{2+} > Cu^+ > Cr^{2+} > Cr^{3+}$ (3) $Cr^{2+} > Cr^{3+} > Cu^{2+} > Cu^+$ (4) $Cr^{3+} > Cr^{2+} > Cu^+ > Cu^{2+}$

Q9. Consider the following reactions



Little
amount



The number of terminal 'O' present in the compound 'C' is _____.

Q10. The incorrect relationship in the following pairs in relation to ionisation enthalpies is :

(1) $\text{Mn}^+ < \text{Cr}^+$

(2) $\text{Mn}^+ < \text{Mn}^{2+}$

(3) $\text{Fe}^{2+} < \text{Fe}^{3+}$

(4) $\text{Mn}^{2+} < \text{Fe}^{2+}$

Q11. KMnO_4 acts as an oxidising agent in acidic medium. 'X' is the difference between the oxidation states of Mn in reactant and product. 'Y' is the number of 'd' electrons present in the brown red precipitate formed at the end of the acetate ion test with neutral ferric chloride. The value of $X + Y$ is _____.

Q1. The d-orbital electronic configuration of the complex among $[\text{Co}(\text{en})_3]^{3+}$, $[\text{CoF}_6]^{3-}$, $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ that has the highest CFSE is :

(1) $t_{2g}^6 e_g^0$

(2) $t_{2g}^6 e_g^4$

(3) $t_{2g}^3 e_g^2$

(4) $t_{2g}^4 e_g^2$

Q2. The correct order of $[\text{FeF}_6]^{3-}$, $[\text{CoF}_6]^{3-}$, $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ complex species based on the number of unpaired electrons present is :

(1) $[\text{FeF}_6]^{3-} > [\text{CoF}_6]^{3-} > [\text{Ni}(\text{CN})_4]^{2-} > [\text{Ni}(\text{CO})_4]$

(2) $[\text{Ni}(\text{CN})_4]^{2-} > [\text{FeF}_6]^{3-} > [\text{CoF}_6]^{3-} > [\text{Ni}(\text{CO})_4]$

(3) $[\text{CoF}_6]^{3-} > [\text{FeF}_6]^{3-} > [\text{Ni}(\text{CO})_4] > [\text{Ni}(\text{CN})_4]^{2-}$

(4) $[\text{FeF}_6]^{3-} > [\text{CoF}_6]^{3-} > [\text{Ni}(\text{CN})_4]^{2-} = [\text{Ni}(\text{CO})_4]$

Q3. Match List-I with List-II

List-I Complex		List-II Primary valency and Secondary valency	
(A)	$[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$	(I)	3 6
(B)	$[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$	(II)	3 4
(C)	$\text{Hg}[\text{Co}(\text{SCN})_4]$	(III)	2 6
(D)	$[\text{Mg}(\text{EDTA})]^{2-}$	(IV)	2 4

Choose the correct answer from the options given below :

(1) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(2) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(3) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

Q4. Given below are two statements :

Statement I : A homoleptic octahedral complex, formed using monodentate ligands, will not show

stereoisomerism.

Statement II : cis- and trans- platin are heteroleptic complexes of Pd.

In the light of the above statements, choose the correct answer from the options given below.

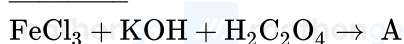
(1) Both statement I and Statement II are false.

(2) Statement I is false but Statement II is true.

(3) Both statement I and Statement II are true.

(4) Statement I is true but Statement II is false.

Q5. The number of optical isomers exhibited by the iron complex (A) obtained from the following reaction is



Q6. Number of stereoisomers possible for the complexes, $[\text{CrCl}_3(\text{py})_3]$ and $[\text{CrCl}_2(\text{ox})_2]^{3-}$ are respectively
(py = pyridine, ox = oxalate)

(1) 3&3

(2) 2&2

(3) 2&3

(4) 1&2

Q7. An octahedral complex having molecular composition $\text{Co} \cdot 5\text{NH}_3 \cdot \text{Cl} \cdot \text{SO}_4$ has two isomers A and B. The solution of A gives a white precipitate with AgNO_3 solution and the solution of B gives white precipitate with BaCl_2 solution. The type of isomerism exhibited by the complex is,

(1) Co-ordinate isomerism

(2) Linkage isomerism

(3) Ionisation isomerism

(4) Geometrical isomerism

Q8. The type of hybridization and the magnetic property of $[\text{MnCl}_6]^{3-}$ are :

(1) d^2sp^3 , paramagnetic with four unpaired electrons

(2) $sp^3 d^2$, paramagnetic with four unpaired electrons

(3) d^2sp^3 , paramagnetic with two unpaired electrons

(4) $sp^3 d^2$, paramagnetic with two unpaired electrons

Q9. The spin-only magnetic moment value of M^{n+} ion formed among Ni, ZnMn and Cu that has the least enthalpy of atomisation is _____. (in nearest integer)

Here n is equal to the number of diamagnetic complexes among $K_2 [NiCl_4]$, $[Zn(H_2O)_6] Cl_2$, $K_3 [Mn(CN)_6]$ and $[Cu(PPh_3)_3]I$

Q10. Match the LIST-I with LIST-II

LIST-I (Molecules/ion)		LIST-II (Hybridisation of central atom)	
A.	PF_5	I.	dsp^2
B.	SF_6	II.	sp^3d
C.	$Ni(CO)_4$	III.	sp^3d^2
D.	$[PtCl_4]^{2-}$	IV.	sp^3

Choose the correct answer from the options given below :

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-I, C-II, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-I, C-IV, D-II

Q11. Which one of the following complexes will have $\Delta_0 = 0$ and $\mu = 5.96$ B.M.?

(1) $[Fe(CN)_6]^{4-}$

(2) $[Co(NH_3)_6]^{3+}$

(3) $[FeF_6]^{4-}$

(4) $[Mn(SCN)_6]^{4-}$

Q12. The number of paramagnetic complex among $[FeF_6]^{3-}$, $[Fe(CN)_6]^{3-}$, $[Mn(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}$, $[MnCl_6]^{3-}$ and $[CoF_6]^{3-}$, which involved d^2sp^3 hybridization is _____

Q13. The number of unpaired electrons responsible for the paramagnetic nature of the following complex species are respectively:



(1) 1, 5, 4, 2

(2) 1, 5, 5, 2

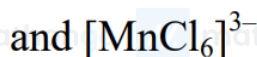
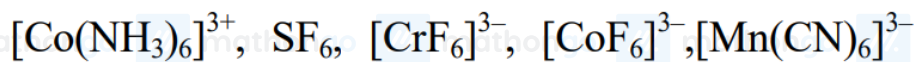
(3) 1, 1, 4, 2

(4) 1, 4, 4, 2

Q14. The number of paramagnetic metal complex species among $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{MnCl}_6]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{CoF}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ with same number of unpaired electrons is _____.

Q15. The number of species from the following that are

involved in sp^3d^2 hybridization is



(1) 5

(2) 6

(3) 4

(4) 3

Q16. Match the LIST-I with LIST-II

	LIST-I (Complex/ Species)		LIST-II (Shape & magnetic moment)
A.	$[\text{Ni}(\text{CO})_4]$	I.	Tetrahedral, 2.8 BM
B.	$[\text{Ni}(\text{CN})_4]^{2-}$	II.	Square planar, 0 BM
C.	$[\text{NiCl}_4]^{2-}$	III.	Tetrahedral, 0 BM
D.	$[\text{MnBr}_4]^{2-}$	IV.	Tetrahedral, 5.9 BM

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-IV, B-I, C-III, D-II

Q17. Given below are two statements :

Statement (I) : In octahedral complexes, when $\Delta_o < P$ high spin complexes are formed. When $\Delta_o > P$ low spin complexes are formed.

Statement (II) : In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but Statement II is incorrect.

(2) Both Statement I and Statement II are incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

Q18. A transition metal (M) among Mn, Cr, Co and Fe has the highest standard electrode potential (M^{3+}/M^{2+}). It forms a metal complex of the type $[M(CN)_6]^{4-}$. The number of electrons present in the e_g orbital of the complex is _____.

Q19. The correct order of the complexes $[Co(NH_3)_5(H_2O)]^{3+}$ (A),

$[Co(NH_3)_6]^{3+}$ (B),

$[Co(CN)_6]^{3-}$ (C) and $[CoCl(NH_3)_5]^{2+}$ (D) in terms wavelength of light absorbed is :

(1) $D > A > B > C$

(2) $C > B > D > A$

(3) $D > C > B > A$

(4) $C > B > A > D$

Q20. Identify the diamagnetic octahedral complex ions from below ;

A. $[Mn(CN)_6]^{3-}$

B. $[Co(NH_3)_6]^{3+}$

C. $[Fe(CN)_6]^{4-}$

D. $[Co(H_2O)_3F_3]$

Choose the correct answer from the options given below :

(1) B and D Only

(2) A and D Only

(3) A and C Only

(4) B and C Only

Q21. 'X' is the number of electrons in t_{2g} orbitals of the most stable complex ion among $[\text{Fe}(\text{NH}_3)_6]^{3+}$, $[\text{Fe}(\text{Cl}_6)]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. The nature of oxide of vanadium of the type V_2O_X is:

(1) Acidic

(2) Neutral

(3) Basic

(4) Amphoteric

Q1. Formation of $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$, a purple coloured complex formed by addition of sodium nitroprusside in sodium carbonate extract of salt indicates the presence of :

(1) Sodium ion

(2) Sulphate ion

(3) Sulphide ion

(4) Sulphite ion

Q2. When a salt is treated with sodium hydroxide solution it gives gas X. On passing gas X through reagent Y a brown coloured precipitate is formed. X and Y respectively, are

(1) $\text{X} = \text{NH}_3$ and $\text{Y} = \text{HgO}$

(2) $\text{X} = \text{NH}_3$ and $\text{Y} = \text{K}_2\text{HgI}_4 + \text{KOH}$

(3) $\text{X} = \text{NH}_4\text{Cl}$ and $\text{Y} = \text{KOH}$

(4) $\text{X} = \text{HCl}$ and $\text{Y} = \text{NH}_4\text{Cl}$

Q3. Choose the correct tests with respective observations.

(A) CuSO_4 (acidified with acetic acid) + $\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$ Chocolate brown precipitate.

(B) $\text{FeCl}_3 + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$ Prussian blue precipitate.

(C) $\text{ZnCl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6]$, neutralised with $\text{NH}_4\text{OH} \rightarrow$ White or bluish white precipitate.

(D) $\text{MgCl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$ Blue precipitate.

(E) $\text{BaCl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6]$, neutralised with $\text{NaOH} \rightarrow$ White precipitate.

Choose the correct answer from the options given below :

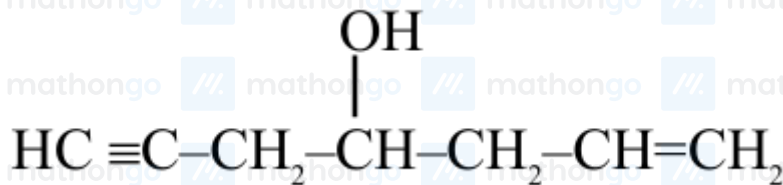
(1) A, D and E only

(2) B, D and E only

(3) A, B and C only

(4) C, D and E only

Q1. The IUPAC name of the following compound is -



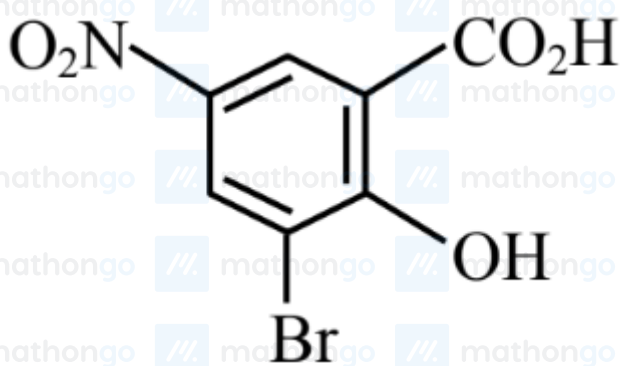
(1) 4-Hydroxyhept-1-en-6-yne

(2) 4-Hydroxyhept-6-en-1-yne

(3) Hept-6-en-1-yn-4-ol

(4) Hept-1-en-6-yn-4-ol

Q2. What is the correct IUPAC name of



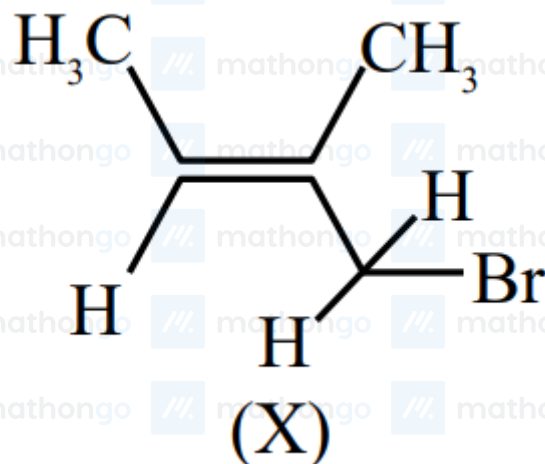
(1) 3-Bromo-2-hydroxy-5-nitrobenzoic acid

(2) 3-Bromo-4-hydroxy-1-nitrobenzoic acid

(3) 2-Hydroxy-3-bromo-5-nitrobenzoic acid

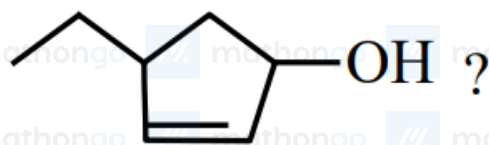
(4) 5-Nitro-3-bromo-2-hydroxybenzoic acid

Q3. Which of the following is the correct IUPAC name of given organic compound (X) ?



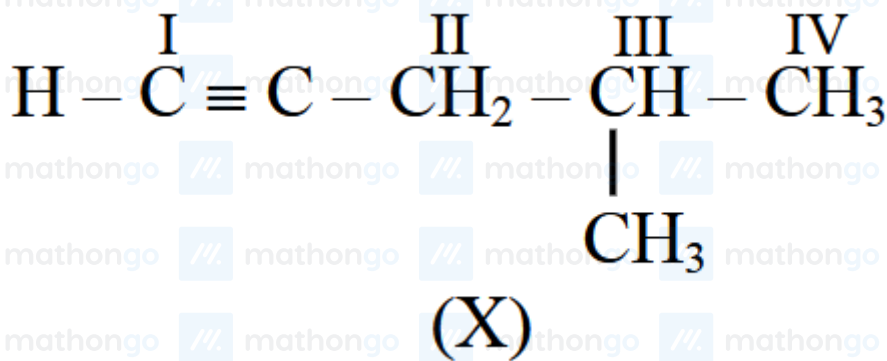
- (1) 2-Bromo-2-methylbut-2-ene
- (2) 3-Bromo-3-methylprop-2-ene
- (3) 1-Bromo-2-methylbut-2-ene
- (4) 4-Bromo-3-methylbut-2-ene

Q4. What is the correct IUPAC name of



- (1) 4-Ethyl-1-hydroxycyclopent-2-ene
- (2) 1-Ethyl-3-hydroxycyclopent-2-ene
- (3) 1-Ethylcyclopent-2-en-3-ol
- (4) 4-Ethylcyclopent-2-en-1-ol

Q5. Consider the following compound (X)



The most stable and least stable carbon radicals, respectively, produced by homolytic cleavage of corresponding C – H bond are :

(1) II, IV

(2) III, II

(3) I, IV

(4) II, I

Q6. Given below are two statements:

Statement I : Hyperconjugation is not a permanent effect.

Statement II : In general, greater the number of alkyl groups attached to a positively charged C -atom, greater is the hyperconjugation interaction and stabilization of the cation.

In the light of the above statements, choose the correct answer from the options given below

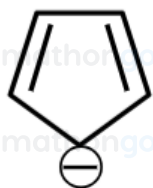
(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are true

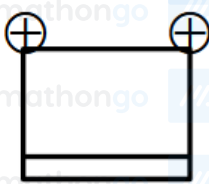
Q7. Designate whether each of the following compounds is aromatic or not aromatic.



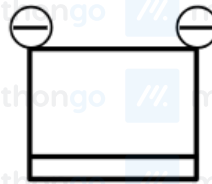
(a)



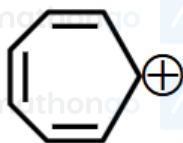
(b)



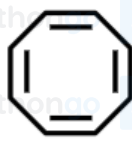
(c)



(d)



(e)



(f)



(g)



(h)

(1) e, g aromatic and a, b, c, d, f, h not aromatic

(2) b, e, f, g aromatic and a, c, d, h not aromatic

(3) a, b, c, d aromatic and e, f, g, h not aromatic

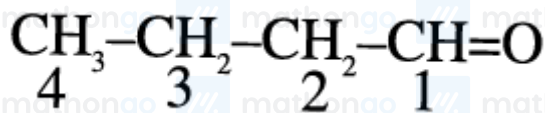
(4) a, c, d, e, h aromatic and b, f, g not aromatic

Q8. Given below are two statements.

Statement I : The dipole moment of $\text{CH}_3 - \overset{4}{\text{C}}\text{H} = \overset{2}{\text{C}} - \overset{1}{\text{H}} - \text{O} = \text{O}$ is greater than

$\text{CH}_3 - \overset{3}{\text{CH}_2} - \overset{\text{H}}{\text{C}} - 2 - \overset{1}{\text{CH}} = \text{O}$

Statement II : $\text{C}_1 - \text{C}_2$ bond length of $\text{CH}_3 - \underset{4}{\text{CH}} = \underset{3}{\text{CH}} = \underset{2}{\text{CH}} - \text{CH} = \text{O}$ is greater than $\text{C}_1 - \text{C}_2$ bond length of



In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

Q9. Match the LIST-I with LIST-II

	LIST-I		LIST-II
A.	Carbocation	I.	Species that can supply a pair of electrons.
B.	C-Free radical	II.	Species that can receive a pair of electrons.
C.	Nucleophile	III.	sp^2 hybridized carbon with empty p-orbital.
D.	Electrophile	IV.	sp^2/sp^3 hybridized carbon with one unpaired electron.

Choose the correct answer from the options given below :

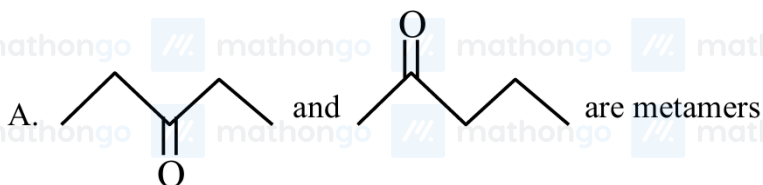
(1) A-IV, B-II, C-III, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-IV, C-I, D-II

Q10. Identify the correct statements from the following



Choose the correct answer from the options given below

(1) C & D only

(2) B & C only

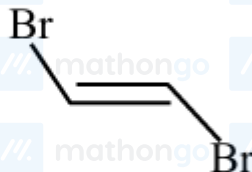
(3) A & B only

(4) A,B & C only

Q11. Given below are two statements :

Statement (I) :  is more polar than



Statement (II) : Boiling point of  is

lower than  but it is more polar than



In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

Q12. Match List-I with List-II

List-I (Purification technique)		List-II (Mixture of organic compounds)	
(A)	Distillation (simple)	(I)	Diesel + Petrol
(B)	Fractional distillation	(II)	Aniline + Water
(C)	Distillation under reduced pressure	(III)	Chloroform + Aniline
(D)	Steam distillation	(IV)	Glycerol + Spent-lye

Choose the correct answer from the options given below :

(1) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(2) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(3) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Q13. Match List-I with List-II -

	List-I (Separation of)		List-II (Separation Technique)
(A)	Aniline from aniline-water mixture	(I)	Simple distillation
(B)	Glycerol from spent-lye in soap industry	(II)	Fractional distillation
(C)	Different fractions of crude oil in petroleum industry	(III)	Distillation at reduced pressure
(D)	Chloroform-Aniline mixture	(IV)	Steam distillation

Choose the correct answer from the options given below :

(1) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(2) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(3) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(4) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Q14. In Dumas' method for estimation of nitrogen, 0.5 gram of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound (Aqueous tension at 300 K = 15 mmHg) is

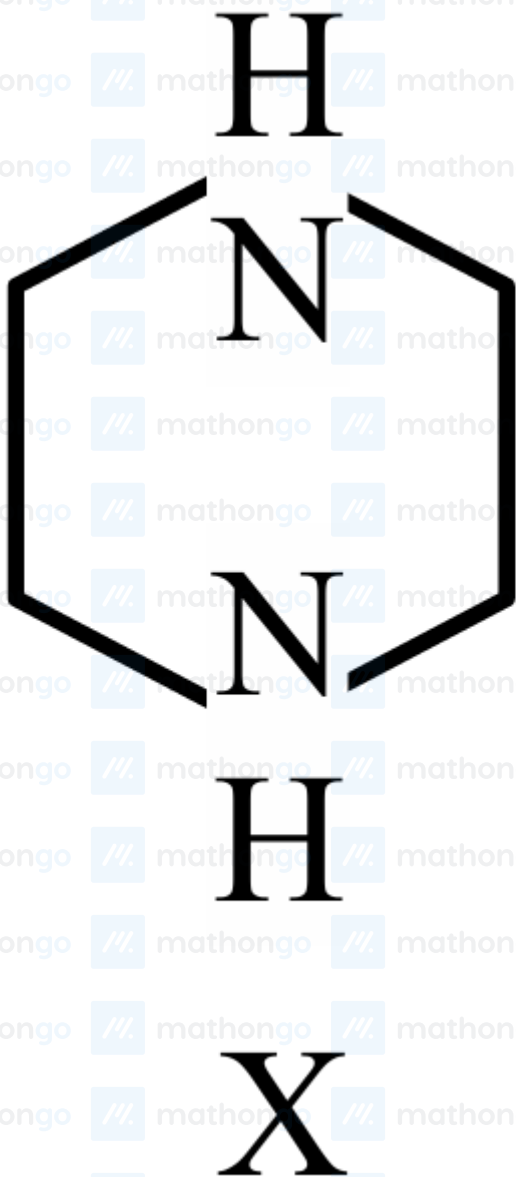
(1) 1.257

(2) 20.87

(3) 18.67

(4) 12.57

Q15. During estimation of nitrogen by Dumas' method of compound X(0.42 g) :



_____ mL of N_2 gas will be liberated at STP. (nearest integer)

(Given molar mass in gmol^{-1} : C : 12, H : 1, N : 14)

Q16. In Dumas' method for estimation of nitrogen 0.4 g of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is

(Given : Aqueous tension at 300 K = 15 mmHg)

(1) 15.71%

(2) 20.95%

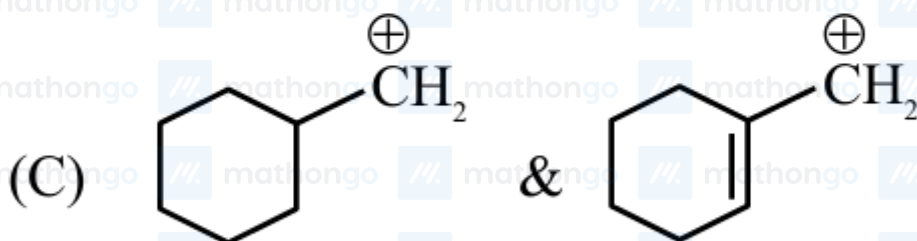
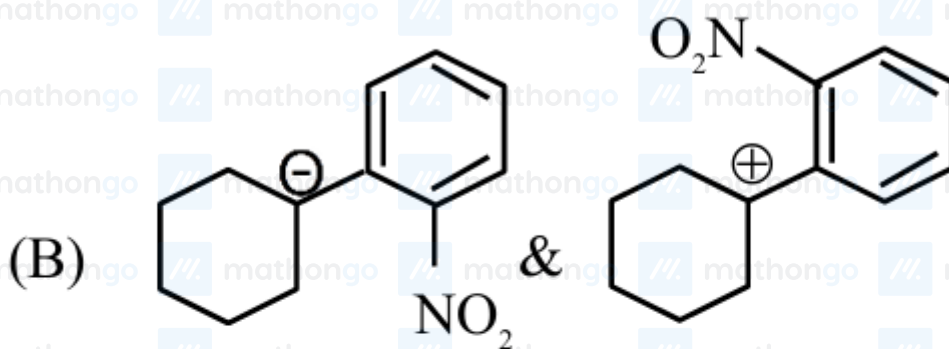
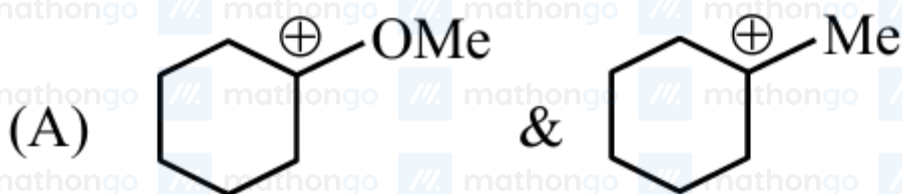
(3) 17.46%

(4) 7.85%

Q17. In Dumas' method for estimation of nitrogen 1 g of an organic compound gave 150 mL of nitrogen collected at 300 K temperature and 900 mm Hg pressure. The percentage composition of nitrogen in the compound is _____ % (nearest integer).
(Aqueous tension at 300 K = 15 mmHg)

Q18. In Dumas' method 292 mg of an organic compound released 50 mL of nitrogen gas (N_2) at 300 K temperature and 715 mm Hg pressure. The percentage composition of 'N' in the organic compound is _____ % (Nearest integer)
(Aqueous tension at 300 K = 15 mmHg)

Q19. In which pairs, the first ion is more stable than the second?



(1) (B) & (D) only

(2) (A) & (B) only

(3) (B) & (C) only

(4) (A) & (C) only

Q20. The total number of structural isomers possible for the substituted benzene derivatives with the molecular formula C_9H_{12} is _____.

Q1. 0.5 g of an organic compound on combustion gave 1.46 g of CO_2 and 0.9 g of H_2O . The percentage of carbon in the compound is _____. (Nearest integer) [Given : Molar mass (in gmol^{-1}) C : 12, H : 1, O: 16]

Q2. Given below are two statements :

Statement (I) : Neopentane forms only one monosubstituted derivative.

Statement (II) : Melting point of neopentane is higher than n-pentane

In the light of the above statements, choose the most appropriate answer from the options given below :

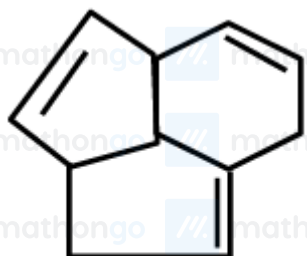
(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are correct

(3) Both Statement I and Statement II are incorrect

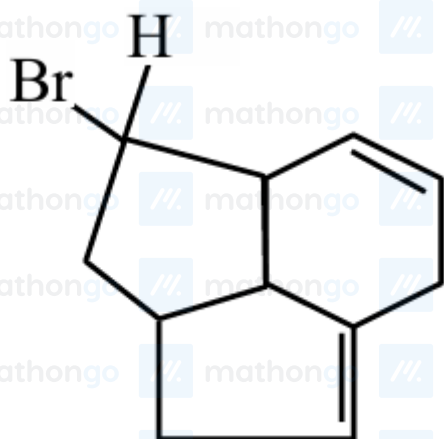
(4) Statement I is incorrect but Statement II is correct

Q3. Consider the following molecule (X).

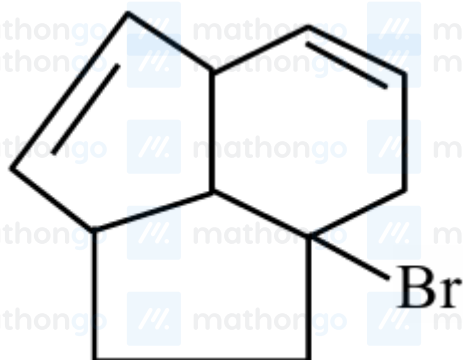


(X)

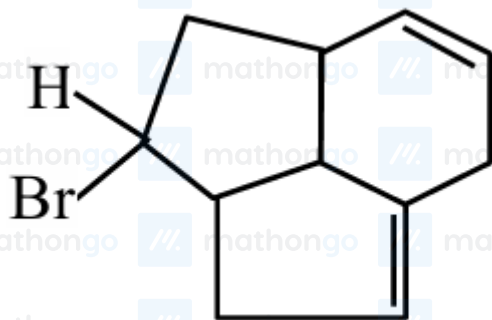
The structure of X is



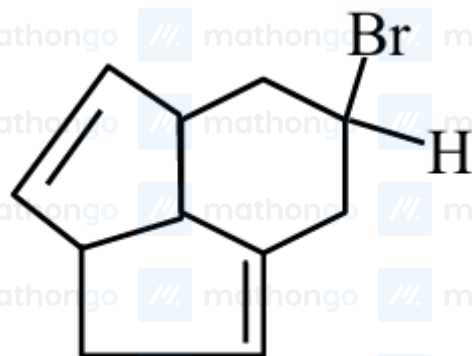
(1)



(2)

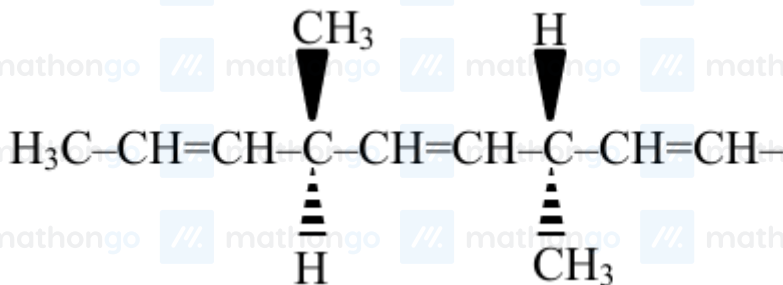


(3)



(4)

Q4. The number of optically active products obtained from the complete ozonolysis of the given compound is :



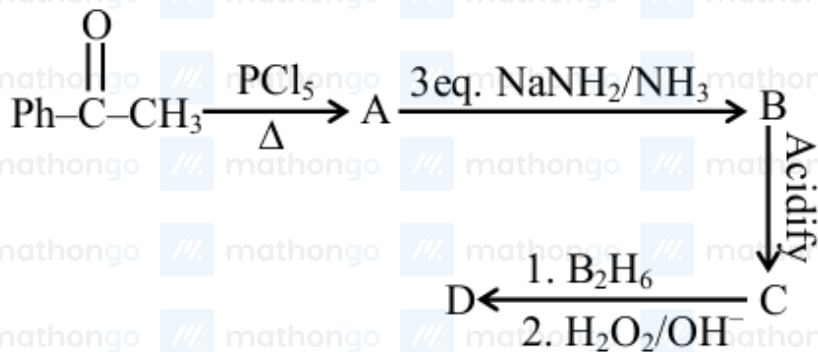
(1) 2

(2) 0

(3) 1

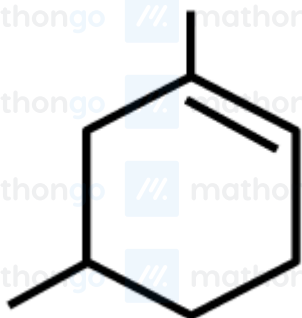
(4) 4

Q5. Identify the structure of the final product (D) in the following sequence of the reactions :



Total number of sp^2 hybridised carbon atoms in product D is.

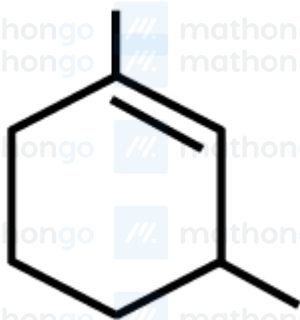
Q6. Which compound would give 3-methyl-6-oxoheptanal upon ozonolysis?



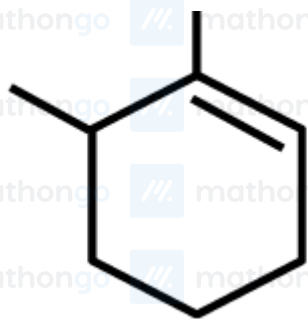
(1)



(2)



(3)



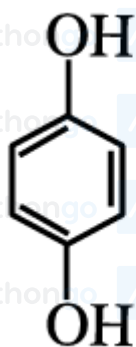
(4)

Q7. X g of nitrobenzene on nitration gave 4.2 g of m-dinitrobenzene.

X = _____ g. (nearest integer)

[Given : molar mass (in gmol^{-1}) C : 12, H : 1, O : 16, N : 14]

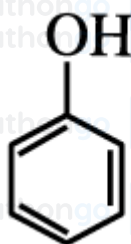
Q8. Benzene is treated with oleum to produce compound (X) which when further heated with molten sodium hydroxide followed by acidification produces compound (Y). The compound Y is treated with zinc metal to produce compound (Z). Identify the structure of compound (Z) from the following option.



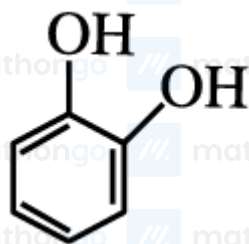
(1)



(2)

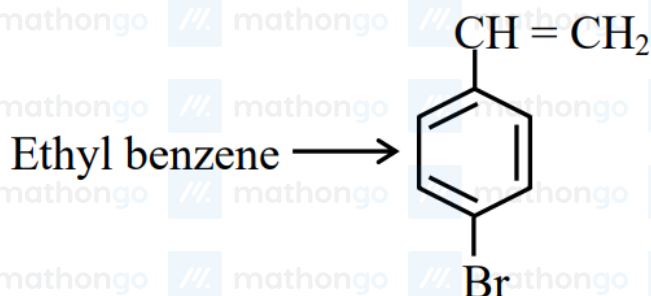


(3)



(4)

Q9. Choose the correct set of reagents for the following conversion.



(1) Br_2/Fe ; Cl_2, Δ ; alc. KOH

(2) Cl_2/Fe ; $\text{Br}_2/\text{anhy. AlCl}_3$; aq. KOH

(3) $\text{Br}_2/\text{anhy. AlCl}_3$; Cl_2, Δ ; aq. KOH

(4) $\text{Cl}_2/\text{anhy. AlCl}_3$; Br_2/Fe ; alc. KOH

Q10. Given below are two statements :

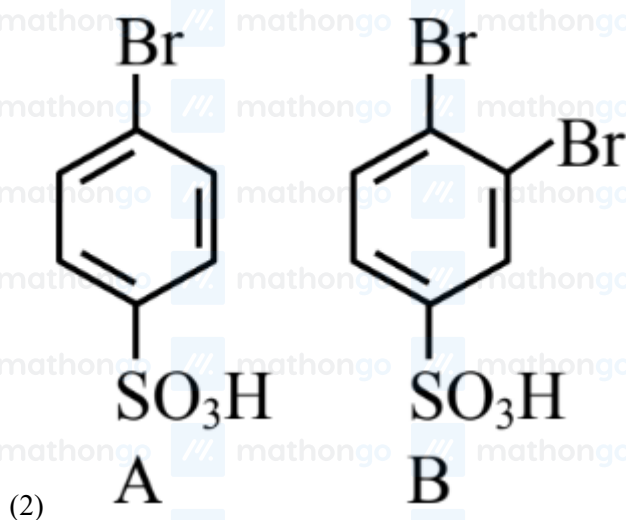
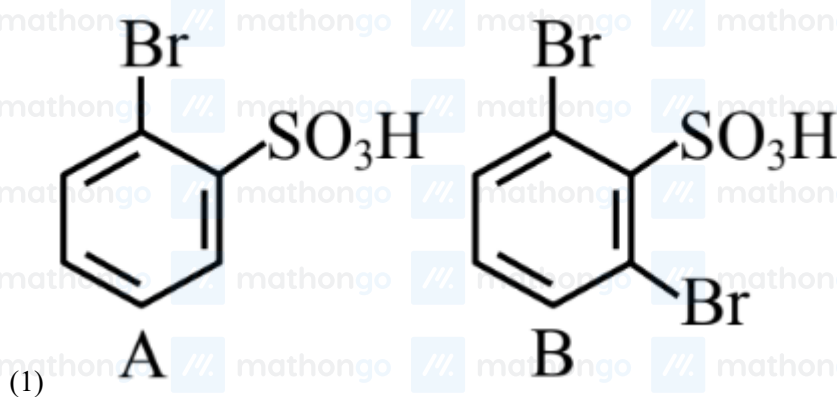
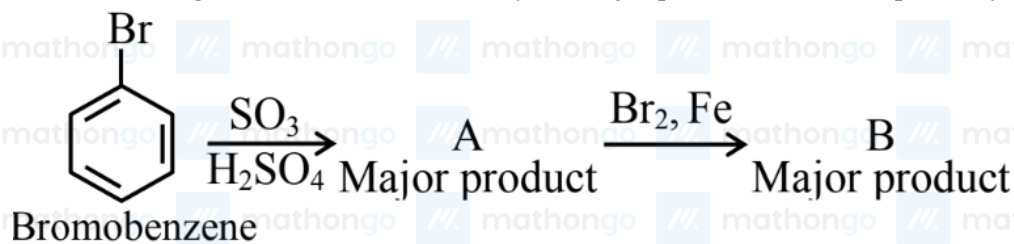
Statement I : Ozonolysis followed by treatment with $\text{Zn}, \text{H}_2\text{O}$ of cis-2-butene gives ethanal. **Statement II :** The production obtained by ozonolysis followed by treatment with $\text{Zn}, \text{H}_2\text{O}$ of 3, 6-dimethyloct-4-ene has no chiral carbon atom.

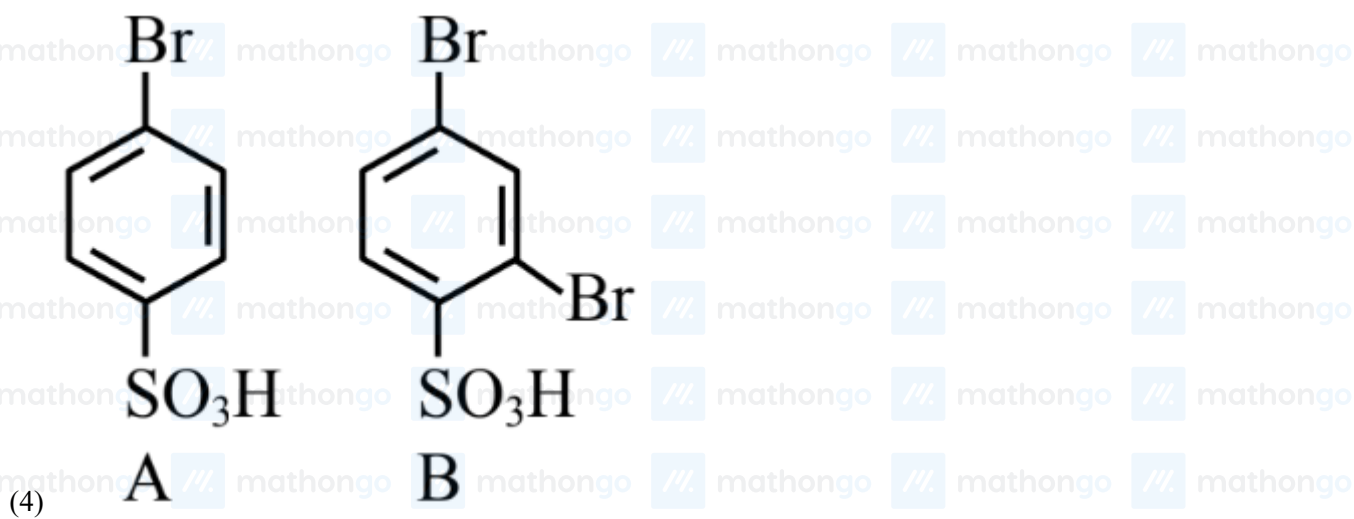
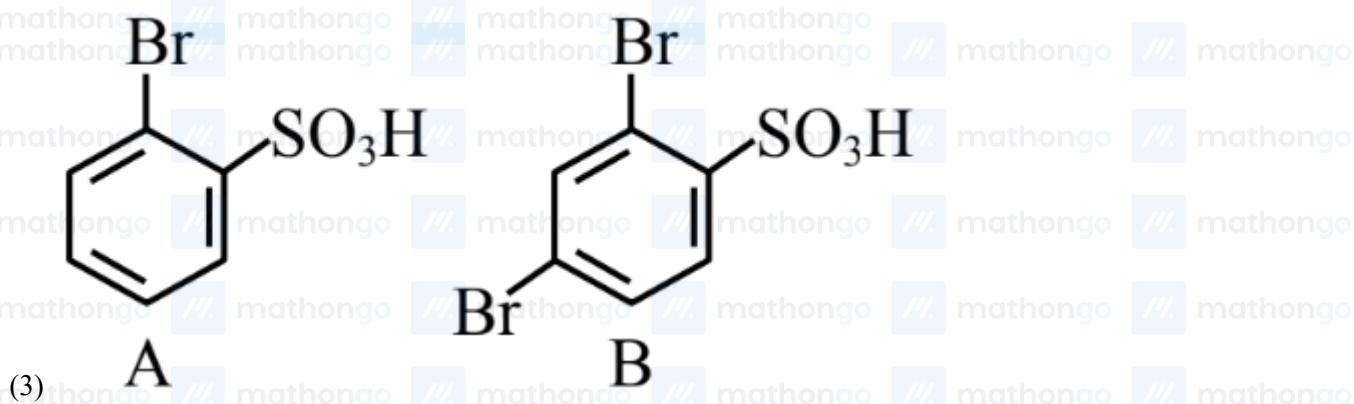
In the light of the above statements, choose the **correct** answer from the options given below

(1) Both Statement I and Statement II are true

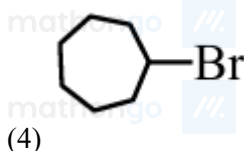
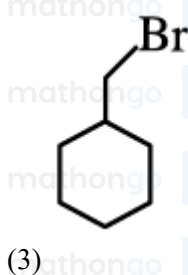
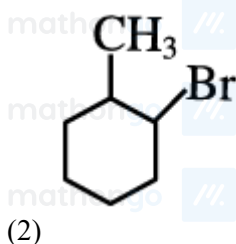
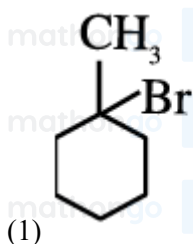
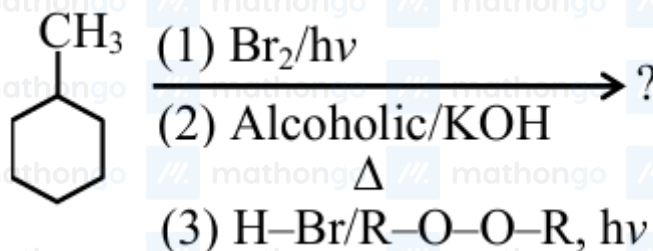
- (2) Statement I is false but Statement II are true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

Q11. In the following series of reactions identify the major products A & B respectively.





Q1. Predict the major product of the following reaction sequence :-



Q2. Given below are two statements :

Statement (I) : Alcohols are formed when alkyl chlorides are treated with aqueous potassium hydroxide by elimination reaction.

Statement (II) : In alcoholic potassium hydroxide, alkyl chlorides form alkenes by abstracting the hydrogen from the β -carbon.

In the light of the above statements, choose the most appropriate answer from the options given below :

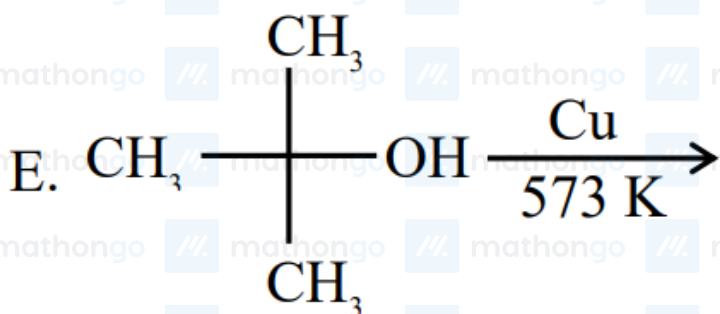
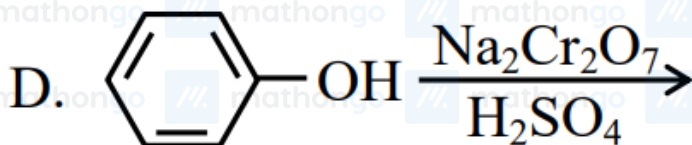
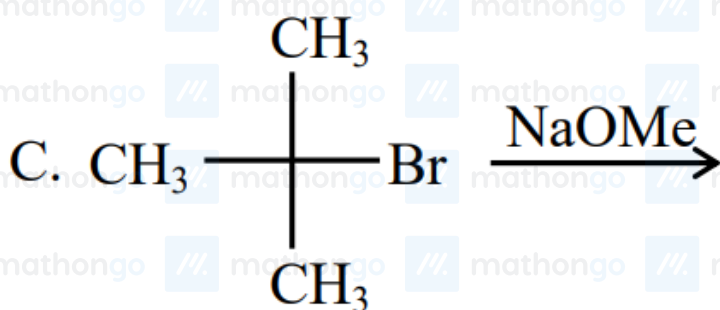
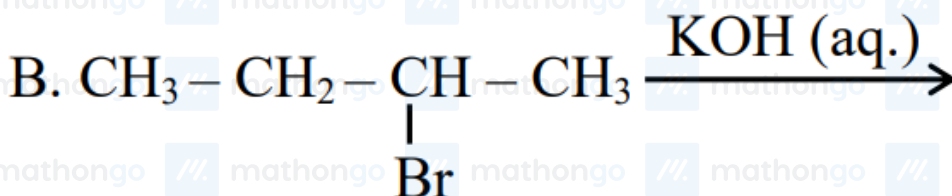
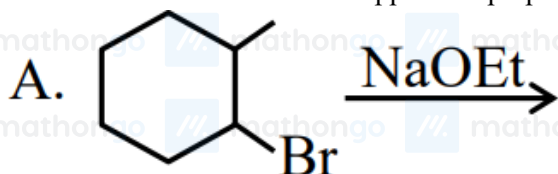
(1) Both Statement I and Statement II are incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Statement I is correct but Statement II is incorrect

(4) Both Statement I and Statement II are correct

Q3. The reactions which cannot be applied to prepare an alkene by elimination, are



Choose the **correct** answer from the option given below :

(1) B & E Only

(2) B, C & D Only

(3) A, C & D Only

(4) B & D Only

Q4. Match List-I with List-II

List-I (Reaction)		List-II (Name of reaction)	
(A)	$2 \text{C}_6\text{H}_5\text{X} + 2\text{Na} \xrightarrow[\text{Ether}]{\text{Dry}} \text{C}_6\text{H}_5\text{C}_6\text{H}_5 + 2\text{NaX}$	(I)	Lucas reaction
(B)	$\text{ArN}_2^+\text{X}^- \xrightarrow[\text{HCl}]{\text{Cu}} \text{ArCl} + \text{N}_2 \uparrow + \text{CuX}$	(II)	Finkelstein reaction
(C)	$\text{C}_2\text{H}_5\text{Br} + \text{NaI} \xrightarrow[\text{Acetone}]{\text{Dry}} \text{C}_2\text{H}_5\text{I} + \text{NaBr}$	(III)	Fittig reaction
(D)	$\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)_2 \xrightarrow[\text{ZnCl}_2]{\text{HCl}} \text{CH}_3\text{C}(\text{Cl})(\text{CH}_3)_2$	(IV)	Gatterman reaction

Choose the correct answer from the options given below :

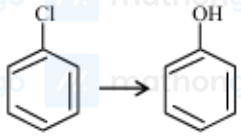
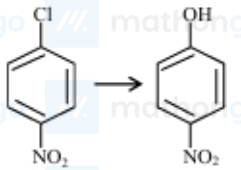
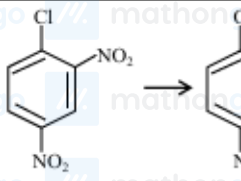
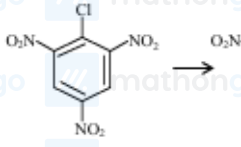
(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(3) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(4) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Q5. Match List-I with List-II

List-I Conversion		List-II Reagents, Conditions used	
(A)		(I)	Warm, H ₂ O
(B)		(II)	(a) NaOH, 368 K ; (b) H ₃ O ⁺
(C)		(III)	(a) NaOH, 443 K ; (b) H ₃ O ⁺
(D)		(IV)	(a) NaOH, 623 K, 300 atm ; (b) H ₃ O ⁺

Choose the correct answer from the options given below :

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Q1. Given below are two statements :

Statement I : Dimethyl ether is completely soluble in water. However, diethyl ether is soluble in water to a very small extent. **Statement II :** Sodium metal can be used to dry diethyl ether and not ethyl alcohol. In the light of given statements, choose the **correct** answer from the options given below

(1) Statement I is false but Statement II are true

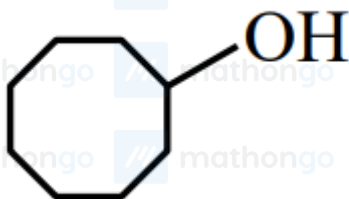
(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

Q2. 1, 2-dibromocyclooctane $\xrightarrow[\text{(iv) Zn-Hg/H}^+]{\text{(i) KOH (alc.)}, \text{(ii) NaNH}_2, \text{(iii) Hg}^{2+}/\text{H}^+}$ **P**
(Major product)

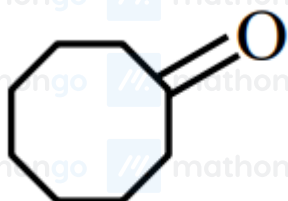
'P' is



(1)



(2)



(3)



(4)

Q3. Match the LIST-I with LIST-II

	LIST-I (Reagent)		LIST-II (Functional Group detected)
A.	Sodium bicarbonate solution	I.	double bond/unsaturation
B.	Neutral ferric chloride	II.	carboxylic acid
C.	ceric ammonium nitrate	III.	phenolic - OH
D.	alkaline KMnO	IV.	alcoholic - OH

Choose the correct answer from the options given below :

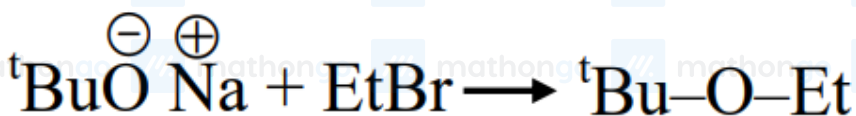
(1) A-II, B-III, C-IV, D-I

(2) A-II, B-III, C-I, D-IV

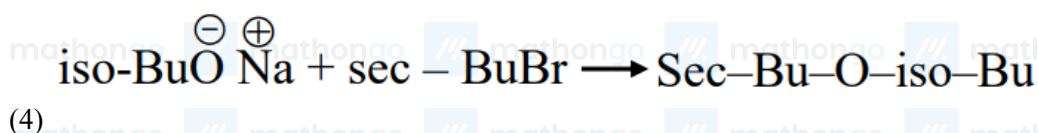
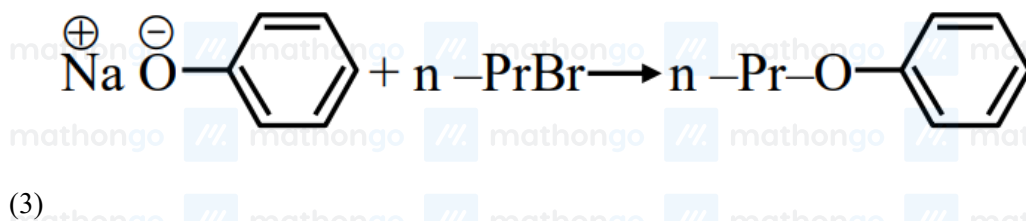
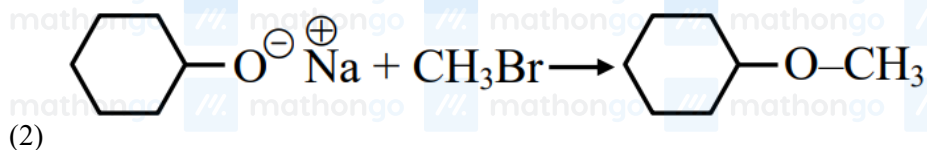
(3) A-III, B-II, C-IV, D-I

(4) A-II, B-IV, C-III, D-I

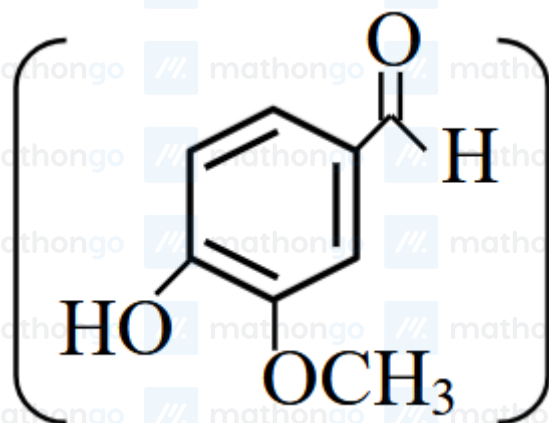
Q4. Which one of the following reactions will not lead to the desired ether formation in major proportion?

(iso-Bu $\Rightarrow$ isobutyl, sec-Bu $\Rightarrow$ sec-butyl,nPr $\Rightarrow$ n-propyl, ^tBu $\Rightarrow$ tert-butyl, Et $\Rightarrow$ ethyl)

(1)

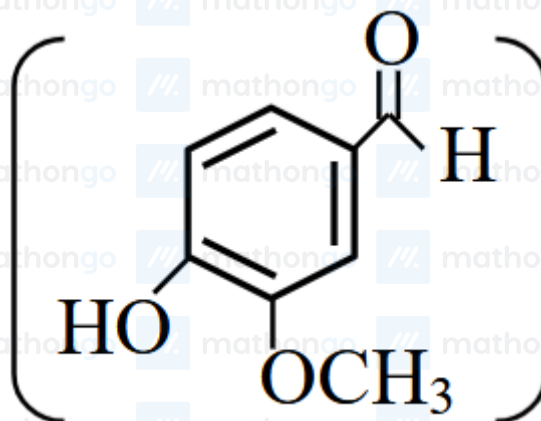


Q5. Given below are two statements :



Statement (I) : Vanillin

will react with NaOH and also with Tollen's reagent.



Statement (II) : Vanillin

will undergo self aldol condensation very easily.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is incorrect but Statement II is correct

- (2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are incorrect
(4) Both Statement I and Statement II are correct

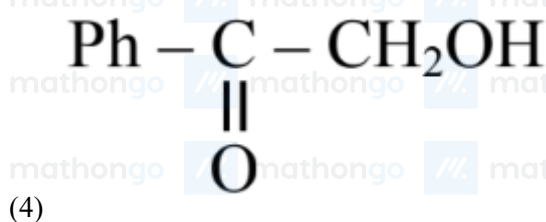
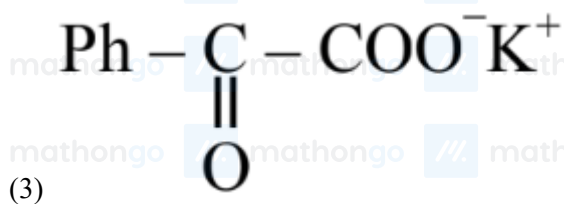
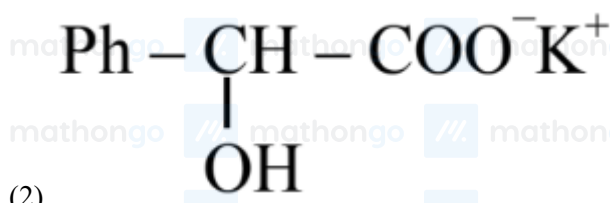
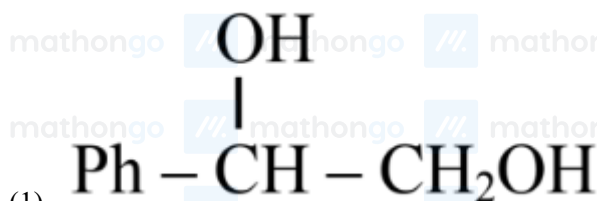
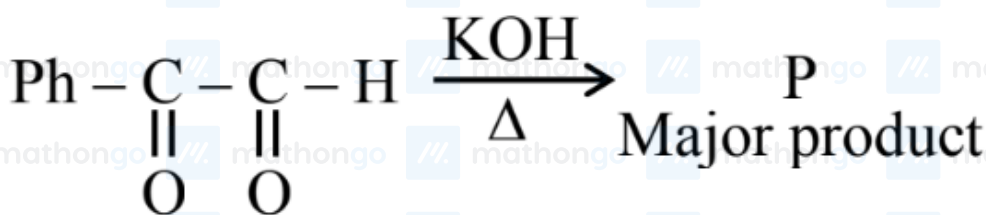
Q6. An organic compound (X) with molecular formula C_3H_6O is not readily oxidised. On reduction it gives C_3H_8O (Y) which reacts with HBr to give a bromide (Z) which is converted to Grignard reagent. This Grignard reagent on reaction with (X) followed by hydrolysis give 2,3-dimethylbutan-2-ol. Compounds (X), (Y) and (Z) respectively are :

- (1) CH_3COCH_3 , $CH_3CH_2CH_2OH$, $CH_3CH(Br)CH_3$
(2) CH_3COCH_3 , $CH_3CH(OH)CH_3$, $CH_3CH(Br)CH_3$
(3) CH_3CH_2CHO , $CH_3CH_2CH_2OH$, $CH_3CH_2CH_2Br$
(4) CH_3CH_2CHO , $CH_3CH=CH_2$, $CH_3CH(Br)CH_3$

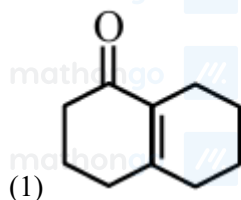
Q7. A toxic compound "A" when reacted with NaCN in aqueous acidic medium yields an edible cooking component and food preservative 'B'. "B" is converted to "C" by diborane and can be used as an additive to petrol to reduce emission. "C" upon reaction with oleum at $140^\circ C$ yields an inhalable anesthetic "D". Identify "A", "B", "C" and "D", respectively.

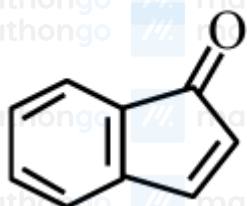
- (1) Methanol; formaldehyde; methyl chloride; chloroform
(2) Ethanol; acetonitrile; ethylamine; ethylene
(3) Methanol; acetic acid; ethanol; diethyl ether
(4) Acetaldehyde; 2-hydroxypropanoic acid; propanoic acid; dipropyl ether

Q1. The major product (P) in the following reaction is :

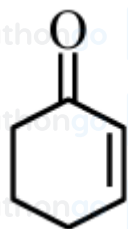


Q2. Aldol condensation is a popular and classical method to prepare α, β -unsaturated carbonyl compounds. This reaction can be both intermolecular and intramolecular. Predict which one of the following is not a product of intramolecular aldol condensation?

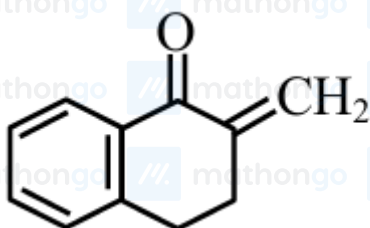




(2)



(3)



(4)

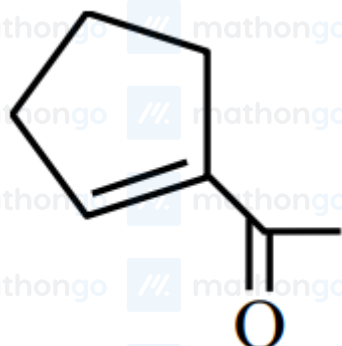
Q3.

When

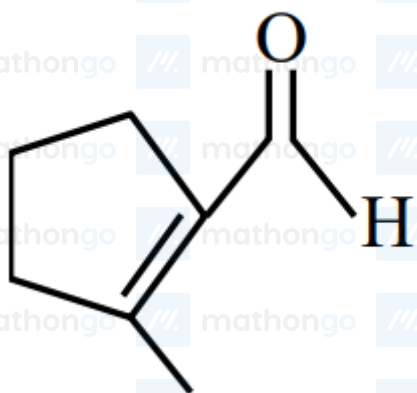


undergoes

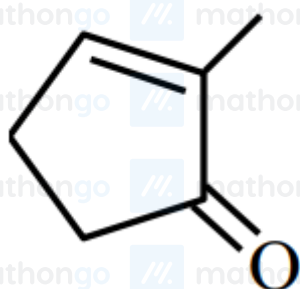
intramolecular aldol condensation, the major product formed is :



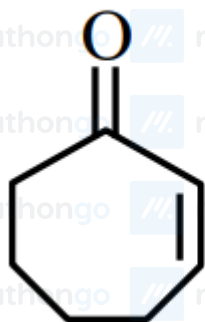
(1)



(2)

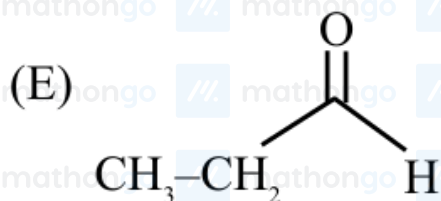
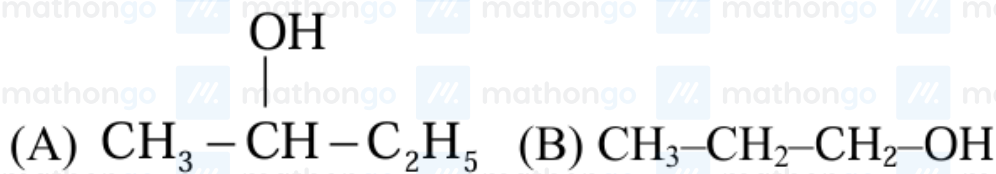


(3)



(4)

Q4. Which among the following compounds give yellow solid when reacted with NaOI/NaOH ?



Choose the correct answer from the options given below :

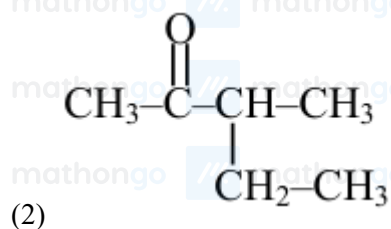
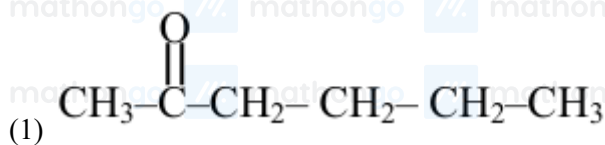
(1) (B), (C) and (E) Only

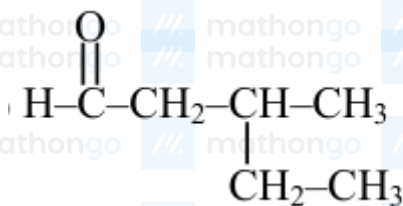
(2) (A) and (C) Only

(3) (C) and (D) Only

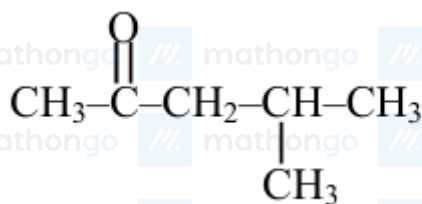
(4) (A), (C) and (D) Only

Q5. "P" is an optically active compound with molecular formula $\text{C}_6\text{H}_{12}\text{O}$. When "P" is treated with 2,4-dinitrophenylhydrazine, it gives a positive test. However, in presence of Tollens reagent, "P" gives a negative test. Predict the structure of "P".





(3)



(4)

Q6. An optically active alkyl halide $\text{C}_4\text{H}_9\text{Br}$ [A] reacts with hot KOH dissolved in ethanol and forms alkene [B] as major product which reacts with bromine to give dibromide [C]. The compound [C] is converted into a gas [D] upon reacting with alcoholic NaNH_2 . During hydration 18 gram of water is added to 1 mole of gas [D] on warming with mercuric sulphate and dilute acid at 333 K to form compound [E]. The IUPAC name of compound [E] is :

(1) But-2-yne

(2) Butan-2-ol

(3) Butan-2-one

(4) Butan-1-al

Q7. Number of molecules from below which cannot give iodoform reaction is :

Ethanol, Isopropyl alcohol, Bromoacetone, 2-Butanol, 2-Butanone, Butanal, 2-Pentanone, 3-Pentanone, Pentanal and 3-Pentanol

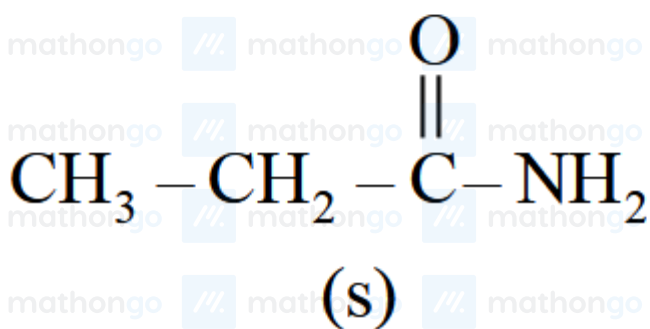
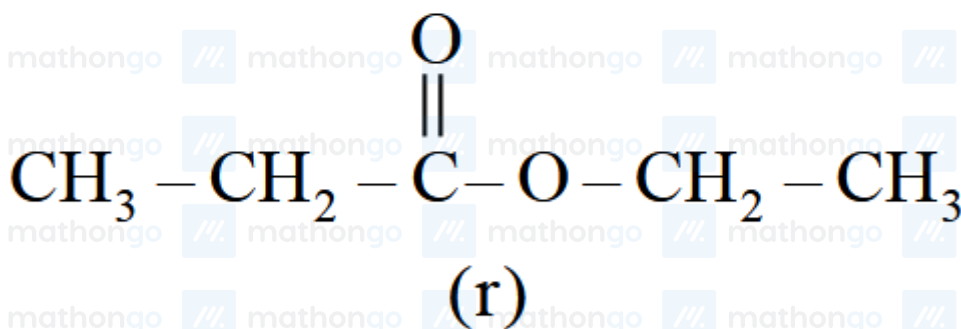
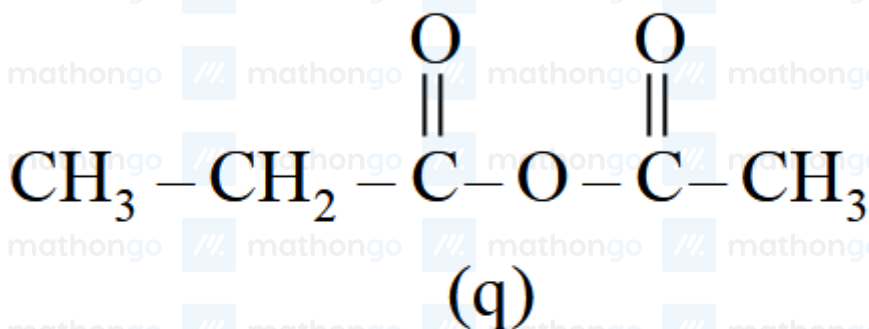
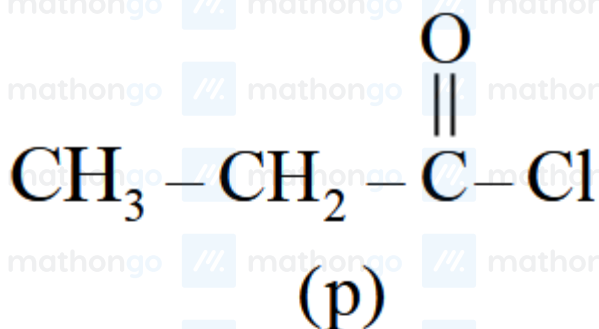
(1) 5

(2) 4

(3) 3

(4) 2

Q1. Consider the following molecules :



The correct order of rate of hydrolysis is :

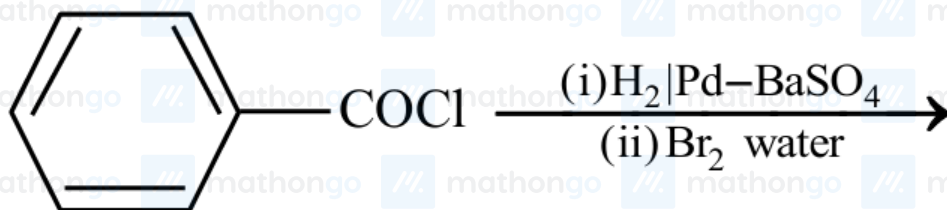
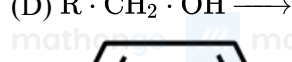
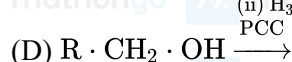
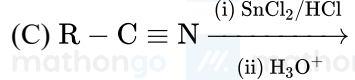
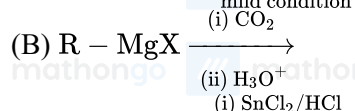
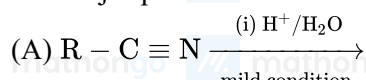
(1) $r > q > p > s$

(2) $q > p > r > s$

(3) $p > r > q > s$

(4) $p > q > r > s$

Q2. Consider the following reactions. From these reactions which reaction will give carboxylic acid as a major product ?



(E)

Choose the correct answer from the options given below:

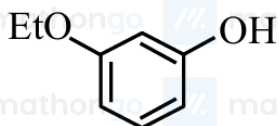
(1) A and D only

(2) A, B and E only

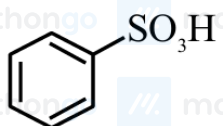
(3) B, C and E only

(4) B and E only

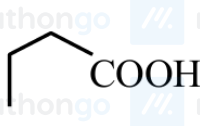
Q3. The least acidic compound, among the following is :



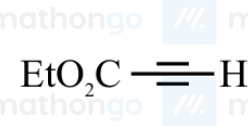
(A)



(B)



(C)



(D)

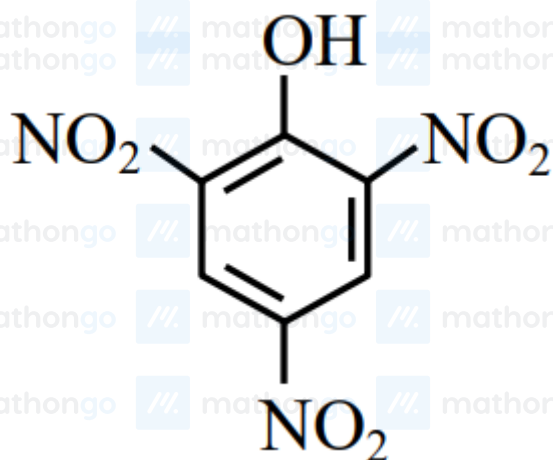
(1) D

(2) A

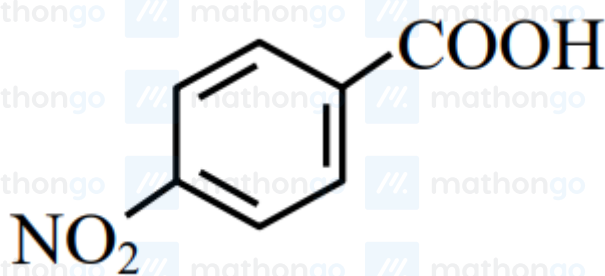
(3) B

(4) C

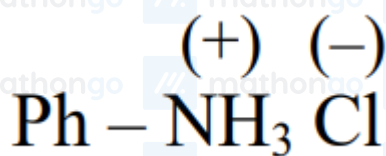
Q4. Which of the following compounds is least likely to give effervescence of CO_2 in presence of aq. $NaHCO_3$?



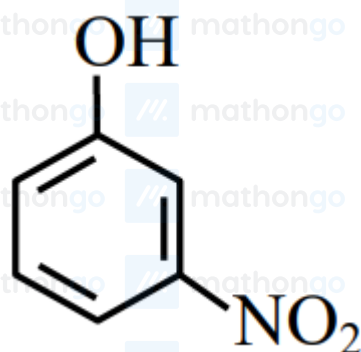
(1)



(2)



(3)



(4)

Q5. Mixture of 1 g each of chlorobenzene, aniline and benzoic acid is dissolved in 50 mL ethyl acetate and placed in a separating funnel, 5 M NaOH (30 mL) was added in the same funnel. The funnel was shaken vigorously and then kept aside. The ethyl acetate layer in the funnel contains :

(1) benzoic acid

(2) benzoic acid and aniline

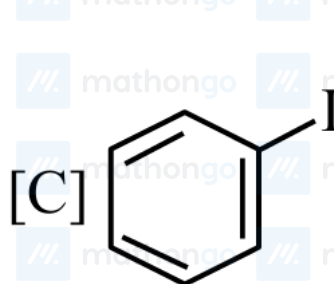
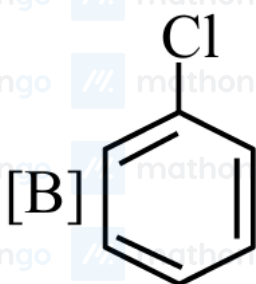
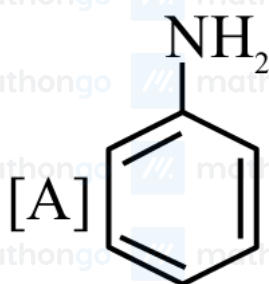
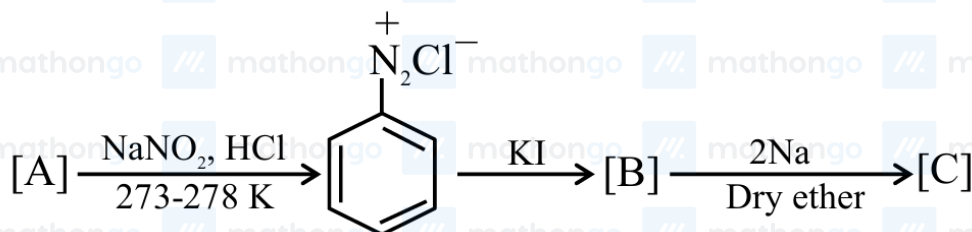
(3) benzoic acid and chlorobenzene

(4) chlorobenzene and aniline

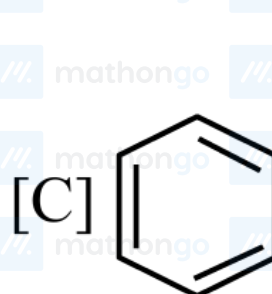
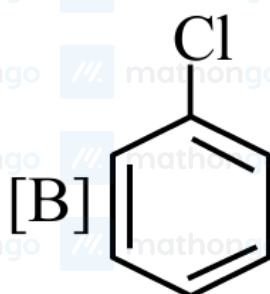
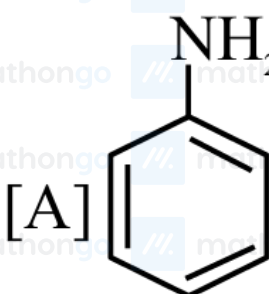
Q1. The correct order of basic nature on aqueous solution for the bases NH_3 , $\text{H}_2\text{N}-\text{NH}_2$, $\text{CH}_3\text{CH}_2\text{NH}_2$, $(\text{CH}_3\text{CH}_2)_2\text{NH}$ and $(\text{CH}_3\text{CH}_2)_3\text{N}$ is :

- (1) $\text{NH}_3 < \text{H}_2\text{N}-\text{NH}_2 < (\text{CH}_3\text{CH}_2)_3\text{N} < \text{CH}_3\text{CH}_2\text{NH}_2 < (\text{CH}_3\text{CH}_2)_2\text{NH}$
- (2) $\text{NH}_3 < \text{H}_2\text{N}-\text{NH}_2 < \text{CH}_3\text{CH}_2\text{NH}_2 < (\text{CH}_3\text{CH}_2)_2\text{NH} < (\text{CH}_3\text{CH}_2)_3\text{N}$
- (3) $\text{H}_2\text{N}-\text{NH}_2 < \text{NH}_3 < (\text{CH}_3\text{CH}_2)_3\text{N} < \text{CH}_3\text{CH}_2\text{NH}_2 < (\text{CH}_3\text{CH}_2)_2\text{NH}$
- (4) $\text{H}_2\text{N}-\text{NH}_2 < \text{NH}_3 < \text{CH}_3\text{CH}_2\text{NH}_2 < (\text{CH}_3\text{CH}_2)_3\text{N} < (\text{CH}_3\text{CH}_2)_2\text{NH}$

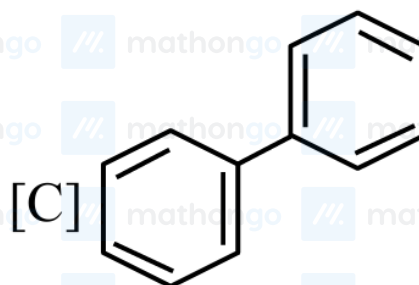
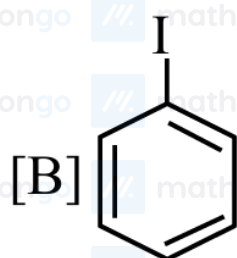
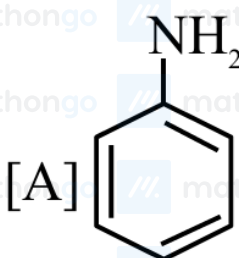
Q2. Identify [A], [B], and [C], respectively in the following reaction sequence :



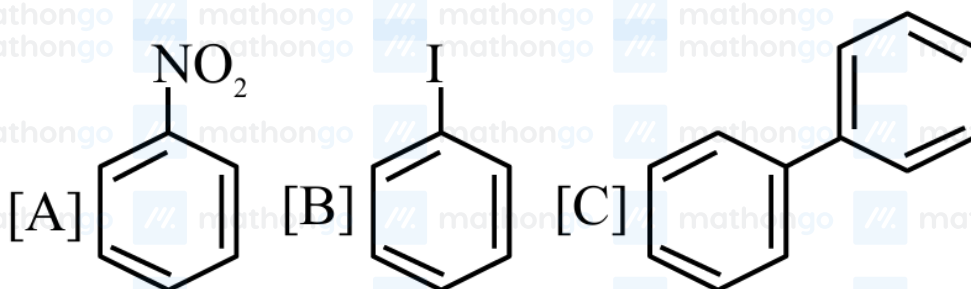
(1)



(2)

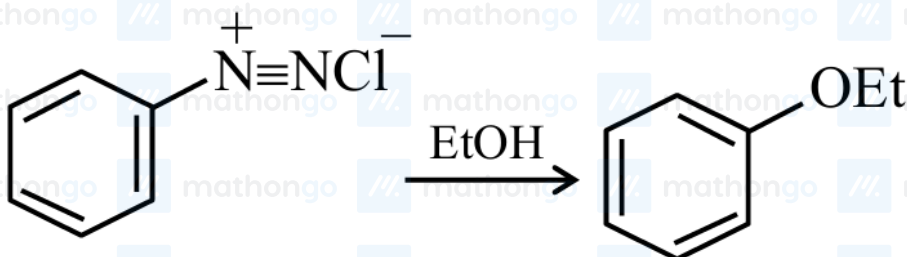


(3)

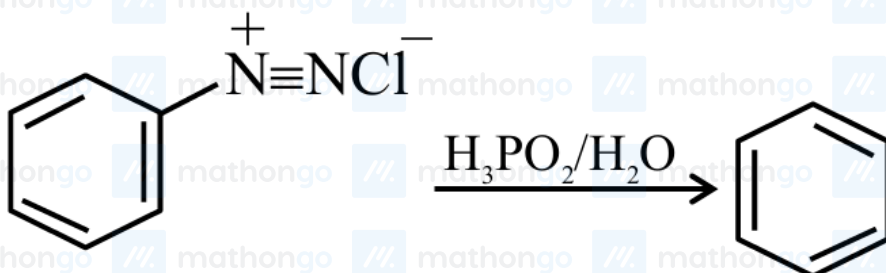


(4)

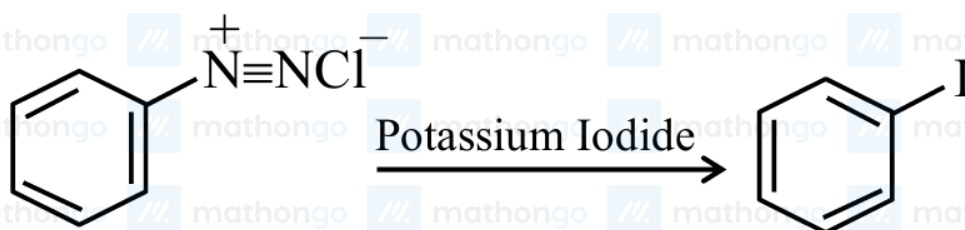
Q3. In the following reactions, which one is NOT correct?



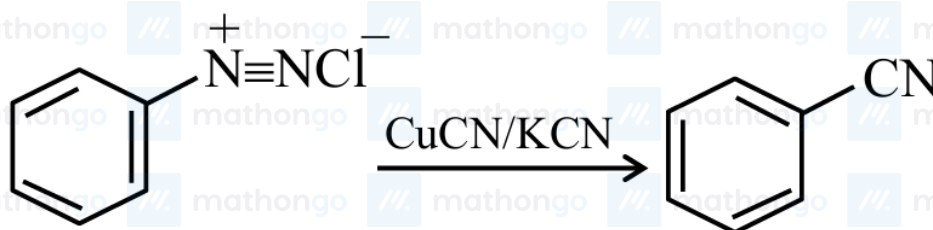
(1)



(2)

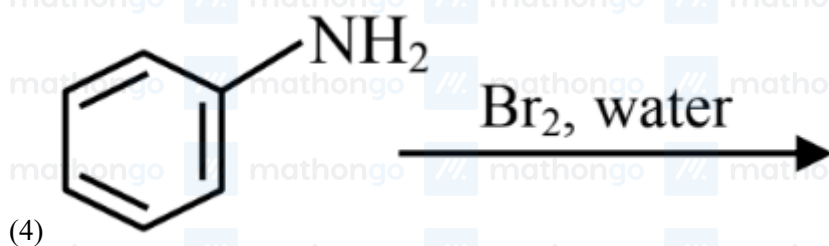
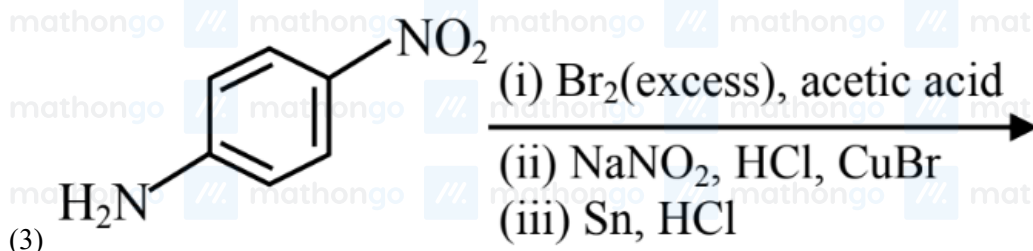
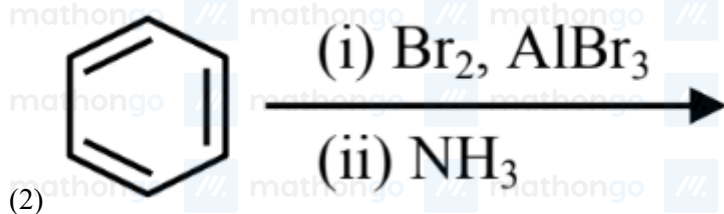
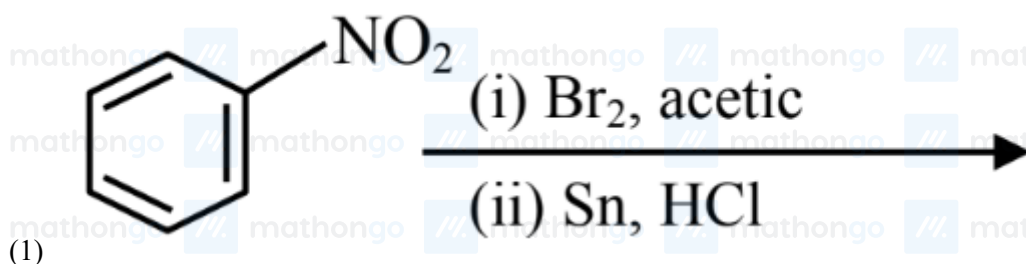


(3)

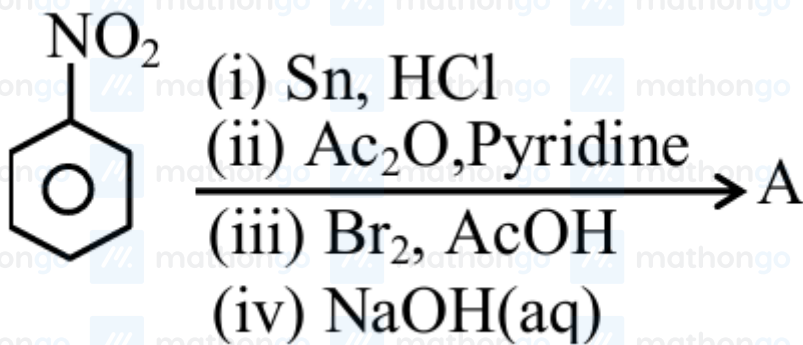


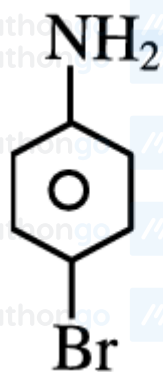
(4)

Q4. The sequence from the following that would result in giving predominantly 3, 4, 5-Tribromoaniline is :

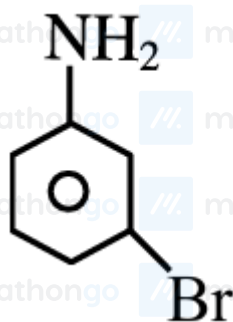


Q5. The major product (A) formed in the following reaction sequence is

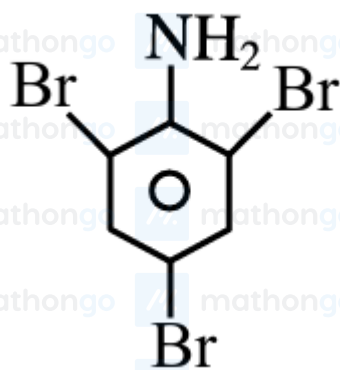




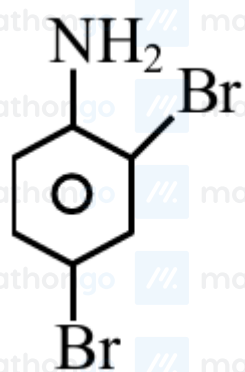
(1)



(2)

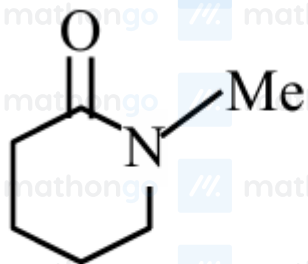


(3)

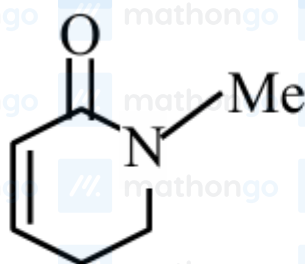


(4)

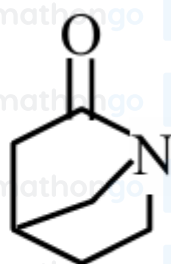
Q6. The correct order of basicity for the following molecules is:



(P)



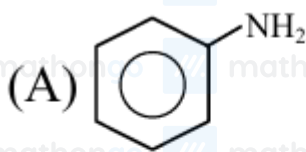
(Q)



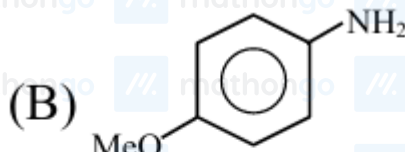
(R)

(1) $P > Q > R$ (2) $R > P > Q$ (3) $Q > P > R$ (4) $R > Q > P$

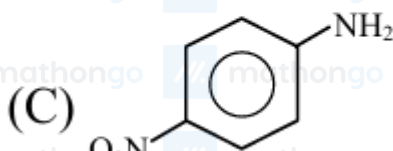
Q7. The descending order of basicity of following amines is:



(A)



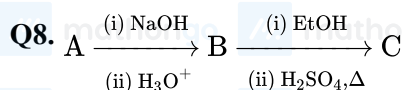
(B)



(C)

(D) CH_3NH_2 (E) $(\text{CH}_3)_2\text{NH}$

Choose the correct answer from the options given below :

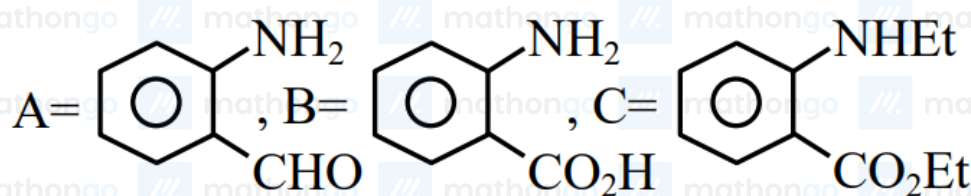
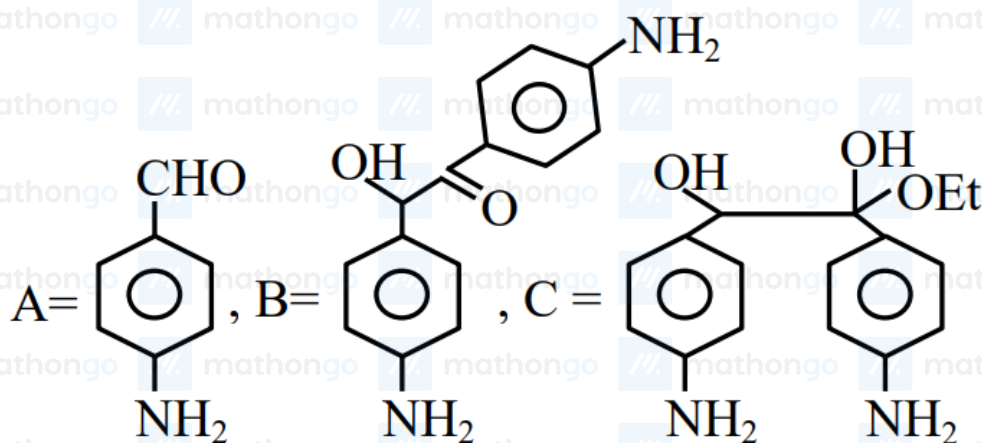
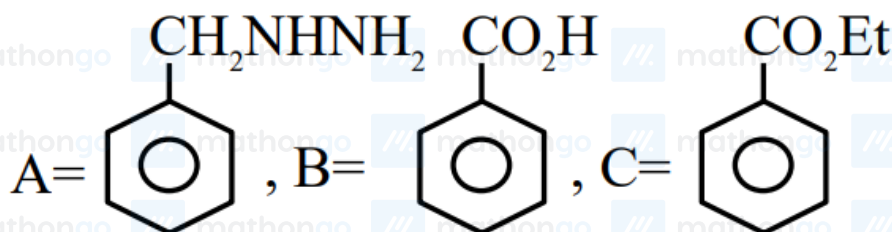
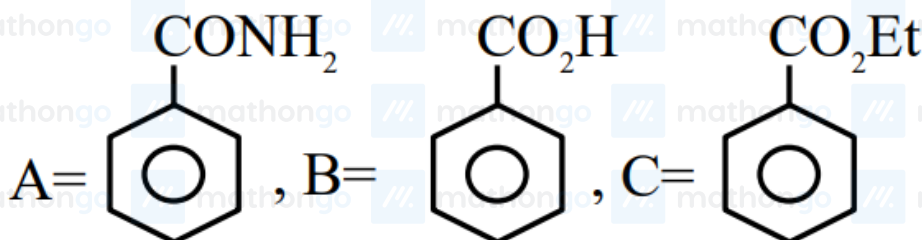
(1) $B > E > D > A > C$ (2) $E > D > B > A > C$ (3) $E > D > A > B > C$ (4) $E > A > D > C > B$ 

'A' shows positive Lassaigne's test for N and its molar mass is 121.

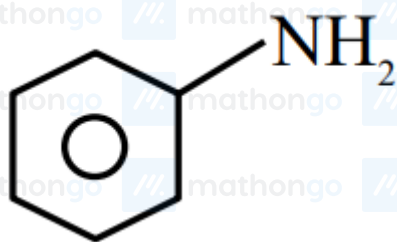
'B' gives effervescence with aq. NaHCO_3 .

'C' gives fruity smell.

Identify A, B and C from the following.

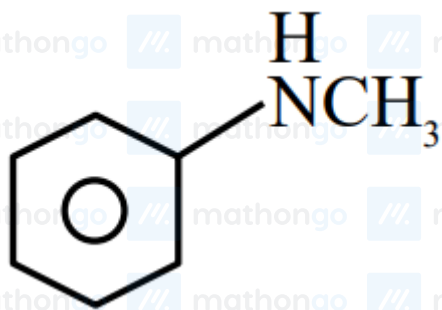


Q9. Which of the following amine (s) show (s) positive carbylamines test?



(A)

(B) $(CH_3)_2NH$

(C) CH_3NH_2 (D) $(\text{CH}_3)_3\text{N}$ 

(E)

(1) A and E Only

(2) C Only

(3) A and C Only

(4) B, C and D Only

Q10. When a concentrated solution of sulphanilic acid and 1-naphthylamine is treated with nitrous acid (273 K) and acidified with acetic acid, the mass (g) of 0.1 mole of product formed is :

(Given molar mass in gmol^{-1} H : 1, C : 12, N : 14, O : 16, S : 32)

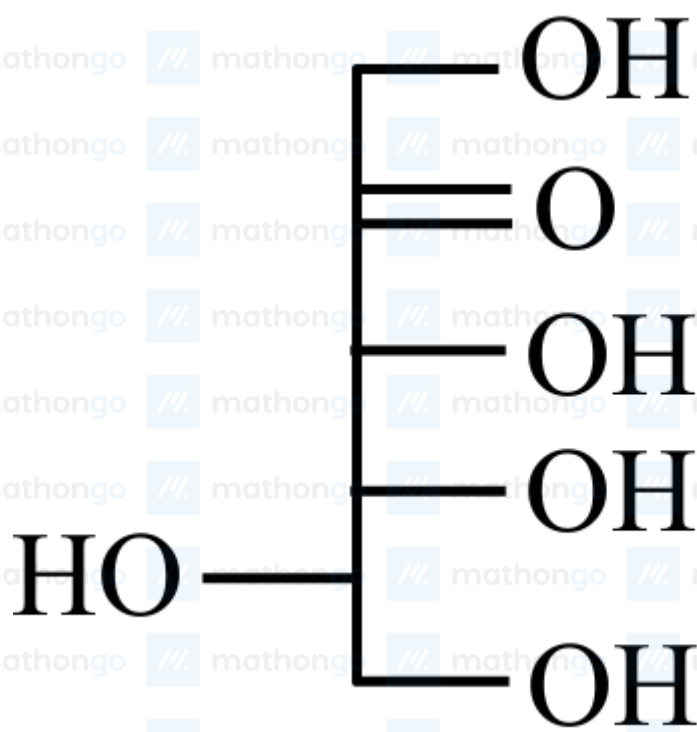
(1) 343

(2) 330

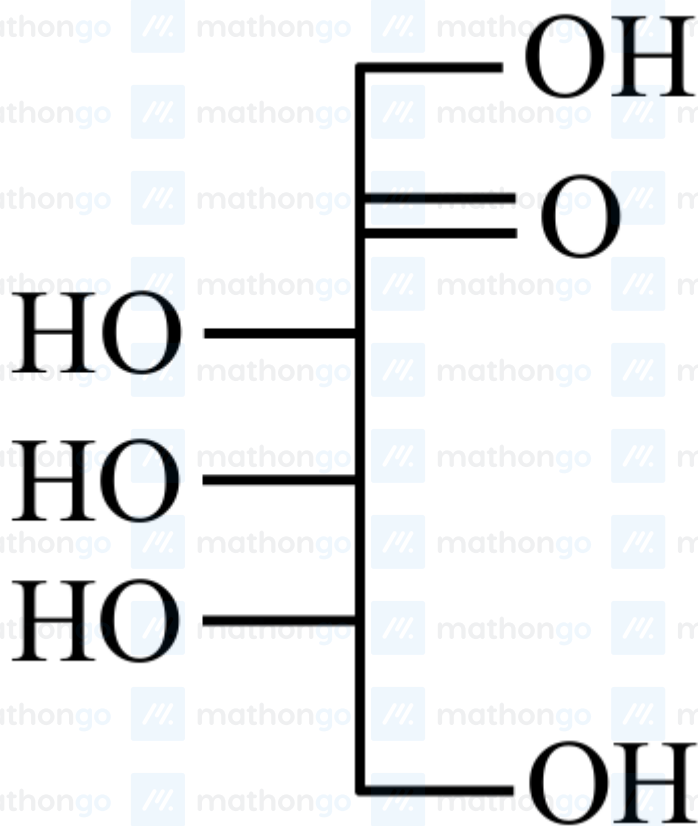
(3) 33

(4) 66

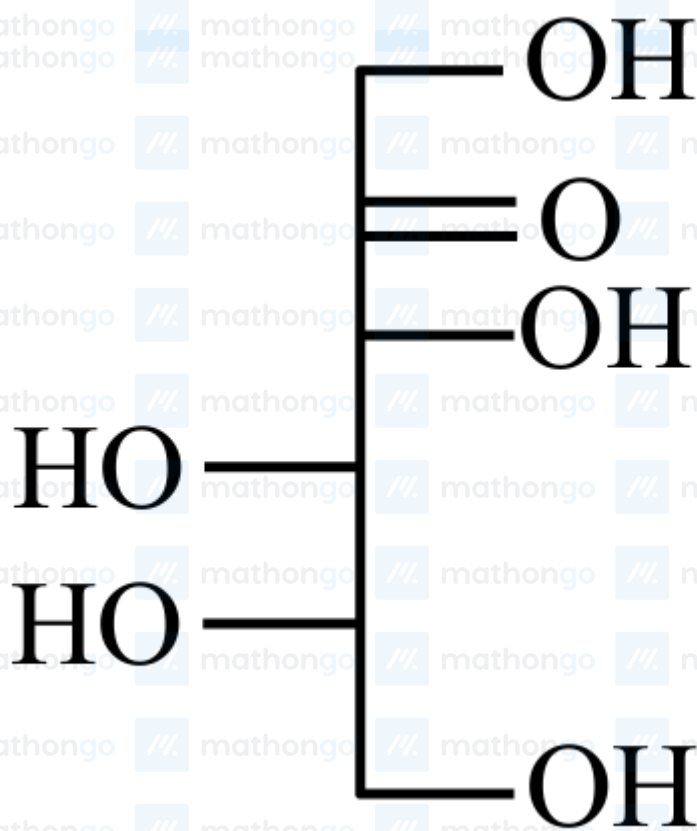
Q1. Which of the following is the correct structure of L-fructose?



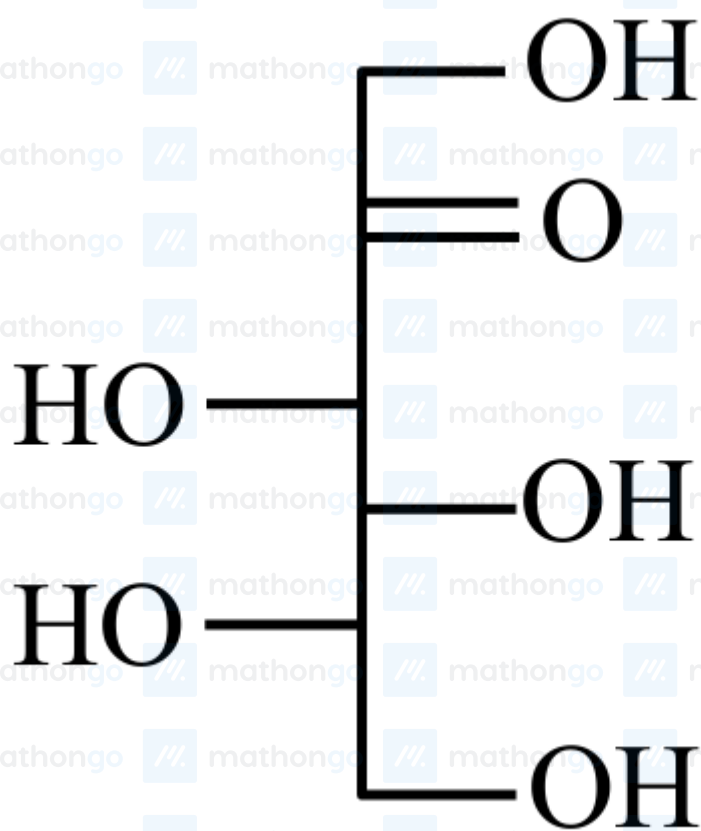
(1)



(2)



(3)



(4)

Q2. Given below are two statements:

Statement I : $D-(+)\text{-glucose} + D-(+)\text{-fructose} \xrightarrow{-H_2O}$

Statement II : Invert sugar is formed during sucrose hydrolysis.

In the light of the above statements, choose the *correct* answer from the options given below -

(1) Both Statement I and Statement II are true.

(2) Statement I is false but Statement II are true.

(3) Statement I is true but Statement II is false.

(4) Both Statement I and Statement II are false.

Q3. Fat soluble vitamins are :

A. Vitamin B_1

B. Vitamin C

C. Vitamin E

D. Vitamin B_{12}

E. Vitamin K

Choose the correct answer from the options given below :

(1) C & D Only

(2) A & B Only

(3) B & C Only

(4) C & E Only

Q4. Given below are two statements :

Statement I : Wet cotton clothes made of cellulose based carbohydrate takes comparatively longer time to get dried than wet nylon polymer based clothes.

Statement II : Intermolecular hydrogen bonding with water molecule is more in nylon-based clothes than in the case of cotton clothes.

In the light of above statements, choose the Correct answer from the options given below

(1) Statement I is false but Statement II is true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Q5. Identify the correct statement among the following:

(1) All naturally occurring amino acids except glycine contain one chiral centre.

(2) All naturally occurring amino acids are optically active.

(3) Glutamic acid is the only amino acid that contains a -COOH group at the side chain.

(4) Amino acid, cysteine easily undergo dimerization due to the presence of free SH group.

Q6. A tetrapeptide "x" on complete hydrolysis produced glycine (Gly), alanine (Ala), valine (Val), leucine (Leu) in equimolar proportion each. The number of tetrapeptides (sequences) possible involving each of these amino acids is

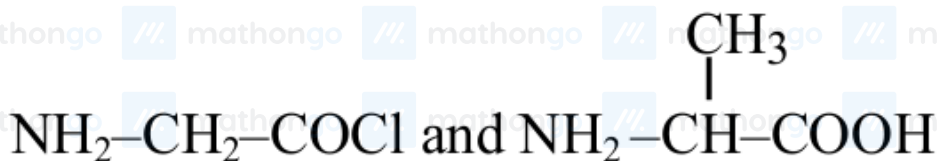
(1) 16

(2) 32

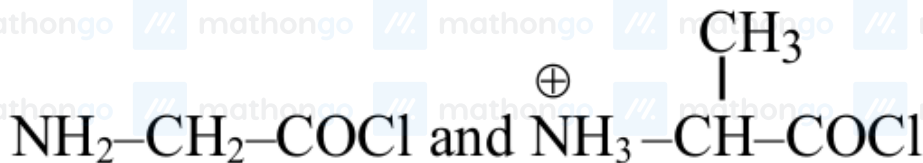
(3) 8

(4) 24

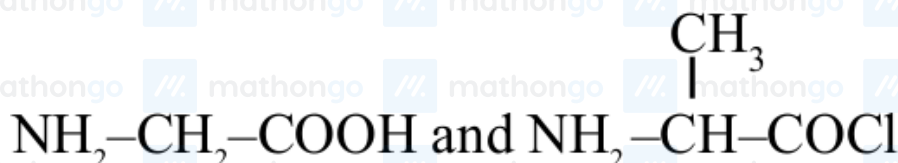
Q7. Identify the pair of reactants that upon reaction, with elimination of HCl will give rise to the dipeptide Gly-Ala.



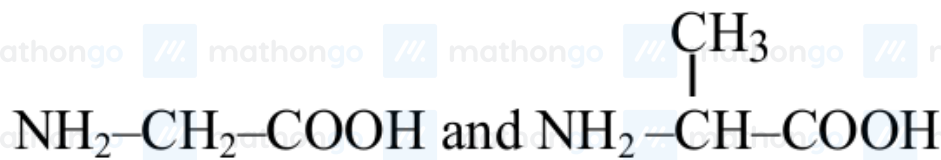
(1)



(2)

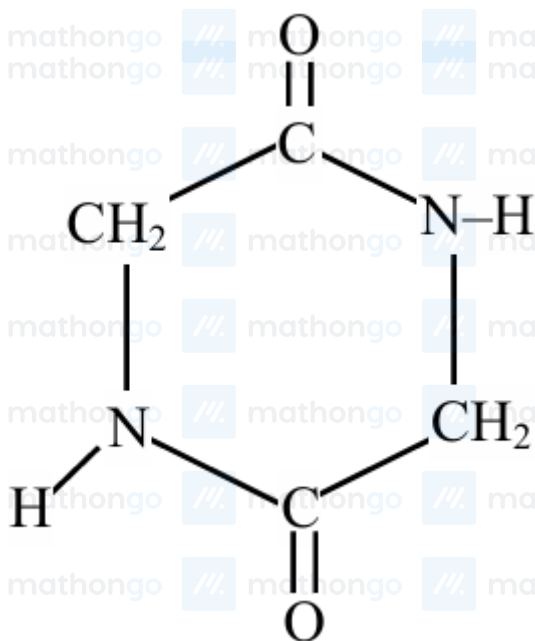


(3)



(4)

Q8. A dipeptide, "x" on complete hydrolysis gives "y" and "z". "y" on treatment with aq. HNO_2 produces lactic acid. On the other hand "z" on heating gives the following cyclic molecule.



Based on the information given, the dipeptide X is:

- (1) valine-glycine
- (2) alanine-glycine
- (3) valine-leucine
- (4) alanine-alanine

Q9. Given below are two statements :

Statement (I) : On hydrolysis, oligo peptides give rise to fewer number of α -amino acids while proteins give rise to a large number of β -amino acids.

Statement (II) : Natural proteins are denatured by acids which convert the water soluble form of fibrous proteins to their water insoluble form.

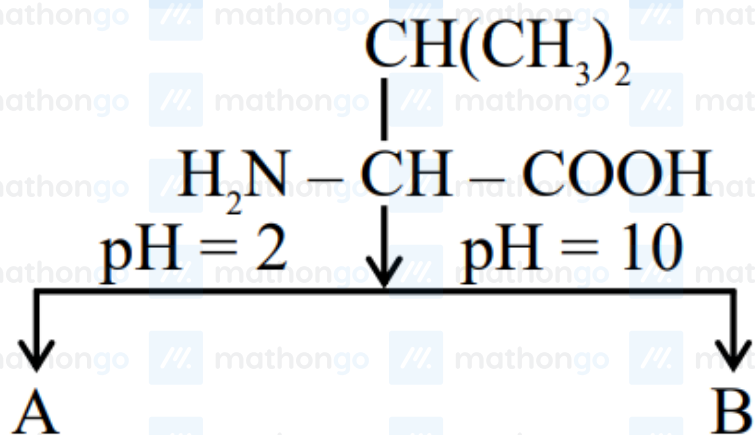
In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct
- (2) Statement I is incorrect but Statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Statement I is correct but Statement II is incorrect

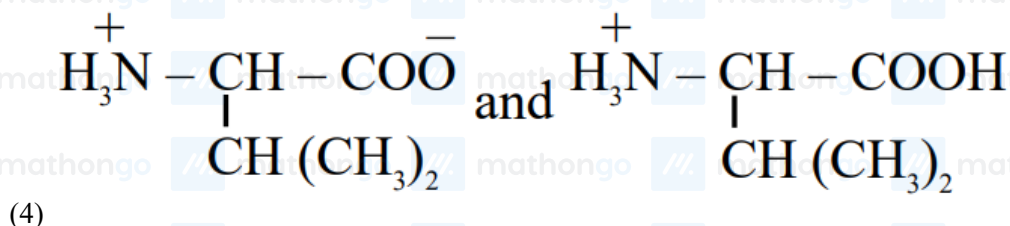
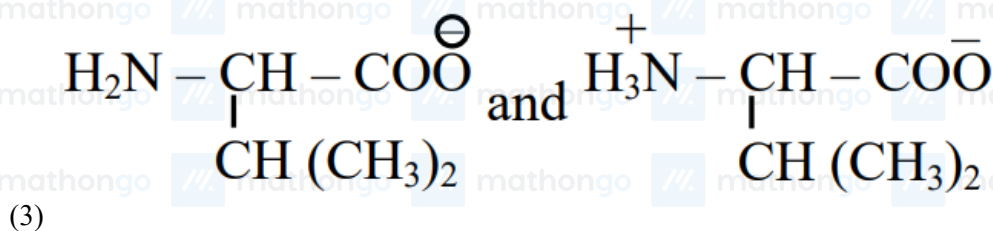
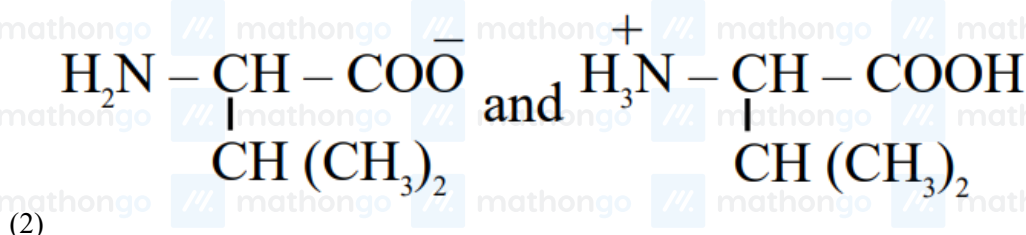
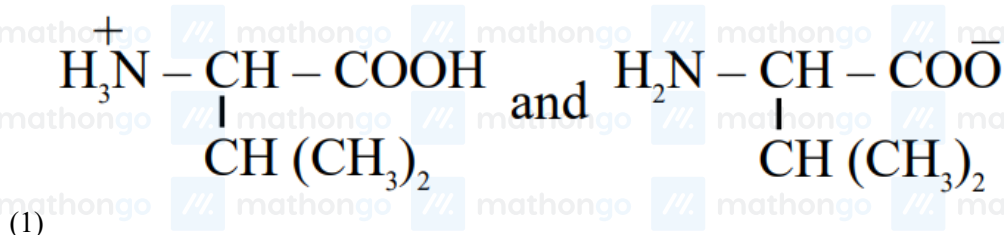
Q10. The total number of hydrogen bonds of a DNA-double Helix strand whose one strand has the following sequence of bases is _____.

5' - G - G-C-A-A-A-T-C-G-G-C-T-A-3'

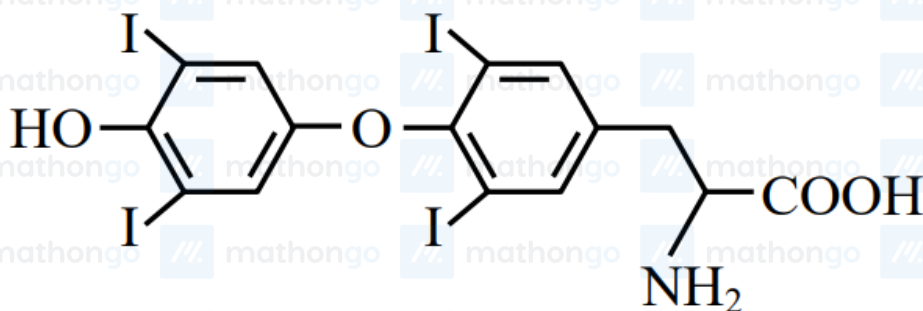
Q11.



Choose the correct option for structures of A and B, respectively.



Q1. Thyroxine, the hormone has given below structure



The percentage of iodine in thyroxine is ____ %. (nearest integer)

(Given molar mass in gmol^{-1} C : 12, H : 1, O : 16, N : 14, I : 127)